Backlash or Bonding: Effect of an Integration Policy in the Context of Affirmative Action*

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Abstract

Inter-group contact can reshape existing social networks. Do policies that promote cross-group connections expand access to new economic opportunities, or do they come at the expense of support from within-group networks? I examine this trade-off in the context of caste integration at a selective public university in India, exploiting random assignment of first-year roommates. I use original survey data from 2,400 students across two cohorts, combined with administrative data on roommate assignments. I develop a model in which students allocate social time between bonding social capital, which supports well-being, and bridging social capital, which expands access to economic opportunities. Results show that, one year after assignment, lower-caste students in mixed-caste rooms have better grades but worse mental health, while upper-caste students have lower grades, relative to peers of the same caste in same-caste rooms. Lower-caste academic gains appear to be driven by higher-performing roommates, while upper-caste losses are consistent with reduced same-caste collaboration. Social frictions in mixed rooms reduce well-being for both groups. Three years later, lower-caste students' academic gains persist and the mental health effects fade, but they are less likely to receive third-year internship offers. Upper-caste students earn lower internship stipends, and their grade losses persist. After the first year, both groups become more supportive of affirmative action (AA), each updating attitudes toward the policy dimension concerning the other group. By the third year, upper-caste students become less supportive of AA, particularly in employment. Taken together, the findings show that while cross-caste contact can reduce academic and labor inequalities, but also impose different costs on both groups.

Keywords: Affirmative action; caste integration; intergroup contact; academic outcomes; mental health
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1 Introduction

Affirmative action (AA) is a widely used policy tool for addressing historical marginalization (Gisselquist et al., 2024). India, one of its early adopters, relies mainly on caste-based quotas (reservations) in education, public employment, and politics. These quotas increase the representation of disadvantaged groups (A. Deshpande, 2013) and are associated with better educational attainment and earnings (Bagde et al., 2016; Bertrand et al., 2010), political empowerment (Chin and Prakash, 2011; Gulzar et al., 2020), and upward mobility (Asher et al., 2024). The effects of AA may extend beyond admission. College is a formative period in which students build social connections that shape later economic opportunities. By bringing students from different castes together, AA may help lower-caste students form ties with upper-caste peers and gain access to their social capital (Chetty et al., 2022a, 2022b). Yet cross-caste ties may be limited by identity-based frictions (Akerlof and Kranton, 2000) or by upper-caste students' view that AA is unfair. Backlash may also erode support for the policy and create political and legal challenges, threatening both access to selective institutions and the opportunities they create for disadvantaged students.[1]

Intergroup contact can reduce prejudice and promote social integration, a premise rooted in the contact hypothesis (Allport, 1954; Pettigrew and Tropp, 2006). In selective institutions, where large quotas operate alongside competitive admissions, contact may affect academic and labor-market outcomes as well as attitudes. Students from different castes often enter with different levels of academic preparation and access to social capital,[2] so the costs and benefits of integration are theoretically ambiguous and may differ across groups. Lower-caste students may benefit from better-prepared peers and from networks that help them enter high-paying jobs (Rivera, 2015). I call these benefits bridging social capital, following (Granovetter, 1973; Putnam, 2000). For upper caste students, contact may weaken stereotypes and reduce opposition to affirmative action. Yet building cross-group ties can come at the expense of within-caste relationships, which provide access to economic opportunities (Munshi, 2019) as well as psychological support and a sense of belonging. I call these benefits bonding social capital (Putnam, 2000).

This paper studies these trade-offs by exploiting the random assignment of first-year roommates at a highly selective engineering institution in India. I ask three questions. First, what are the costs and benefits of inter-caste contact for the academic and labor-market outcomes of upper- and lower-caste students? Does contact affect mental health in either group? Third, how does this contact shape support for affirmative action among upper- and lower-caste students over time? The Indian context is particularly well suited to studying these questions for two reasons. First, caste-based affirmative action reserves roughly 50% of seats, creating an opportunity to study the effects of integration on both students from historically advantaged and disadvantaged groups within the same selective academic environment. Second, admission combines a highly compet-

[1] Examples include protests by upper-caste groups in India, such as those at Jantar Mantar in August 2026 (ThePrint, 2026), and the US Supreme Court's 2023 decision ending race-conscious admissions in *Students for Fair Admissions, Inc. v. President and Fellows of Harvard College*.

[2] Social capital refers to the resources individuals can access through their networks (Bourdieu, 1986; Coleman, 1988).

itive national entrance examination with caste-based quotas, and caste-specific admission cutoffs are publicly known. Students therefore enter college with large differences (almost 1 s.d.) in academic performance, and AA links these differences to caste identity. This creates scope for cross-group contact to generate benefits through the exchange of resources and knowledge, but also frictions through academic competition, social comparison, and opposition to affirmative action.

I collect original survey data from two cohorts of approximately 1,200 students each, entering a highly selective Indian engineering institute in 2023 and 2024, and combine it with administrative records on roommate assignments, entrance examination scores, and caste status. Both cohorts were surveyed one year after entering college. Additionally, the 2024 cohort was surveyed at baseline, during the first week of college, while the 2023 cohort was followed up three years after entry to measure medium-run outcomes. Response rates exceeded 90% in every wave.

My empirical strategy exploits the random assignment of first-year roommates to estimate the causal effects of caste integration, comparing students assigned to mixed rooms vs same-caste rooms. I validate this random assignment by comparing observed mixing rates with theoretical probabilities based on hostel-level caste composition and room capacity, and by testing balance in predetermined student characteristics. I estimate treatment effects separately for lower- and upper-caste students to allow the consequences of integration to differ across groups. My primary outcomes are academic performance, internship outcomes[3], and mental health. I also examine attitudes toward affirmative action in the short and medium run.

Results show that for lower-caste students, integration yields academic gains at a psychological cost: first-year GPA rises by 0.091 standard deviations (s.d.), but mental health declines by 0.082 s.d. For upper-caste students, the academic effect is reversed: GPA falls by 0.103 s.d., accompanied by, statistically insignificant decline in mental health (0.062 s.d.). Consequently, while mixed-room assignment reduces the overall caste achievement gap by 22%, this convergence comes at an academic cost to advantaged students and a combined psychological cost.

To interpret these findings, I develop a model of peer transmission and social integration. Academic performance reflects the transmission of academic capital from better- to less-prepared students, alongside a fixed cost of social mixing. This transmission is asymmetric: students benefit from academically stronger peers, while having a less-prepared peer does not impose an equivalent academic loss. The model therefore predicts that lower-caste students benefit academically from better-prepared roommates in mixed rooms but may also incur social costs. For upper-caste students, the model predicts unambiguously negative effects: they are paired with less-prepared peers and may also bear the costs of social mixing.

In the model individuals allocate scarce social time between building bonding and bridging capital, shaping their mental well-being and labor-market outcomes. Social alignment increases the bonding returns to roommate interactions, while social distance can create opportunities to accumulate bridging capital through access to new information and networks, with these returns

[3]Internship outcomes provide a proxy measure of labor-market outcomes, as students have not yet entered the full-time labor market. The internships are relatively prestigious in this context, with average and median stipends equivalent to approximately 70% and 55% of the corresponding average and median first-job salaries, respectively.

depending on the roommate's social advantage.[4] Individuals choose between in-room interactions and outside social environments, such as student clubs, based on the returns to bonding and bridging capital in each setting. This choice depends on caste alignment, students' own advantage, which shapes their access to alternative social environments, and their roommate's advantage, which affects the bridging returns to in-room interactions. The model predicts that lower-caste students rely more on roommates for bonding capital, whereas upper-caste students are better able to diversify their social networks. Caste misalignment can therefore generate a trade-off between mental well-being and internship outcomes, with the balance depending on the roommate's advantage and students' access to outside options. Finally, I model bridging capital, through weak ties, as facilitating access to lower-stakes opportunities such as early college internships, while bonding and bridging capital are complements in accessing more competitive, higher-stakes opportunities such as elite internships.

My analysis on mechanisms is guided by the model predictions. The academic gains for lower-caste students appear to reflect peer transmission. The mixed-room effect falls from 0.091 to 0.02 standard deviations and becomes statistically insignificant when controlling for roommates' entrance-exam scores, consistent with gains from exposure to more academically prepared peers. This interpretation is further supported by lower-caste students reporting more academic help from upper-caste roommates, while I find little evidence of increased study effort or group work. By contrast, the academic losses for upper-caste students are not explained by exposure to less-prepared roommates. Instead, upper-caste students in mixed rooms report receiving less help from their roommates and exhibit fewer collaborative study interactions, suggesting that the losses may reflect reduced academic collaboration rather than peer quality. I cannot, however, rule out other channels, including effects of mental health or caste-related social distance on academic performance.

I next examine social interactions in mixed- and same-caste rooms to understand the mechanisms underlying the adverse mental well-being effects of caste mixing. Descriptive patterns support two key features of the model. First, lower-caste students interact more with roommates in same-caste rooms and participate less in student clubs than upper-caste students in same-caste rooms, suggesting greater dependence on in-room relationships. Consistent with this pattern, lower-caste students in mixed rooms report fewer social interactions with roommates, including eating together and going out, as well as lower-quality interactions, reflected in less psychological and financial support. Second, upper-caste students in mixed rooms experience frictions associated with reduced access to within-caste networks, consistent with their reporting greater exclusion from study groups. Their mental well-being effects disappear in three-person rooms when another upper-caste roommate is present, suggesting that access to same-caste peers can buffer the psychological costs of mixing. Finally, I examine roommate choices in the second year. Consistent with social frictions in mixed rooms, students become more likely to sort into same-caste

[4]Social advantage refers to family income, parental education, and schooling quality, among other factors. It is distinct from, but correlated with, academic performance.

rooms following the initial randomized assignment. The share of mixed rooms declines by 8.3 percentage points.

These short-run results suggest that mixing imposes academic costs on upper-caste students, while lower-caste students' academic gains come at a cost to their mental health. As students adjust to the institute's environment, expand their social networks beyond their assigned roommates, and transition to specialized coursework, the initial effects of mixing may attenuate or compound. To examine these dynamics, I conducted a follow-up survey of the 2023 cohort three years after entry, two years after roommate randomization ended.

I estimate the medium-run intent-to-treat effects of the initial roommate assignment. Two findings emerge. First, the effects of caste integration on grades largely persist through the third year, while the adverse treatment effects on mental health fade for both upper- and lower-caste students. However, mental health deteriorates over time among students assigned to both mixed-caste and same-caste rooms, with a larger decline among lower-caste students. Consequently, the mental-health gap between upper- and lower-caste students widens from near zero at entry to 0.15 standard deviations by the third year. Second, the initial mixed-room assignment has persistent consequences for labor-market outcomes. By the third year, upper-caste students initially assigned to mixed-caste rooms earn INR 32,556 less in internships (a 20% reduction, approx 310 euros) than their counterparts initially assigned to same-caste rooms. These internships are consequential: median stipends are five to six times larger than in the second year and often provide a pathway to permanent employment. By the third year, lower-caste students in mixed rooms are 7.3 percentage points less likely to receive a summer internship offer. Their internship stipends are also lower, by INR 9,466 (14.5%, 90 euros), although this estimate is imprecise.

Finally, I examine whether caste integration affects attitudes toward affirmative action (AA). In the short run, support for AA increases among both lower- and upper-caste students, with each group revising its beliefs about how the policy affects the other group. Among lower-caste students, support increases by 0.110 SD ($p < 0.05$), driven by the belief that quotas are not unfair to upper-caste students. Among upper-caste students, support increases by 0.115 SD ($p < 0.05$), driven by the belief that affirmative action is justified on social justice grounds. I conduct three heterogeneity tests to examine whose beliefs change following caste mixing. The results suggest that mixing induces greater belief updating among students whose prior beliefs or social experiences differ from those of their roommates. First, students who enter college believing that entrance-exam performance reflects merit rather than circumstances are more likely to revise their views. Second, among upper-caste students, the effects are strongest for those with the fewest pre-college cross-caste friendships and are concentrated in support for policies benefiting reserved groups—the dimension of AA about which they were least informed at baseline. Third, higher-performing students become more supportive of AA following caste mixing. In the medium run, the gains fade among upper-caste students, with negative point estimates for both beliefs about AA's costs and its social justice justification, while effects among lower-caste students remain positive but only marginally significant. Across policy domains, caste integration significantly reduces

upper-caste support for employment reservations, while the negative effect on support for political reservations is imprecisely estimated. Among lower-caste students, estimates are positive but statistically insignificant across employment, politics, and education.

This paper contributes to four strands of the literature. First, I contribute to the literature on social capital and economic mobility. Classical work theorizes that social capital shapes human capital accumulation and labor-market opportunities (Coleman, 1988; Granovetter, 1973), raising concerns about its decline (Putnam, 2000; Putnam et al., 2003). Recent work decomposes cross-class friendship gaps into differences in exposure and tie formation (Chetty et al., 2022b). Higher education institutions shape social connections that affect career outcomes (Michelman et al., 2022; S. D. Zimmerman, 2019), long-run mobility (Barrios-Fernández et al., 2026), and access to higher-status friends and partners (Londoño-Vélez et al., 2026). With tertiary education enrollment rising rapidly across many low- and middle-income countries, social capital formation during these formative years can have long-term consequences for economic growth and inequality. Yet evidence on how social capital forms in higher education and transmits across groups remains limited in low- and middle-income countries. I extend this literature by studying the returns to cross-caste exposure and interactions in India, where cross-group relationships are shaped by not only by socioeconomic differences but also identity-based frictions. Randomized roommate assignment at a highly selective engineering institution isolates the returns to cross-caste peer exposure from the broader benefits of selective university access, including institutional resources and signaling benefits (Barrera-Osorio and Bayona-Rodríguez, 2019; Sekhri, 2020). Randomized roommate designs have also been used to study academic outcomes in developing-country settings (Ahimbisibwe et al., 2026; Corno et al., 2022), highlighting mechanisms such as reduced prejudice, cultural similarity, and observable signals of ability. I extend this work by examining how cross-caste ties transmit academic advantages and create access to opportunities when groups differ substantially in prior academic preparation, and how identity-based frictions limit these returns. Disadvantaged students improve their grades through exposure to better-prepared roommates, and come at a cost to mental health and lower third year summer internship offers. I also show that integration can impose costs on advantaged students. While prior roommate-assignment studies find limited adverse effects on this group, I document lower academic performance and internship stipends among upper-caste students assigned to mixed rooms.

Second, I contribute to research on peer effects and network formation (Banerjee et al., 2024; Bramoullé et al., 2016; Jackson et al., 2022; Sacerdote, 2011; D. J. Zimmerman, 2003). Peer effects depend not only on assigned peers' characteristics but also on the relationships through which their inputs are transmitted (Carrell et al., 2013; List and Uchida, 2024), and recent work incorporates endogenous network responses into peer-matching designs (Bhargava, 2025). Yet less is known about what resources flow through assigned relationships and how their transmission varies across groups. Different ties transmit different resources: connections to higher-SES individuals may improve economic outcomes, while within-group ties support psychological well-being (Fleischman, 2026), reflecting the distinction between bridging and bonding social capital

(Putnam, 2000). Using detailed data on roommate support exchanges and actual interaction. I show how caste and prior academic preparation shape resource transmission. Lower-caste students in mixed rooms receive more academic help but less psychological and financial support from upper-caste roommates and engage in fewer shared social activities. Upper-caste students receive less academic help and psychological and financial help. These findings reveal an asymmetry in integration: academic support flows primarily from better- to lower-performing students across caste groups, while psychological support appears more closely tied to within-group relationships.

Third, I contribute to the literature on how intergroup contact shapes support for redistributive policies that bring groups together. Beliefs about fairness and social mobility shape redistributive preferences (Alesina and Giuliano, 2011; Alesina and La Ferrara, 2005; Alesina et al., 2018), while contact can reduce prejudice and improve social cohesion, particularly when interactions are cooperative (see Lowe, 2025 for a review; Lowe, 2021; Mousa, 2020). Much of this work examines attitudes toward outgroups rather than support for the redistributive policies that bring groups into contact. The closest study, Rao, 2019, examines affirmative action (AA) admissions to private primary schools in India and finds that having poorer classmates makes wealthier students more prosocial and less likely to discriminate against poorer students. I examine how randomized cross-caste roommate assignment shapes support for affirmative action (AA) among lower- and upper-caste students, distinguishing between different dimensions of AA's costs and benefits rather than treating support as a single attitude—a distinction that matters because both groups revise their beliefs about AA on dimensions concerning the other group in the short run. Three years later, treatment effects on upper-caste students' support for AA on the measures that increased in the short run attenuate, while their support for employment and political reservations declines. Positive effects among lower-caste students persist, though they are only marginally significant. These findings show that contact can reshape beliefs about redistributive policy, but its effects on policy support may differ across groups and over time.

Fourth, I contribute to the literature on the socioeconomic effects of caste in India by examining how affirmative action (AA) shapes social interactions and economic opportunities within higher education. Studies of AA have largely focused on access to higher education (Bagde et al., 2016; Bertrand et al., 2010; Cassan, 2019; Frisancho and Krishna, 2016). A related literature documents how caste shapes economic interactions, including water trading (Anderson, 2011), technology diffusion (Emerick, 2018), schooling and occupational choices through caste networks (Munshi, 2019; Munshi and Rosenzweig, 2006), and labor supply through caste identity (Oh, 2023). Recent work further highlights the role of affirmative action and within-caste networks in narrowing educational and occupational gaps between historically disadvantaged and upper-caste groups in India (Gupta et al., 2025). I complement this work by studying another channel through which AA may shape economic mobility: the formation of cross-caste connections within educational institutions. Using randomized roommate assignment and following both beneficiaries and non-beneficiaries of AA, I show that mixed-caste room assignments reduce social interaction and affect

academic performance, internship outcomes, and psychological well-being in both groups. These findings highlight how pre-existing inequalities and identity-based frictions shape the returns to cross-caste connections, even when students gain access to the same educational environment.

The rest of the paper is organized as follows. Section 2 describes the institutional context and data. Section 3 presents descriptive statistics. Section 4 describes the empirical specification. Section 5 presents the effects of roommate assignment on the primary outcomes. Section 6 develops a model and derives its key predictions. Section 7 uses the predictions of the model to examine the mechanisms underlying the empirical results. Section 8 studies the medium-run effects of integration. Section 9 examines short- and medium-run effects on support for affirmative action (AA), and Section 10 concludes.

2 Context and Data

2.1 Affirmative action, elite colleges, and merit

The institute is an elite public engineering school that enrolls approximately 1,200 undergraduate students each year. Affirmative action in India's public higher education is largely caste-based and through fixed quotas. Central educational institutions reserve 27% of seats for Other Backward Classes (OBC, non-creamy-layer), 15% for Scheduled Castes (SC), and 7.5% for Scheduled Tribes (ST). OBCs comprise castes classified as socially and educationally disadvantaged, while SCs and STs include groups that have historically faced severe caste-based discrimination and social exclusion.[5] An additional 10% of seats are reserved under the Economically Weaker Sections (EWS) category for students who are not eligible for the OBC, SC, or ST quotas. I classify EWS students together with non-reserved students as upper-caste throughout the analysis. The admissions system also includes supernumerary seats for women, introduced in elite engineering institutions in 2018 to increase women's representation without reducing seats available through other admission categories. All the quotas are implemented at specialization-college level.

Selective engineering colleges provide an important setting in which affirmative action intersects with highly competitive academic selection. Caste-based quotas have been part of the institutional environment of India's most selective engineering colleges since 1973 and remain among the country's most debated social policies (A. Deshpande, 2013; Henry and Ferry, 2017). Admission is based on the Joint Entrance Examination (JEE), one of India's most competitive national examinations, taken at the end of grade 12 in mathematics, physics, and chemistry. Each year, approximately 1.6 million candidates register for the examination, competing for around 16,000 undergraduate seats across these public engineering colleges. The JEE provides a common metric for ranking applicants and is central to the meritocratic model of selection in India's engineering institutes (Subramanian, 2015). However, reservation policies alter how examination performance

[5] I refer to them collectively as lower-caste students for brevity, although these categories are institutionally defined and internally heterogeneous. OBCs, in particular, are not stigmatized in the same way as SCs. Pooling these groups is therefore a modeling choice that allows me to analyze the effects on students who enter through caste-based quotas.

translates into admission: applicants compete within admission categories with category-specific cutoffs published annually. Students often use previous years' cutoffs to assess their admission chances.[6] Consequently, students entering the same specialization can have substantially different entrance-examination scores across admission categories (Bagde et al., 2016; Bertrand et al., 2010). This combination of common academic ranking and category-specific admission thresholds is a defining feature of admission to public higher education in India.

These differences in examination performance reflect, in part, the unequal conditions under which students prepare for the JEE. Preparation depends on prior schooling, family resources, coaching, and educational and social networks, while caste and economic status remain closely related in India (A. Deshpande, 2011). The JEE thus provides a common measure of academic achievement while also reflecting differences in prior opportunities. This relationship between social background and achievement is central to scholarship on merit and engineering education in India (S. Deshpande, 2013). As Subramanian, 2015 argues, the institutional history of these colleges illustrates how advantages associated with caste, education, and social networks can be converted into achievements understood as individual merit. This perspective does not imply that the JEE fails to measure academic ability; rather, it highlights that achievement is produced under unequal conditions and that the interpretation of merit is shaped by social structures. These inequalities may also matter for peer interactions. Caste is not phenotypically ascriptive (A. Deshpande, 2011), and students from different parts of India may not know each other's caste when they first meet. However, conversation and social interaction can reveal signals of caste and social background that subsequently shape economic interactions and access to employment (A. Deshpande and Newman, 2007; Shukla, 2025; Thorat and Newman, 2010).

The stakes of these interactions extend beyond the classroom. Selective engineering institutes provide pathways into competitive labor markets, where students compete for internships, jobs, and occupations, and where college networks can facilitate access to information and opportunities. Evidence from other settings shows that attending elite colleges affects access to top jobs and incomes (Chetty et al., 2026; S. D. Zimmerman, 2019), while cross-group social connections can influence economic mobility (Chetty et al., 2022b). In India, formal-sector hiring is also shaped by social networks and caste (A. Deshpande and Newman, 2007; Jodhka and Newman, 2007). Peer interactions during college may therefore shape both students' academic experiences and the networks through which they access subsequent economic opportunities.

2.2 Room Allocation and the Natural Experiment

The institute is fully residential, with undergraduates assigned to one of 13 single-gender hostels for the duration of their degree[7]. Ninety-seven percent of students are assigned to two- or three-person rooms; I exclude the remaining 3% in single or four-person rooms from the analysis.

[6] After receiving their ranks, students submit preferences over colleges and specializations, and seats are allocated using a modified deferred acceptance algorithm.

[7] 11 male hostels and 2 female hostels

Room assignment follows a two-stage process. First, central administration assigns students to hostels: for each specialization, an allocation matrix based on hostel vacancies and number of mentors from that specialization. Once the number of vacancies is decided, students are distributed randomly across hostels, stratified by caste category. Second, within each hostel, administrators assign students to specific rooms, either sequentially from a college-ID-sorted list or in order of move-in arrival, a method that is consistent within each hostel across cohorts. College IDs encode the year of entry and specialization, together with a randomly assigned, unique four-digit number for each student. The list sent to hostels from college administration does not include caste information or entrance exam score ranks. In the baseline survey for the cohort, only 1% of students reported knowing their roommate before college, and Figure 1 shows the observed share of mixed-caste rooms closely matches the theoretical share under random assignment in every hostel, regardless of allocation method. Approximately 5% of students changed rooms during the first year. I also show balance of observables explained in detail in Section 4.1.

2.3 Data and Outcomes

I collected data from two cohorts, entering in 2023 and 2024, in three phases: baseline data from the 2024 cohort during orientation week in July 2024; a cross-sectional survey of the 2023 cohort between August and October 2024, roughly one year after their own roommate assignment; and a one-year follow-up with the 2024 cohort in August 2025. I later returned to the 2023 cohort in April 2026 for an additional wave, described below, to measure medium-run outcomes.

All surveys were self-administered online, with in-person sessions and subsequent tracking to facilitate participation and follow-up. Students were recruited through hostel orientation sessions and subsequent hostel-level follow-up. The study was described as a study of "student life in college and how it changes over time," without reference to caste or roommate assignment. Students received INR 200 for the baseline survey and INR 300 for the endline and follow-up surveys. The survey protocol, sampling and tracking procedures, construction of student and hostel lists, respondent identification, and procedures for detecting and resolving duplicate submissions are described in Appendix Section A.

The 2024 cohort's baseline survey was administered during orientation week, when students had minimal roommate contact: 90% of responses were collected within two to seven days of move-in, the remainder within one additional week.[8] I report detailed survey response rates under alternative definitions of response in Table A1. The 2024 cohort achieved an 85.1% baseline response rate (1,031 of 1,212 students), with 980 students (80.9%) completing the full survey. The one-year follow-up achieved a 91.1% response rate, with 89.4% completing the full survey. The 2023 cohort was surveyed using a different protocol because of logistical constraints. The initial response rate was 71.8%; I then drew a simple random sample of 25% of the initial nonrespondents for intensive follow-up. Of these selected nonrespondents, 73% completed the survey, yielding an

[8]This timing is directly relevant to interpreting the baseline measures in Fact 3 and several mechanism tests in Section 7; see the discussion there.

effective response rate of 92.8% (90.2% full completion) following Kling et al., 2007. This two-tier sampling procedure is the basis for the survey weights used in the analysis.[9]

Medium-run follow-up (2023 cohort). Because the 2023 cohort entered one year ahead of the 2024 cohort, I returned to survey them once more, at approximately three years after their first-year roommate assignment. These waves add two outcomes not available in the first-year data. First, third-year summer internship placement, both completion and monthly stipend, capturing outcomes organized primarily through the institute's placement cell and substantially higher-stakes than the first-year internship measure. Second, a broader affirmative-action attitudes module, extending the first-year AA items to test whether attitude shifts generalize beyond the campus-specific setting to diversity and inclusion policies in the workplace.

Administrative and institutional data. I complement the survey data with JEE examination records published by the institute, containing exam ID, JEE score and rank, admission category, institute, specialization, and examination centre. I match survey respondents to these records using their self-reported JEE rank within cohort and specialization, with examination-centre location, gender, and category-specific cutoff ranks used to resolve ambiguous or inconsistent matches. For remaining unmatched observations, I use publicly available rank information, including coaching-institute rank lists and students' public professional profiles. This procedure identifies JEE records for 99% of the 2024 cohort and 95% of the 2023 cohort.

I obtain first-year room-allocation lists separately and identify roommates using room numbers and student names, linking each student to their caste category through JEE records. For the 2024 cohort, all room-allocation data come from administrative lists. For the 2023 cohort, administrative lists cover 9 of the 13 hostels. For the remaining four hostels, I reconstruct room assignments using students' reported room numbers and roommate names, following up to clarify flagged cases. I use the initial assignment from administrative records and The construction and validation of the JEE measures, caste categories, and hostel-room assignments are described in Appendix Section A.

2.4 Primary Outcomes

2.4.1 Academic and Labor-Market Outcomes

Academic outcomes The primary short-run academic outcome is first-year GPA. In the medium-run follow-up, I observe second-year GPA and GPA at the time of the survey, corresponding to cumulative GPA through the fifth semester.

Labor-market outcomes I measure labor-market outcomes using summer internship offers and internship stipends in the summers following the first, second, and third years of study.

[9] Among the initial nonrespondents, each student had a 25% probability of being selected for intensive follow-up. Accordingly, initial respondents receive weight 1, while selected initial nonrespondents receive weight 4 in analyses that account for this sampling design.

2.4.2 Mental Health and Attitudes

Mental health I measure mental health using three modules: the PHQ-4 clinical screener, comprising two GAD-2 anxiety items and two PHQ-2 depression items; a relative well-being module comparing current and pre-college well-being across four dimensions; and a college-worries module covering academics, internships, discrimination, cultural and language differences, and homesickness. I construct a separate index for each module and combine the three module-level indices into a composite Mental Health Index. Higher values indicate better mental health.

Support for affirmative action I measure attitudes toward affirmative action using five-point Likert-scale items on the perceived costs and benefits of affirmative action for reserved- and non-reserved-category students. I construct separate indices capturing support based on perceived costs and benefits for each group, as well as a composite index combining both. Higher values indicate greater support for affirmative action.

All indices in the paper follow the procedure in Kling et al., 2007. I standardize each component using the mean and standard deviation among students assigned to non-mixed rooms, the control group. I then construct each index as an unweighted average of its standardized components. For composite indices, I first construct the component indices and then take an unweighted average across the available component indices. When a respondent is missing one or more components, I calculate the index using the non-missing components.

2.5 Secondary Outcomes

Social interactions I measure interactions with roommates along several dimensions, including time spent studying together and participation in social activities such as eating together, going out, and playing sports. I combine these measures into an index of social interaction. I also ask students whether they have received help from or provided help to each roommate with exam preparation, obtaining internships, dealing with stress, financial assistance, and access to social and professional networks. I construct separate indices of help received and help provided, as well as measures of the types of help exchanged. I additionally measure help received from other hostel mates.

Social networks I measure the number of close friends within the institute, the number of close friends at other selective institutions, and the share of close friends belonging to a different caste.

Study effort and student activities I measure self-reported study time, participation in student clubs, and whether students hold leadership positions in clubs.

Roommate choice From the second year onward, students choose their own roommates. I observe these choices in hostel-allocation records and examine whether first-year mixed- versus same-caste roommate assignments predict subsequent roommate choices along caste lines.

3 Descriptive Stats

Table 1 presents summary statistics on students' economic background, social capital, college preparation, pre-college test scores, mental health, and attitudes toward affirmative action. These characteristics also provide key descriptive facts that motivate the model[10].

Fact 1: Upper-caste students have socioeconomic advantage. From Panel A of Table 1, upper-caste students are more likely to have parents with higher levels of education, to know someone who attended an elite engineering college or studied engineering, and to come from wealthier households, as measured by income and assets. They are also less likely to take out education loans.

Fact 2: Upper-caste students enter college with stronger academic performance. From Panel B of Table 1, their entrance-exam scores are about 60 points (1.04 standard deviations) higher than those of lower-caste students. They are also 13.6 percentage points less likely to take an additional year to clear the exam. Upper-caste students begin preparing for the exam approximately 0.3 years earlier, although spending on coaching, which is relatively standardized across students, is similar across groups. Lower-caste students do not appear to exert less effort in terms of study hours. Upper-caste students also have small but higher academic performance in high school, as measured by their Class 10 and Class 12 scores.

Fact 3: Upper-caste students enter college with more advantaged friendship networks. From Panel C of Table 1, Upper-caste students report more close friends and fewer friends who did not clear the entrance examination. They also have slightly more friends who attended IITs. In contrast, lower-caste students have more caste-diverse friendship networks.

Fact 4: Lower-caste students hold more favorable views of affirmative action. Panel D of Table 1 shows that lower-caste students have higher scores on both a reserved group index, capturing perceived costs and benefits of affirmative action for reserved-group students, and a non-reserved-group index, capturing perceived costs and benefits for non-reserved-group students (0.675 and 0.827 s.d., respectively). The same pattern shows up in a policy preference: lower-caste students favor allocating a larger share of seats to reserved groups, 24 out of 100 on average, compared with 10 among upper-caste students, though both groups favor a share well below the actual reservation rate of 50%. Lower-caste students are also less likely to endorse the view that the entrance examination is cleared solely through merit. Both groups are equally likely to believe opportunities inside college are equal.

Fact 5: Mental health at the start of the academic year is broadly similar across caste groups, but differs in specific dimensions. Panel E of Table 1 shows that the overall Mental health index is similar across caste groups at the start of the academic year. Lower-caste students report higher relative well-being compared to their time preparing for the entrance exam, but also higher levels of worry, particularly about the social aspects of college life[11]. They are also more satisfied with

[10] Panels A and B draw on both the 2023 and 2024 cohorts; the variables in Panels C, D, and E were collected only for the 2024 cohort.

[11] Because this measure was collected after room assignment, it may already reflect early worries induced by the

their specialization or department. There is no statistically significant difference in the PHQ-4 index at the start of the academic year.

3.1 Roommate Interactions and Academic and Social Support

Table 2 documents frequent academic and social interactions among students in same-caste rooms, with some differences by caste.

Substantial Academic and social interactions exist between roommates. Studying together at least once a week is reported by 46.2% of upper-caste and 57.4% of lower-caste students. 66% of students in both groups report receiving substantial help from a roommate with exam preparation. Social interactions are also common: 59.7% of upper-caste and 65.1% of lower-caste students report eating together at least once a week, while 30.4% and 34.3%, respectively, report going out together. The caste differences in these social activities are small and not statistically significant.

Lower-caste students report greater exchanges of support. Studying together at least once a week is reported by 46.2% of upper-caste and 57.4% of lower-caste students. About 66% of students in both groups report receiving substantial help from a roommate with exam preparation. Lower-caste students are more likely to provide emotional support and report greater exchanges of financial support. Social interactions, such as eating and playing sports together, are also more common among lower-caste students. I report support exchanges at the extensive and intensive margins in Table A2. The differences are driven by the extensive margin: lower-caste students are more likely to both seek and provide support. Conditional on being asked for help, upper-caste students are more likely to provide job-related assistance, but there are no significant caste differences in emotional or financial support.

Upper-caste students are more active in clubs and hold more positions Upper-caste students participate more in student clubs that provide opportunities to build networks and skills (Table A3). They are significantly more likely to be active members of cultural clubs, technical clubs, institute student bodies, and hostel student bodies, and to hold leadership positions in these organizations. Sports and volunteering clubs are exceptions: participation shows no clear caste differences, consistent with institutional requirements that students complete 100 hours of activities through sports, volunteering, or scouts.[12]

4 Empirical Specification

I estimate the effect of mixed-caste room assignment on a range of outcomes using the following baseline specification, estimated on the full sample and separately on the subsamples of upper-

treatment, I therefore do not use it as a baseline control, to avoid conditioning on a post-treatment variable.

[12]Cultural festivals at these institutes are large-scale events that require broad student participation. Top leadership positions are typically held by club office bearers, while lower-level positions are also available to newer members.

and lower-caste respondents:

$$Y_{ikcst} = \beta_0 + \beta_1 \text{Mixed}_{ikc0} + \beta_2 Y_{ikcs0} + \gamma_{kc} + \delta_{sc} + X_{ikcs0} + \varepsilon_{ikcst} \tag{1}$$

where Y_{ikcst} is the outcome for student i, in hostel k, cohort c, specialization s, at follow-up time t, and Y_{ikcs0} is the baseline value of the same variable.[13] The treatment variable Mixed_{ikc0} equals one if the student is assigned a room with at least one different-caste roommate. γ_{kc} is a vector of hostel-by-cohort fixed effects, controlling for systematic differences across hostels, such as room size, gender composition, and hostel size, and for differences in student composition across cohorts. δ_{sc} is a vector of specialization-by-cohort fixed effects, controlling for differences in students' academic backgrounds. X_{ikcs0} is a vector of controls, which includes the number of roommates. Standard errors are clustered at the room level, the level at which roommate assignment is determined following Abadie et al., 2017. As a robustness check against correlation in outcomes that extends beyond the room, I additionally report p-values clustered at the wing level.

The regression is estimated separately for upper-caste and lower-caste students, allowing the effect of roommate assignment to vary across groups. The coefficient of interest, β_1, is the intent-to-treat effect of inter-caste mixing on Y, given random assignment of roommates within the hostel and the cohort. I report ITT estimates.

I use additional variation from three-sharing rooms, distinguishing partially and fully mixed assignments:

$$Y_{ikcst} = \beta_0 + \beta_1 \text{Fully Mixed}_{ikc0} + \beta_2 \text{Partial Mixed}_{ikc0} + \gamma_{kc} + \delta_{sc} + X_{ikcs0} + \varepsilon_{ikcst} \tag{2}$$

where $\text{Partial Mixed}_{ikc0}$ equals one for a three-person room with exactly one different-caste roommate, and $\text{Fully Mixed}_{ikc0}$ equals one if all roommates are a different caste; two-person rooms are fully mixed by definition when different-caste. The omitted category is a same-caste room. This specification tests whether having at least one same-caste roommate moderates the effects of diversity in mixed rooms.

4.1 Balance Tables

Appendix Tables A4, A5 and A6 examine balance on pre-treatment characteristics. Table A4 presents balance on observable socioeconomic indicators and pre-college characteristics for mixed versus non-mixed rooms. There are no systematic differences between mixed and non-mixed rooms for either lower- or upper-caste students, with joint F-test p-values of 0.456 and 0.240 for lower-caste and upper-caste students, respectively[14]. Table A6 reports the analogous test for the

[13] Baseline measures of mental health and affirmative-action attitudes are available only for the 2024 cohort. Where a baseline measure is unavailable, I replace it with zero and include a dummy for missingness.

[14] The only individual covariate showing imbalance in the balance table (Table A4) is the family income dummy, and this imbalance disappears when income is measured continuously. It appears to be driven by a small fraction of upper-caste students classified as economically weaker section (EWS), who are 6 percentage points more likely to be assigned to mixed rooms relative to a baseline mixing share of 73% (equivalent to roughly 14 students), a small share of

alternative partial-mixed and fully-mixed specification, restricted to the subsample of students in three-person rooms. There are no systematic differences between same-caste and partially mixed rooms (p-values of 0.789 and 0.408 for lower- and upper-caste students, respectively) or between same-caste and fully mixed rooms (p-values of 0.515 and 0.600, respectively).

Table A5 presents balance on pre-college networks, measured using baseline data for the 2024 cohort only, and similarly shows no systematic differences between mixed and non-mixed rooms for either group. Taken together with Figure 1, which shows that the observed proportion of mixed rooms closely matches the theoretical probability under random assignment, weighted by hostel size, these results support the identification assumption that roommate allocation within the hostel is effectively random.

4.1.1 Attrition:

To address the concern that survey non-response may be related to room assignment, potentially biasing estimates even under random assignment, I examine whether item-level refusals and overall survey non-response are associated with mixed-room assignment, separately for upper- and lower-caste students. Table A7 reports attrition on baseline socioeconomic indicators and JEE preparation variables; Table A8 reports attrition on the mental health (PHQ-4) and academic (CGPA) outcomes used in the main analysis; and Table A9 reports attrition on affirmative-action attitude items. Across all three tables, differences in refusal and non-response rates between mixed and non-mixed rooms are small and almost uniformly statistically insignificant for both upper- and lower-caste students. These results suggest that attrition is not systematically related to treatment assignment.

5 Results

5.1 Main Outcomes: Short run 1 year after entering college

Table 3 reports the effects of mixed-caste room assignment on first-year grades, a composite mental health index (higher values indicate better mental health), and two dimensions of support for affirmative action: support in line with the policy's benefits and costs for reserved-category students, and support in line with its costs and benefits for non-reserved students. I find that caste mixing lowers grades for upper-caste students and improves them for lower-caste students, and it imposes mental health costs on both groups. Mixing also increases affirmative action support, but each group updates on the dimension concerning the other group's policy stake.

the 1,100 upper-caste students in the sample. This pattern is driven largely by small hostel-by-gender-by-specialization strata: nearly all of the imbalance comes from strata among girls, who typically have only one or two EWS students per specialization, together with a small number of strata among boys in three-sharing rooms where specialization size is small. There is no systematic income difference between general-category upper-caste students and EWS upper-caste students across mixed and non-mixed rooms. I show robustness of main results to reservation category fixed effects and also explicitly controlling for family income, parents' education and high-school test scores.

Academic outcomes: Upper-caste students assigned to mixed rooms score 0.103 s.d. lower at the end of the first year ($p < 0.05$) than upper-caste students assigned to same-caste rooms. In contrast, lower-caste students assigned to mixed rooms score 0.091 s.d. ($p < 0.1$) higher than their counterparts in same-caste rooms. Overall, mixing reduces the upper-lower caste achievement gap by 22%, relative to the 0.919 s.d. gap between lower- and upper-caste students in homogeneous rooms. This reduction in the gap doesn't come at the expense of average academic achievement: the pooled effect of mixing on grades is small, negative, and statistically indistinguishable from zero.

As additional checks, I find no significant heterogeneity in the grade effect of mixed-room assignment by gender (Table A10). For room composition, point estimates for upper-caste students suggest a somewhat smaller GPA reduction in partially mixed rooms (-0.085) than in fully mixed rooms (-0.112), but these estimates are not statistically different from each other. For lower-caste students, effects do not differ significantly by room composition.

Mental health: Mixed-room assignment worsens mental health for both caste groups, although the estimates are statistically significant only for lower-caste students (Column 2 of Table 3). The composite Mental Health Index falls by 0.082 standard deviations for lower-caste students ($p < 0.05$), and by 0.062 standard deviations for upper-caste students ($p = 0.162$). The difference between these estimates is not statistically significant ($p = 0.509$). Pooling across caste groups, mixed-room assignment reduces the index by 0.073 standard deviations($p < 0.05$). Mental health, on average, is worse in mixed-caste rooms compared to the same-caste rooms.

The pooled effect is driven primarily by the PHQ-4 index, which measures symptoms of anxiety and depression (Table A11). The estimates are positive for both upper-caste students (0.092 standard deviations) and lower-caste students (0.084 standard deviations), although neither is statistically significant individually. The pooled estimate is 0.098 standard deviations ($p < 0.05$), indicating an increase in anxiety and depressive symptoms on average. No single component accounts for the overall pattern. Depressive-symptom components (Columns 4 and 5 of Table A11) show somewhat larger effects for upper-caste students, whereas anxiety-symptom components (Columns 2 and 3) show somewhat larger effects for lower-caste students. However, these differences are too small to make meaningful conclusions about differential effects by symptom type. The effects of mixing on individual components of the worry index are imprecisely estimated, but their pattern suggests that increased worry may operate through different channels across caste groups. For upper-caste students, the pattern points to concerns about being far from home and time-management pressures, whereas for lower-caste students, it points to concerns about discrimination, cultural adaptation, and academic competition (Table A12). [15].

[15]Mixed-room assignment has no detectable effect on relative well-being (Table A13), measured against students' pre-college well-being; the estimates are precisely estimated zeros for both groups. Descriptive comparisons nevertheless reveal different patterns across caste groups. Relative to their pre-college levels, lower-caste students report greater happiness but also greater stress in same-caste rooms, consistent with the possibility that entry into an selective institution improves their overall well-being while the more competitive environment increases stress. Upper-caste students exhibit the opposite pattern in same-caste rooms. These descriptive patterns provide context for the treatment effects but don't make causal that entry into the institution itself caused the changes (Table A13)

Additionally, I find no significant heterogeneity in the mental health effect of mixed-room assignment by gender: the mixed-room-by-female interaction is small (-0.107) and insignificant for lower caste students and small and insignificant for upper caste students. This null interaction sits alongside a large baseline gender gap in mental health levels: in same-caste rooms, upper-caste females report mental health 0.215 s.d. lower than upper-caste males ($p < 0.01$), and lower-caste females similarly report lower levels than lower-caste males (-0.137 s.d., not significant), and in other words, there is a substantial gender gap in mental health levels for both upper and lower caste females independent of room assignment, but mixed-room assignment does not appear to widen or narrow that gap.

Affirmative Action attitudes. Columns 3 and 4 of Table 3 report effects on two dimensions of support for affirmative action (AA): the reserved-group index (Column 3), which captures beliefs about the costs and benefits of AA for reserved-category students, and the non-reserved-group index (Column 4), which captures beliefs about its costs and benefits for non-reserved-category students. All indices are coded so that positive values indicate greater support for AA. The results reveal an asymmetric pattern: each group shifts primarily on the dimension concerning the other group rather than its own. Upper-caste students become more supportive of AA on the reserved-group dimension (0.115 standard deviations), while lower-caste students become more supportive on the non-reserved-group dimension (0.110 standard deviations). Correspondingly, mixing reduces the cross-caste gap in support for AA by 21.5% on the dimension that AA is beneficial to the targeted groups, whereas the gap in costs and benefits to the non-targeted group does not change significantly.

For upper-caste students, the increase in support on the reserved-group dimension appears to reflect learning about the disadvantage of lower-caste students. To examine the underlying beliefs driving this change, I analyze the individual items composing the reserved-group index (Table A14). Upper-caste students update in the direction of greater support across most items, although the individual estimates are generally statistically insignificant. These include beliefs about social justice, fostering diversity, and whether AA leads to mismatch, with a somewhat smaller change in beliefs about whether AA reduces incentives to work hard. Upper-caste students also become more likely to believe that AA does not reduce institutional quality (0.142 standard deviations, $p < 0.1$). Overall support for AA also increases among students assigned socially distant roommates. However, there is no detectable change in the belief that AA is unfair relative to a purely merit-based admissions policy; the estimate is precisely zero. For lower-caste students, the increase in support on the non-reserved-group dimension is driven primarily by a shift in beliefs about the fairness of AA toward upper-caste students (Table A15). The point estimate indicates that lower-caste students become more likely to believe that AA is not unfair to upper-caste students relative to a purely merit-based admissions policy (0.114 standard deviations), although the estimate is not statistically significant. By contrast, their point estimates for beliefs about the costs and benefits of AA for their own group are negative but statistically insignificant (Table A14). Thus, mixing in the short run helps students learn about the other groups'

advantages and disadvantages and update their support for AA policy accordingly.

The main results are robust to two alternative specifications. First, I add controls for family income, parental education, and high school graduation scores (Table A16). The results remain almost unchanged. One exception is the effect on lower-caste students' support for AA on the non-reserved-group dimension. The p-value increases from 0.040 to 0.053. The estimate is no longer statistically significant at the 5% level. Second, I add reservation-category fixed effects (Table A17). This specification compares General and EWS students within the upper-caste group, and OBC and SC/ST students within the lower-caste group. It is more conservative as it absorbs variation across caste groups with-in upper and lower caste categories. For upper-caste students, the estimated effects on grades and AA attitudes remain similar in size. However, the estimates become less precise. The p-values increase to 0.066 and 0.060, respectively, from below 0.05 in the main specification. For lower-caste students, the estimated effects on mental health and AA support remain similar. The estimated effect on GPA falls from 0.091 to 0.083 standard deviations and is no longer statistically significant ($p = 0.11$). When I add demographic controls to this specification, the GPA effect becomes statistically significant at the 10% level. Overall, the estimates remain robust to alternative specifications.

6 Model

Facts 1, 3, and 4 in Section 3 show that integration brings together groups with different levels of socioeconomic background, social networks, and attitudes toward policies such as affirmative action. The short-run results pose a puzzle: why do grades and mental health diverge for lower-caste students, while upper-caste students bear costs on both dimensions?

In residential elite universities, students spend a substantial share of their time in hostels, but also in student clubs, some tied to hostel life and others operating at the broader college level, spanning social and technical activities. These spaces can provide an important opportunity for social mixing (Obminski, 2026), but access to and membership in them is often shaped by preexisting preparation (Michelman et al., 2022), such as having attended elite private schools. If frictions arise inside the room, students with better preparation may be able to substitute toward these alternative social spaces, but access to that substitute may itself be asymmetric across groups.

I use this hostel-and-club setting to develop a model of social integration, in which interactions between groups shape the accumulation of social and academic capital both within the room and in outside spaces such as clubs, with advantaged students better able to substitute toward the latter when frictions arise. These in turn affect grades, mental health, and labor-market outcomes. A companion model of belief updating, which accounts for the shift in attitudes toward affirmative action, is developed in Appendix B.

6.1 Environment and Social Capital

Let $\mathcal{N} = \{1, \ldots, n\}$ index an entering cohort. Student i has caste type $c_i \in \{U, L\}$, academic preparation θ_i (entrance score), and social advantage $A_i \geq 0$ (family income, parental education, schooling quality), distinct from but correlated with academic preparation. A room assignment $\rho \colon \mathcal{N} \rightrightarrows \mathcal{N}$ is imposed, $\rho_i \subseteq \mathcal{N} \setminus \{i\}$, with random assignment used for identification: within the strata used in the empirical analysis, a student's roommate is drawn independently of the student's own characteristics.

Definition 1 (Social distance). *$d_{ij} \geq 0$ is the social distance between roommates i and j, with $d_{ij} = 0$ for a same-caste pair and $d_{ij} = d_M > 0$ for a mixed pair.*

Assumption 1 (Endowment gap). *Affirmative action admits lower-caste students at a lower entrance threshold: $\theta \mid U$ first-order stochastically dominates $\theta \mid L$, and likewise $A \mid U$ dominates $A \mid L$ (Fact 1, Fact 2). Write $\Delta\theta \equiv \mathbb{E}[\theta \mid U] - \mathbb{E}[\theta \mid L] > 0$ and $\Delta A \equiv \mathbb{E}[A \mid U] - \mathbb{E}[A \mid L] > 0$. Let*

$$p \equiv \Pr(\theta_j > \theta_i \mid i \text{ lower-caste, room mixed}) \tag{3}$$

denote the probability that a mixed pairing runs in the expected direction. First-order stochastic dominance implies $p > 1/2$.

Assumption 2 (Two aligned relationships). *Roommate alignment $\lambda_{ij} = \lambda(d_{ij}) \in (0, 1]$ is decreasing in social distance d_{ij} (Definition 1), with $\lambda(0) = 1$ for same-caste roommates; in a mixed room $\lambda_{ij} = \lambda_M \equiv \lambda(d_M) < 1$.*

Club alignment $\iota_i \equiv \iota(c_i, A_i) \in [0, 1)$ is a student's alignment with the institution's student clubs, a function of caste c_i and social advantage A_i, nondecreasing in A_i for fixed c_i: within a caste, more advantaged students fit the club culture more closely. The caste ordering holds at every advantage level x, $\iota(U, x) > \iota(L, x)$, since returns to club participation are higher for upper-caste students at any given level of advantage. Alignment is strictly below one[16]*, so that in a same-caste room, where $\lambda_{ij} = 1$, time with the roommate always produces more belonging than time in clubs.*

Definition 2 (Belonging and bridging capital). *Social time is scarce: $s_i + a_i = 1$, where s_i is time with the roommate and a_i is time in clubs. Time investment converts into belonging and bridging capital,*

$$B_i = \lambda_{ij}\, s_i + \iota_i\, a_i, \qquad R_i = 1 - B_i. \tag{4}$$

Time with socially aligned peers generates belonging (strong ties, B_i). The remainder of social time, spent across social groups, generates bridging capital (weak ties, R_i), which provides access to new information and networks.

Definition 3 (Preparation gap and utility, cost, and return functions). *$\Delta_i \equiv (\theta_j - \theta_i)_+ \geq 0$ is the positive preparation score gap between roommates, generally positive even in same-caste rooms (same-caste*

[16] Student clubs contain a mix of upper- and lower-caste students, so a student's alignment with club peers is below the alignment with a same-caste roommate, $\lambda(0) = 1$.

expectation $\frac{1}{2}\delta_c$, Proposition 1); for a lower-caste student paired above in a mixed room, Δ_i is typically larger. $v(B_i)$ is the direct utility of belonging. $c(R_i;\Delta_i)$ is the cost of holding bridging capital R_i, evaluated at the preparation gap Δ_i. $\pi_{ij} \equiv \pi(A_j)$ is the network return to bridging with roommate j, a function of j's advantage A_j. π_g is the return to grades.

Assumption 3 (Curvature and returns)**.** *$v(\cdot)$ is increasing and strictly concave. $c(\cdot\,;\Delta)$ is twice continuously differentiable, increasing and convex in its first argument, with $c(0;\Delta) = c_R(0;\Delta) = 0$, and is increasing in Δ with $c_{R\Delta} \geq 0$, so that the marginal cost of bridging capital rises with the preparation gap. $\pi(\cdot) \geq 0$ is nondecreasing, so π_{ij} weakly rises in the roommate's advantage A_j. $\pi_g \geq 0$.*

Assumption 4 (Same-caste benchmark)**.** *The return to bridging satisfies $\pi(A) \leq v'(1)$ on the support of lower-caste advantage. Lower-caste students in same-caste rooms then choose full contact, $B_i = 1$ and $R_i = 0$. No such restriction is imposed for upper-caste students, who bridge in a same-caste room when their roommate's network value exceeds $v'(1)$.*

Assumption 5 (Social distance costs)**.** *Cross-caste roommate time carries a psychic cost $\chi_{c_i}(d_{ij}) \geq 0$, a function of social distance d_{ij} alone. In a mixed room $d_{ij} = d_M$ and I write $\chi_c(d_M)$ or simply $\chi_c(d)$. The slope $\gamma_c \equiv \chi_c'(d_M)$ satisfies $\gamma_L > 0$; γ_U is left unrestricted, so the model does not presuppose whether upper-caste responses reflect discomfort or competition. In a same-caste room, $d_{ij} = 0$ gives $\chi_c(0) = 0$.*

Assumption 6 (Club access)**.** *$\underline{A}_c \geq 0$ is a caste-specific advantage threshold for participation in the institution's clubs and societies for cross-group contact outside the room, reflecting institutional barriers to entry. Student i is* club-capable *if $A_i \geq \underline{A}_{c_i}$, and* club-excluded *otherwise. Club-excluded students derive no belonging from time outside the room, $\iota_i = 0$. Thresholds do not favor lower-caste students: $\underline{A}_L \geq \underline{A}_U$.*

6.2 Grades as Academic Transmission

Grades are determined first and enter the placement decision (§6.4). Learning is directional: a student benefits only from a better-prepared roommate. Grades are

$$g_i = \theta_i + \beta\,\lambda_{ij}\,(\theta_j - \theta_i)_+ - \mu_{c_i}\,d_{ij} + \varepsilon_i, \qquad \mathbb{E}[\varepsilon_i \mid \theta_i, \theta_j, A_i, A_j] = 0, \tag{5}$$

with $\beta > 0$ the transmission rate and $\mu_{c_i} \geq 0$ a constant capturing costs of mixing to grades beyond the transmission channel itself. Potential explanations include mental wellbeing acting as a mediator for grade decline, forgone same-caste collaboration, particularly for upper-caste students, or a direct cost of navigating social distance that reduces study effort. The model does not distinguish between these; I explore them in the mechanisms section.

Proposition 1 (Academic transmission)**.** *Let $\delta \equiv \mathbb{E}[\theta_j - \theta_i \mid \theta_j > \theta_i,$ mixed$]$ and $\delta' \equiv \mathbb{E}[\theta_i - \theta_j \mid \theta_i > \theta_j,$ mixed$]$ denote the cross-caste score gaps conditional on each ordering, and let $\delta_c \equiv \mathbb{E}[\theta_j - \theta_i \mid \theta_j > \theta_i,$ both caste $c$$]$ denote the within-caste upward gap. Under Assumption 1, in a same-caste room each student receives baseline transmission $\beta \cdot \frac{1}{2} \cdot \delta_c$ (since within caste $\lambda_{ij} = 1$, $d_{ij} = 0$, and the probability that the roommate scores higher is $1/2$ by symmetry). Relative to this baseline:*

(i) *A lower-caste student's expected grade change is* $\beta[\lambda_{ij}\, p\, \delta - \delta_L/2] - \mu_L\, d_{ij}$.

(ii) *An upper-caste student's expected grade change is* $\beta[\lambda_{ij}\, (1-p)\, \delta' - \delta_U/2] - \mu_U\, d_{ij}$.

Corollary 1 (Net transmission effects when $p = 1$)**.** *Take the limiting case* $p = 1$ *(empirically* $p \in [0.85, 0.90]$*, Appendix C.2). Since* p *is common to both parts of Proposition 1, the same limit yields sharply asymmetric net effects for the two castes in a mixed room:*

(i) *For a lower-caste student, part (i) reduces to* $\beta[\lambda_{ij}\delta - \delta_L/2] - \mu_L d_{ij}$*, which is positive if and only if*

$$\lambda_{ij} > \frac{\delta_L}{2\delta} + \frac{\mu_L\, d_{ij}}{\beta\delta}. \tag{6}$$

That is, the discounted return to cross-caste transmission, $\beta\lambda_{ij}\delta$*, must exceed the sum of the opportunity cost of forgone within-caste transmission,* $\beta\delta_L/2$*, and the social-mixing cost,* $\mu_L d_{ij}$*.*

(ii) *For an upper-caste student, setting* $p = 1$ *sets the reversal probability* $1 - p$ *in part (ii) to zero, so the expected grade change collapses to* $-\beta\delta_U/2 - \mu_U d_{ij}$*. It is negative for every* $\lambda_{ij} \in (0, 1]$ *whenever* $\delta_U > 0$ *or* $\mu_U d_{ij} > 0$*: both the forgone within-caste transmission and the social-mixing cost are weakly positive costs, so upper-caste grades decline in mixed rooms.*

6.3 The Social Life of Integration

Students allocate their social time between roommates and campus clubs, generating two forms of social capital. Belonging capital is a consumption good that captures mental well-being and is generated through time spent with socially aligned peers, both in the room and in campus clubs. Bridging capital is an investment good that accumulates through time spent across social groups and with returns depending on the socioeconomic advantage of the roommate. Internship outcomes depend on bridging capital, students' own socioeconomic advantage, and grades, while grades are determined by roommate assignment rather than social time spent together. Final utility combines mental well-being with the returns to academic performance and internship outcomes.

$$S_i = v(B_i) - c(R_i; \Delta_i) - \chi_{c_i}(d_{ij})\, s_i, \qquad Y_i = N_i + \pi_g\, g_i, \qquad N_i \equiv A_i + \pi_{ij}\, R_i, \qquad U_i = S_i + Y_i, \tag{7}$$

with Δ_i, v, c, π_{ij}, and π_g as defined in Definition 3.[17] [18]

[17] Own advantage A_i is predetermined and therefore shifts the level of Y_i without affecting the student's choice. The additive specification of consumption and investment is a simplifying assumption, not a claim that students value them equally: students who place less weight on the wellbeing costs of mixing, or greater weight on placement, may choose different contact levels. The specification abstracts from this heterogeneity, which is relevant for the welfare analysis (Appendix C.6).

[18] The model values all bridging capital at the roommate-specific return π_{ij}. If bridging capital generated in student clubs instead earns a constant return π_c, the effective return governing time allocation, the network value gained per unit of belonging given up, is $\pi_{\text{eff},i}$, which equals π_{ij} when $\pi_c = \pi_{ij}$. If club peers yield, on average, a higher return

Timing. (i) ρ is drawn at random. (ii) Each student announces desired roommate time $s_i^* \in [\underline{s}, 1]$, the maximizer of their payoff (7) if realized; the floor $\underline{s} \in [0, 1)$ is the unavoidable contact of co-residence. (iii) Realized interaction is $\tilde{s}_{ij} = \min\{s_i^*, s_j^*\}$: either roommate can limit contact, and neither can force it beyond the other's willingness.[19] (iv) Social outcomes are evaluated at $\tilde{s}_{ij}$, while grades are fixed by the assignment (equation (5)).

6.3.1 Belonging and club alignment

Rearranging Definition 2 with $a_i = 1 - s_i$, belonging is affine in roommate time,

$$B_i(s_i) = \iota_i + \tau_i \, s_i, \qquad \tau_i \equiv \lambda_{ij} - \iota_i,$$

where ι_i is club alignment (Assumption 2) and τ_i is the bonding pass-through: the marginal belonging return to shifting time from clubs into the room. Because club alignment rises with advantage within caste, τ_i varies within each caste rather than splitting by caste label.

Definition 4 (Room-oriented and club-oriented students)**.** *In a mixed room, student i is* room-oriented *if $\tau_i = \lambda_M - \iota_i > 0$ and* club-oriented *otherwise. A club-excluded student ($\underline{A}_{c_i} > A_i$, Assumption 6) derives no belonging from time outside the room, so $\iota_i = 0$ and the student is room-oriented. In a same-caste room $\lambda_{ij} = 1$, so every student can attain full belonging through the roommate.*

Proposition 2 (Club alignment and the composition of wellbeing costs)**.** *Let $C_i \equiv v(1) - S_i^{mixed}$ denote the wellbeing cost of a mixed room relative to full belonging, which equals the cost relative to a same-caste room when same-caste students attain full belonging, comprising three components: belonging loss (through v), bridging costs (through c, including the preparation-gap premium $c(R; \Delta_i) - c(R; 0)$), and friction $\chi_{c_i}(d_M)\tilde{s}_{ij}$. Let $\underline{\pi}_i$ denote the largest return to bridging at which student i still chooses the belonging-maximizing contact in a mixed room (Appendix C.6 gives the expression). Then:*

(i) Incidence: *Given $\underline{A}_L \geq \underline{A}_U$ (Assumption 6), first-order stochastic dominance of advantage (Assumption 1) implies that a weakly larger share of lower-caste than upper-caste students are room-oriented.*

(ii) Belonging-seeking regime: *Suppose $\pi_{ij} \leq \underline{\pi}_i$ for every student in a mixed room, so that each chooses the contact level that maximizes their belonging; this holds automatically for club-oriented*

to bridging than lower-caste roommates and a lower return than upper-caste roommates, so that $\bar{\pi}_L \leq \pi_c \leq \bar{\pi}_U$, the effective return is weakly lower than π_{ij} for lower-caste room-oriented and upper-caste club-oriented students, which makes the belonging-seeking regime easier to satisfy for them. These are the types that are relatively more common in each caste than in the other (Proposition 2(i)). For the other two types, the effective return is higher and can induce a trade-off between belonging and bridging capital. Appendix C.4 examines the implications for regime conditions, wellbeing, and placement.

[19] Under Assumption 3, payoffs are concave, hence single-peaked, in realized contact, and depend on the partner's choice only through $\tilde{s}_{ij}$. Announcing one's own peak s_i^* is then weakly dominant under the min rule. I assume students play it, so the outcome depends only on the pair's types and there is no fixed point to solve for. Equilibria in which both announce below their peaks exist, but they require a student to announce less than they want even though doing so can never help them.

students whose roommate satisfies $\pi_{ij} \leq v'(1)$. Realized contact is $\tilde{s}_{ij} = \min\{s_i^*, s_j^*\}$. Within caste and for a given preparation gap, C_i is weakly larger for room-oriented than for club-oriented students, and is largest for a room-oriented student paired with a club-oriented roommate. For room-oriented students, C_i is therefore weakly increasing in the roommate's club alignment, and hence in the roommate's advantage, with a single upward jump where ι_j reaches λ_M and the roommate substitutes club-based interactions for room contact.

(iii) Bridging-seeking regime: *Suppose a student's return to bridging exceeds the threshold, $\pi_{ij} > \underline{\pi}_i$, so that desired contact is interior, and the roommate is willing to match it. Then wellbeing decreases and network capital increases with π_{ij}, and at the margin the wellbeing loss equals the value of the additional bridging capital, $dS_i = -\pi_{ij}\, dR_i$. For roommates with $\pi_{ij} \leq v'(1)$, only room-oriented students can be in this regime, and only if friction is large relative to the bonding pass-through, $\chi_{c_i}(d_M)/\tau_i > v'(\lambda_M) + c_R(1 - \lambda_M; \Delta_i) - \pi_{ij} > 0$: interior contact in a mixed room reflects costly contact as well as valuable bridging.*

(iv) Academic comparison and wellbeing: *In the regime of part (ii), wellbeing is weakly decreasing in the roommate's entrance score only when the roommate outscores the student ($\Delta_i > 0$). This condition holds with probability p for lower-caste students and $1 - p$ for upper-caste students, so the effect is concentrated among lower-caste students. The decline is steeper for room-oriented students, who bear greater wellbeing costs from academic comparisons with stronger roommates.*

The belonging-seeking regime of part (ii) coincides with bonding constraint: students would choose more belonging than is attainable. Its welfare implications are taken up in Appendix C.6.

6.4 Social Capital and Internships

Network capital $N_i = A_i + \pi_{ij} R_i$, which adds the value of bridging capital to own advantage, and grades together determine the internship index $Y_i = N_i + \pi_g g_i$ (equation (7)). Network capital measures the value of network access, not the number of contacts.

Let j and j' denote the roommates in the mixed room and the counterfactual same-caste room, respectively, and let Δ denote the difference in outcomes between the two assignments (mixed minus same-caste). Then $\Delta Y_i = \Delta V_i + \pi_g \Delta g_i$, where $\Delta V_i \equiv \pi(A_j) R_i - \pi(A_{j'}) R_i^{\text{same}}$ is the change in the value of bridging capital under mixed assignment, with R_i and R_i^{same} denoting bridging capital in the mixed and same-caste rooms, respectively, and Δg_i is the change in grades (Proposition 1). Let $\mathbb{E}_c$ denote the expectation over caste-c students and their roommate draws, and define $\bar{g}_c \equiv \mathbb{E}_c[\Delta g_i]$ as the expected grade change. In a same-caste room, a student desires to bridge if and only if the return to bridging exceeds the marginal value of full belonging, $\pi(A_{j'}) > v'(1)$, and realized contact is the lower of the two roommates' desired levels (Lemma 3, Appendix C.3). The condition requires at least one student to be sufficiently advantaged and is therefore more likely to hold in upper-caste rooms, where socioeconomic advantage is higher (Assumption 1). By contrast, lower-caste students in same-caste rooms devote all their social time to bonding with their roommate and do not gain bridging capital (Assumption 4).

Proposition 3 (Internship effects of mixed assignment)**.** *Compare mixed-room with same-caste-room assignment under random assignment (§6.1) and Assumptions 1, 2, 3, and 4. Then:*

(i) Lower-caste students. *Bridging capital in same-caste rooms is zero, so mixed assignment raises network capital through bridging capital, and*

$$\mathbb{E}_L[\Delta Y_i] = \mathbb{E}_L[\pi(A_j)R_i] + \pi_g \bar{g}_L.$$

The expected change in the placement index is therefore nonnegative whenever $\bar{g}_L \geq 0$ (Corollary 1(i)).

(ii) Upper-caste students. *The change in the value of bridging capital, ΔV_i, has no sign, since a more advantaged same-caste roommate raises the value of bridging capital in same-caste rooms, and the grade effect is nonpositive, $\bar{g}_U \leq 0$ (Corollary 1(ii)):*

$$\mathbb{E}_U[\Delta Y_i] = \mathbb{E}_U[\Delta V_i] + \pi_g \bar{g}_U,$$

which is negative if and only if $\mathbb{E}_U[\Delta V_i] < -\pi_g \bar{g}_U$.

Remark (Competition for elite internships)**.** *The proposition characterizes the effects of mixing on internship outcomes. How changes in the internship index translate into access to internships depends on the competitiveness of the recruitment process, which Appendix C.5 examines. As competition for elite internships increases, the returns to bridging operate through three distinct channels. First, in organized recruitment, the placement cell makes information about openings available to all students, reducing the value of bridging capital as a source of information. Instead, bridging matters through access to the hidden curriculum of recruitment, including preparation for selection tests and interviews. Second, the returns to bridging depend on the complementarity between a student's own advantage and that of a closely aligned roommate. Third, eligibility to apply is restricted by grades and specialization, which makes changes in academic performance particularly consequential for students near the grade cutoff. These channels have distinct implications by caste: public information limits the competitive returns to bridging for lower-caste students, complementarity in recruitment preparation favors students who combine their own advantage with an advantaged, closely aligned roommate, and grade cutoffs amplify the consequences of grade declines for upper-caste students near the eligibility threshold. The appendix formalizes these implications.*

7 Mechanisms

The model in Section 6 makes three predictions that I take to the data in this section. First, grade improvement for the lower caste is due to being paired up with academically better-prepared peers. Second, mental health costs are a result of social frictions in the room, and only the advantaged group can asymmetrically substitute with other spaces in the institute to integrate with their own caste networks. Third, there exist complementarities within caste networks when competing for elite jobs or leadership positions.

7.1 Grades effects are due to directional transmission of help:

In this subsection, I test Proposition 1. I examine whether roommate academic preparation explains the effects of mixed-room assignment on grades. I also test whether there is any diversity premium or penalty for roommate academic preparation by interacting treatment with roommates' entrance exam scores. Column 4 of Table 4 shows that the effect of mixed-room assignment on grades disappears after I control for roommates' entrance-exam scores. This result suggests that the effect on grades operates through exposure to better-prepared roommates. Column 6 of Table 4 shows no evidence of an additional diversity premium in academic transmission. If anything, the estimate suggests a small penalty when students are paired with better-prepared roommates from a different caste, but this is not statistically significant.

The evidence on academic interactions is consistent with this pattern. Lower caste students in mixed rooms report having similar study interaction intensity compared to their counterparts in same-caste rooms (Table 5). However, they report receiving (0.12 s.d., $p = 0.104$) more help preparing for exams in mixed rooms in col 5 of Table A18, which suggests that with better-prepared roommates, study interactions would be more helpful to improve GPA. Overall, the empirical results show that the net effect of mixing on academic transmission is positive for lower-caste students, resolving the ambiguity in Corollary 1. The discounted returns to cross-caste transmission exceed the returns forgone from within-caste transmission and the social costs of mixing, as specified in Equation (6) and Corollary 1(i).

For upper-caste students, the negative effect on grades does not appear to result from being paired with lower-performing roommates. In particular, the level of lower caste students' exam scores doesn't hurt upper caste students. This finding is consistent with the asymmetric effects proposed in Proposition 1. Controlling for roommate's exam score doesn't change the effect on mixed room in Column 2 of Table 4 for upper caste students. Column 3 of Table 4 also shows no evidence of an additional penalty from being paired with a lower-performing student from a different caste. If anything, being paired with a lower-performing, lower-caste roommate appears to improve upper-caste students' grades, but this effect is not statistically significant.

Three patterns suggest that upper-caste students lose access to reciprocal academic support from similarly prepared peers in mixed rooms relative to same-caste rooms. First, they report receiving less exam-related support (-0.163 SD, $p < 0.1$, Column 1 of Table A18) than upper-caste students in same-caste rooms. Columns 3 and 7 of Table A18 show that upper-caste students provide academic help to their roommates in mixed rooms but do not receive comparable help from their lower-caste roommates. Second, study interactions are slightly lower for upper-caste students in mixed rooms, although the difference is not statistically significant. Third, in Table A19, I examine whether upper-caste students in mixed rooms substitute help from other hostel mates for the support they would otherwise receive from their roommates and I find no evidence of such substitution. The point estimates are, in fact, negative, although statistically insignificant, suggesting the possibility of crowd-out. Whereas upper castes in same-caste rooms report getting more help from other hostel-mates compared to lower-caste students in same caste rooms (diff is 0.16

s.d., mean diff in academic help in Table A19). Overall, the losses for upper-caste students appear to stem from reduced access to academic support from better-prepared peers. However, the evidence is suggestive, and, as discussed in Corollary 1(ii), I cannot disentangle the costs of forgone academic networks from other social and psychological frictions arising from caste mixing.

7.1.1 Is help leading to learning or cheating?

To investigate further whether the grade gains for lower-caste students in mixed rooms reflect actual learning and, consequently, an increase in human capital, rather than grade inflation arising from shared assignments or copying from better-prepared roommates, I examine whether these effects vary with academic overlap. I define academic overlap as being assigned to the same course track as one's roommates. All first-year students are assigned to one of two course tracks, independently of their specialization and orthogonal to mixed-room assignment. Students take the same courses regardless of specialization. Students follow different course sequences in the first semester, switching sequences in the second semester.

If grade gains arise from copying or shared assignments, I would expect them to be larger when roommates follow the same course track and have greater overlap in their coursework. However, a positive effect in the presence of overlap would not, by itself, establish cheating, since students taking the same courses may also exchange knowledge and help each other learn. Conversely, the absence of differential effects by academic overlap would be inconsistent with cheating/grade inflation being the primary explanation for the grade gains.

There are also reasons why academic overlap may not be necessary for knowledge transmission. First, first-year courses build on high-school mathematics and physics and cover foundational subjects, including calculus, linear algebra, and applied physics. Roommates with stronger prior preparation may therefore help one another understand these subjects even when they follow different course tracks. Second, because the course sequences switch between semesters, roommates who take a course earlier may be able to help one another when the sequence changes. I cannot directly test this second channel because I do not observe grades by semester.

I find no evidence that academic overlap explains the grade gains for lower-caste students in mixed rooms (column 5 of Table A20). If these gains were driven by copying or grade inflation through shared assignments, I would expect them to be larger when roommates follow the same course track. Instead, the interaction between mixed-room assignment and academic overlap is negative, suggesting that overlap does not amplify the grade gains. In addition to the effects on grades, I find no evidence that students in mixed rooms engage in more group work or exert less study effort overall (Table A21). These aggregate findings do not, however, imply that academic overlap is irrelevant. Table A20 shows that academic overlap is associated with more group work and study interactions in same-caste rooms, for both upper- and lower-caste students. Along with the estimates on grades (column 5), estimates on group work and study interactions (columns 6 and 7) are negative in mixed rooms when roommates share the same course track. These patterns are consistent with the possibility that in mixed rooms, some frictions exist when paired up with

better prepared roommate but level effect of roommate's exam scores dominate the transmission and increase grades for lower caste students. Taken together, these findings provide no support for copying or grade inflation through shared academic work as the primary explanation for lower-caste students' grade gains.

For upper-caste students, academic overlap is associated with higher grades in same-caste rooms, although this effect is statistically insignificant. Point estimates suggest upper-caste students also appear to do less group work in mixed rooms than their counterparts in same-caste rooms. Both these patterns are further suggestive of the forgone academic support channel, explaining lower grades for upper-caste students in mixed rooms. Finally, I find no evidence that self-reported study effort is lower in mixed rooms for either group, reported in Table A21. This finding weighs against explanations based on lower-caste students shirking because they have better-prepared roommates or either group reducing its study effort because of the social costs of caste mixing.

7.2 Social Frictions and Mental Health:

Upper caste in elite institutions diversify their networks as they can access other social spaces like student clubs and depend less on roommates for their socio-emotional needs but social frictions inside the room can affect their well-being. For lower caste students roommate bonding is high and plays a crucial role in their mental well-being and they can't diversify their social networks. When academic differences with upper castes is higher it also increases their worry and would negatively affect their well-being. In this section, I argue that social frictions in mixed rooms are the main driving force for worse mental health in both groups.

7.2.1 Early signs of social frictions:

First, I show in table A22 that social frictions in mixed room arise as early as the first week of college. Exposure produces detectable frictions within the first week of college for both groups, though of a different type. Upper-caste students report lower expected friendship with their roommates and higher concern about cultural and language differences. Lower-caste students show little early reactions, but mixing increases their worry about discrimination specifically. These early signs indicate social frictions start before substantial interactions.

7.2.2 Roommate interactions and mental health

I begin by documenting two descriptive patterns that motivate the mechanisms in the model. First, in same-caste rooms, lower-caste students exhibit greater interaction intensity with their roommates than upper-caste students. Second, upper-caste students appear more engaged in student clubs, suggesting greater diversification of social interactions beyond the roommate relationship. Both patterns are consistent with the alignment described in Definition 4 of the model.

Table 5 shows that, in same-caste rooms, lower-caste students have a composite index of roommate interaction intensity that is 0.23 standard deviations higher than that of upper-caste students. This difference is driven primarily by social interactions, such as eating together, playing sports, and going out, for which the gap is 0.20 standard deviations. The difference in academic interactions is smaller, at 0.13 standard deviations. Similar patterns emerge for emotional, financial, and academic support exchanged between roommates (control-group means in Tables A18 and A23).

Control-group means in Table A19 also suggest that upper-caste students diversify their sources of academic support beyond their roommates. Even in same-caste rooms, they are 0.16 standard deviations more likely to receive academic support from other hostel mates than lower-caste students. Upper-caste students also report greater engagement in student clubs: they are 0.20 standard deviations more likely to be active club members (Table A24) and spend more time in student clubs. Together, these descriptive patterns suggest that lower-caste students rely more heavily on their roommate relationships, whereas upper-caste students have greater access to social and academic connections outside the room.

I next examine how mixed-room assignment affects roommate interactions and whether these changes are consistent with the proposed mental-health mechanisms. Table 5 shows that mixed-room assignment significantly reduces roommate interaction intensity for lower-caste students (composite index: -0.160, $p < 0.05$). This decline is driven by social interactions (-0.182, $p < 0.01$), while the effect on study-related interactions is smaller and statistically insignificant (-0.043). Consistent with this pattern, lower-caste students are also 0.127 standard deviations less likely to receive emotional and financial support from their roommates in mixed rooms than in same-caste rooms, reported in Table A18.

These findings are consistent with Propositions 2(ii) and 2(iii) of the model, which predict that cross-caste interactions can reduce bonding when social frictions weaken the benefits of roommate relationships. Table A25 provides suggestive evidence that this reduction in social interaction is related to the decline in lower-caste students' mental health. The mixed-room coefficient falls from -0.076 s.d. ($p < 0.1$) to -0.020 s.d. when social interaction is included as a control. By contrast, adding study-related interaction does not materially change the estimated mental-health effect.[20]

A further channel may compound the mental-health costs of integration for lower-caste students: competitive pressure arising from differences in academic preparation. Consistent with Proposition 2(iv) of the model, the worry index increases with the academic-preparation gap between roommates for lower-caste students. I report that lower caste students are more worried when gap increases with roommate and I don't see any such effects on PHQ4. This pattern is suggestive of a role for unfavorable academic comparisons in shaping lower-caste students' mental health.

For upper-caste students, the evidence on roommate interactions is weaker, consistent with their greater engagement in social spaces beyond the room. Table 5 shows that social interac-

[20] Because interaction intensity is itself affected by treatment, this exercise does not identify a causal mediating channel. The attenuation of the treatment coefficient is instead suggestive evidence consistent with the model's predictions in Propositions 2(ii) and (iii).

tion declines by 0.08 standard deviations in mixed rooms, although the estimate is not statistically significant. Table A18 shows a similar pattern for emotional and financial support received from roommates. Moreover, unlike for lower-caste students, controlling for social interaction does not meaningfully change the estimated mental-health effect for upper-caste students (Table A25). Taken together, these findings are suggestive of reduced roommate bonding in mixed rooms but its impact on mental health of upper caste students is relatively smaller.

To further investigate this argument, I examine heterogeneity by room type (Table A26, Table A27). For upper-caste students, the mental-health costs are larger in two-person mixed rooms and close to zero in three-person rooms. This pattern is suggestive of a buffering role for alternative social networks. In three-person rooms, the presence of another upper-caste roommate may allow students to maintain same-caste connections, while a more diverse set of lower-caste roommates may offer additional opportunities to diversify social interactions. Either way, larger rooms may reduce dependence on a single cross-caste relationship and mitigate the mental-health costs of integration. For lower-caste students, by contrast, the mental-health effects are similar across two-person mixed rooms, partially mixed three-person rooms (one upper-caste and one lower-caste roommate), and fully mixed three-person rooms. This similarity is consistent with social frictions and unfavorable academic comparisons remaining important channels regardless of room composition.

Finally, I examine heterogeneity by the caste distance between roommates (Table A28). Caste distance is constructed by assigning ordinal ranks to caste groups: upper caste = 1, OBC = 2, and SC/ST = 3. Room-level caste distance is measured as the average absolute difference between a student's caste rank and those of their roommates. For lower-caste students, greater social distance is associated with a significantly larger negative mental-health effect (0.086 s.d., ($p < 0.05$)). For upper-caste students, the corresponding estimate is smaller (0.034 s.d.) and statistically insignificant. This pattern is consistent with social frictions being more consequential for lower-caste students when paired with more socially distant roommates.

7.3 Room transitions and Segregation

At the beginning of the second year, students can choose their roommates from within their hostel cohort, making roommate selection a constrained choice. I use this transition to examine the persistence of initial room types and whether students sort into same-caste rooms after the randomized assignment period. Second year room-choice data come from hostel administrative records, providing an objective measure of transitions that is not subject to self-reporting bias Figure 2 shows substantial persistence in room composition, although the patterns differ by initial room type.

Students initially assigned to same-caste rooms, both upper-caste (UC) and lower-caste (LC), are more likely to remain in same-caste rooms than students initially assigned to mixed rooms are to remain in mixed rooms. Conversely, students initially assigned to mixed rooms are more likely to transition to same-caste rooms. UC students initially assigned to same-caste rooms are

9.7 percentage points more likely to remain in that room type than UC students initially assigned to mixed rooms ($p < 0.01$); the corresponding difference for LC students is 5 percentage points ($p < 0.1$). Overall, the share of mixed rooms is 8.3 percentage points lower than would be expected under random allocation. This reduction is larger in rooms with greater scope for compositional adjustment, particularly three-person rooms. Within these rooms, partially mixed rooms—with two students from one caste group and one from the other—are especially likely to transition to same-caste rooms. I report the detailed transition patterns in Table A29.

7.4 Summary on Mental Health and Grades:

Taken together, the model and empirical results suggest distinct mechanisms underlying the effects of integration across caste groups. Lower-caste students improve their academic performance when paired with better-prepared roommates, but social frictions and unfavorable academic comparisons within mixed rooms may undermine their well-being. For upper-caste students, the evidence is less consistent with specific roommate attributes driving the effects on grades and mental health. Instead, reduced access to same-caste networks and forgone opportunities for collaboration appear to be more important in explaining their negative academic and well-being outcomes. Caste frictions experienced by the lower castes and upper castes also seem different, consistent with all the outcomes above. Lower caste students are more likely to report higher experiences of discrimination (Table A30 and Table A31) and whereas upper-caste students are more likely to feel exclusion from study groups (Table A31) representing the costs they are facing in mixed rooms.

Academic and psychological effects of mixed-room assignment differ across groups. Lower-caste students improve academically at the expense of their mental health, while upper-caste students experience a decline in grades by the end of the first year. These findings raise questions about the persistence of these effects and their implications for longer-run outcomes. Do upper-caste students recover from their initial academic losses, and how do the academic and psychological consequences evolve for lower-caste students? Does roommate mixing affect labor-market outcomes? To address these questions, I follow the 2023 entry cohort shortly before the end of their third year of college, examining medium-run effects on academic performance, mental health, internship outcomes, and attitudes toward affirmative action. I report ITT effects of first year roommate assignment.

8 Medium run consequences

8.1 Main Outcomes in the Medium Run: Three Years after Entry

Table 6 reports the effects of first-year mixed-caste roommate assignment on outcomes measured in the third year of college: cumulative GPA,[21] summer internship offer received, internship stipends, a composite mental health index, and indices of support for affirmative action. I use the same specification as in Table 3.

By the time of the survey, the randomized assignment had ended roughly two years earlier, and students had chosen their own roommates. The estimates therefore capture the medium run effects of atleast one-year exposure.

I document three key findings. First, the initial effects of mixed-room assignment on grades have persisted by the end of third year but estimates are noisy and mental health effect has faded by the third year. Second, the effects on internship outcomes differ by caste. For upper-caste students, there is no detectable effect on whether they received an internship offer, but their internship stipends are significantly lower. For lower-caste students, the effect operates primarily on the extensive margin: they receive fewer internship offers, while the estimated effect on stipend amounts is also negative but imprecisely estimated. Third, the effects on support for affirmative action differ across groups. Among lower-caste students, support increases along both dimensions, with marginally significant effects on beliefs about the benefits and costs of affirmative action for disadvantaged groups. Among upper-caste students, there is no persistent support for affirmative action; the estimates are statistically insignificant and point in negative direction. I examine each outcome below in detail.

Grades: By the end of the fifth semester, upper-caste (UC) students initially assigned to mixed rooms have GPAs that are 0.125 standard deviations lower than those of UC students assigned to same-caste rooms. For lower-caste (LC) students, the estimated effect is positive (0.087 standard deviations) but statistically insignificant. The second-year estimates are similar in magnitude for UC students (column 1 of Table A32), with mixing reducing their GPA by 0.119 standard deviations. For LC students mixing increases GPA by 0.121 standard deviations; both second-year estimates are statistically significant at the 10% level. Mixing has persistent effects on students' GPA. I look at how caste gap changes over time by plotting GPA of upper caste and lower caste students in Figure A1 normalized by first-year GPA, the gap declines from 0.80 standard deviations at the end of the first year to 0.71 standard deviations by the end of the fifth semester, an 11.2% reduction over the subsequent three semesters, after the roommate experiment ended. This narrowing is driven both by decreasing grades among UC students and increasing GPA for LC students. Overall, the results suggest that the academic effects of initial roommate mixing persist beyond the first year and the achievement gap between the two groups reduces over time.

Mental health: Effect of mixing doesn't have persistent effect on mental health of both groups.

[21] The survey was done shortly before sixth-semester exams, so reported GPA is cumulative through the fifth semester, approximately two and a half years after entry.

But Figure A2 indicates caste gap in mental health also emerges over college. In same-caste rooms, lower-caste students score about 0.17 s.d. below upper-caste students by the end of third year, compared with almost no gap at entry (Fact 3). This is consistent with a relative deterioration in lower-caste students' mental health over college. Trends also indicate mental health becomes worse for everyone and it becomes much worse, in particular scheduled caste and scheduled tribes.

Internships Outcomes Third-year summer internships are organized primarily through the institute's placement cell and are very competitive and well-paid opportunities. Among students assigned to same-caste rooms, 78.2% of upper-caste students get an offer for summer internship by the time of survey, compared with 48.7% of lower-caste students, a 30pp difference between upper and lower caste students. Stipends are reported in income brackets and coded at bracket midpoints. This ensures these stipends are already top-coded at 95th percentile.

Upper-caste students earn an average monthly stipend of INR 163,631 (approximately USD 1,703), while lower-caste students earn around INR 66,921 per month, or roughly 40% of the upper-caste mean.[22] Mixed-room assignment has no detectable effect on whether upper-caste students receive an internship offer, but it reduces their monthly stipends by INR 32,541, or about 20% relative to the mean among upper-caste students in same-caste rooms. For lower-caste students, mixed-room assignment reduces internship offers by 7.3 percentage points ($p < 0.1$). The estimated effect on monthly stipends is also negative, implying a 14.5% reduction relative to the same-caste-room mean, but is imprecisely estimated.

8.2 Why are internship outcomes worse for both caste groups?

To understand the mechanisms behind the negative effects of mixed-room assignment on internship outcomes, I go back to the model again and Proposition 3 can shed light on this. Even though my model is static and doesn't account for changes in social or network capital, the levels of network capital remain consistent along caste lines over time and mechanism through which job market outcomes are affected can be studied through its lens. Consistent with Proposition 3(i), the effects of roommate mixing on labor-market outcomes vary with the competitiveness of the opportunity. In the second year, when summer internships are typically low-paying and less competitive, mixing improves outcomes for lower-caste (LC) students, while the estimated effects for upper-caste (UC) students are modestly negative but statistically insignificant. Table A32 shows that mixing increases summer internship stipends for LC students by approximately INR 5,944 (USD 62), significant at the 10% level. The stipend distribution suggests that these opportunities are relatively low-paying: the median stipend is zero and the 75th percentile is INR 15,000. By contrast, third-year internships are substantially more competitive and better paid, with a median

[22] Stipends are converted at USD 1 = INR 96.1. Means include students with a zero stipend. The median upper-caste student in a same-caste room falls in the bracket with a midpoint of INR 125,000 (approximately USD 1,300), while the median lower-caste student falls in the bracket with a midpoint of INR 25,000 (approximately USD 260).

stipend of INR 85,000 (approximately USD 900) and a 75th percentile of INR 125,000[23]. These internships can also lead to placement offers or provide important pathways into subsequent employment.

To examine how these early labor-market gains emerge, I document students' summer activities in the first and second years in Figures A3 and A4. LC students assigned to mixed rooms are more likely to undertake summer internships and academic projects, although the estimated effects on academic projects are smaller and statistically less precise. In particular, mixed-room assignment increases LC students' likelihood of undertaking a summer internship by 8.3 percentage points. These students are also less likely to engage in volunteering or part-time work. This pattern suggests a shift toward activities more directly related to career development. Although I cannot directly establish that these changes result from learning from UC roommates, they are consistent with the idea that cross-caste ties facilitate access to information and opportunities that help LC students build bridging capital and pursue job-market-oriented activities.

For UC students, I find no statistically significant adverse effects on labor-market outcomes during these initial years. One possible explanation is that their existing networks and other advantages help them access less competitive opportunities, limiting the short-run costs of roommate mixing consistent with Proposition 3(i).

The third-year results are consistent with the prediction in Proposition 3(ii): roommate mixing has negative effects on internship outcomes for both groups when opportunities become more competitive. One interpretation is that access to these opportunities depends on complementary advantages, such that students benefit from one another's resources only when their combined network capital exceeds a threshold. For lower-caste students, the absence of internship gains despite improvements in academic performance points and early labor market gains. At lower levels of combined advantage, including when one or both roommates belong to a disadvantaged caste, these complementarities may be insufficient to generate positive returns. In particular, many third year internships impose GPA cutoffs, so the persistence of academic disadvantages may constrain lower-caste students' access to these opportunities. The benefits of broader social connections may therefore be insufficient to overcome barriers to entry into high-paying internships. This contrasts with the early years, when cross-caste ties can expand access to less competitive labor-market opportunities for LC students, while UC students' pre-existing advantages may cushion the costs of mixing.

To further examine this, I look at career preferences, and access to campus networks. The results are consistent with above predictions. Instead, mixed-room assignment significantly reduces leadership positions in student clubs and lowers preferences for high-paying sectors, particularly finance and consulting (Table 7). Both effects are statistically significant at the 5% level. These patterns are consistent with reduced access to same-caste networks shaping students' career trajectories. Club leadership may provide access to valuable professional connections or be a direct

[23]The stipend distribution, calculated across all students, including those who did not undertake an internship and those whose internships were unpaid

differentiator in CVs with already small differences in GPA. These results are consistent with evidence from the literature linking club participation to career preferences and earnings in elite US colleges (Michelman et al., 2022). Importantly, I find no evidence that students assigned to mixed rooms apply less to these sectors. Applications are submitted through a centralized college job portal, making the marginal cost of applying low and largely invariant to roommate assignment. This finding suggests that the differences in internship outcomes are more consistent with differences in career preferences, access to advantaged network for upper caste students and entry barriers and weak ties for lower caste students.

9 Affirmative Action Attitudes

In this section, I examine the mechanisms underlying the short-run improvements in attitudes among both groups, how these treatment effects evolve three years later, and whether they extend to support for affirmative action in employment and political representation.

The model in Section B predicts that a roommate's signal induces larger attitude changes when prior beliefs are less accurate and smaller changes when students already possess independent information conveying similar content. I test these predictions using three heterogeneity analyses, reported in Tables 8, 9, and A33.[24]

Updating is largest among students with stronger meritocratic beliefs. Table 8 interacts mixed-room assignment with students' baseline belief that success in the entrance exam depends entirely on merit. This belief strongly predicts lower baseline support for affirmative action (AA) in both groups. Among upper-caste students, those holding stronger meritocratic beliefs exhibit larger shifts in attitudes toward both the benefits of AA for reserved groups and its costs for non-reserved groups (0.142 and 0.131 standard deviations, respectively; both $p < 0.1$). Among lower-caste students, the interaction is positive and significant only for attitudes toward the costs and benefits of AA for disadvantaged groups (0.147, $p < 0.05$). These patterns are consistent with Bayesian updating in the model (Proposition 4(i)): students whose prior beliefs are less aligned with the information conveyed by cross-caste contact exhibit larger revisions in their attitudes.

Baseline social segregation predicts greater updating. Table 9 interacts mixed-room assignment with students' pre-college share of cross-caste friends, coded so that higher values indicate greater social segregation. Proposition 4 predicts that more socially segregated students have less information about other groups ex ante and hold less supportive views of the benefits and costs of AA for those groups. Consistent with this prediction, more socially segregated upper-caste students exhibit larger increases in support for AA benefiting reserved groups (0.118, $p < 0.1$). Among lower-caste students, the point estimates similarly indicate greater attitude changes among those who were more socially segregated before college, in the direction of greater recognition of the benefits and costs of AA for the other group. However, these estimates are less precise. Overall, the patterns are consistent with substitution (Proposition 4): the roommate's

[24]The first two tests use only the 2024 cohort, for which both baseline and one-year follow-up data are available.

signal carries less new information for students who already have greater cross-caste exposure.

Own academic standing may shape attitudes through ego utility. Table A33 interacts mixed-room assignment with students' standardized entrance-exam scores. Higher-scoring students express lower baseline support for AA. This relationship is consistent with the role of ego utility emphasized by Köszegi, 2006: exam success may reinforce the belief that one's academic standing reflects personal merit rather than circumstances. Academic standing may also moderate how students respond to cross-caste contact. Students who feel less threatened by academic comparisons may be more willing to revise their attitudes in favor of the other group. Alternatively, entrance-exam scores may proxy for differences in prior beliefs and the information students bring to the interaction. Own exam score heterogeneity doesn't disentangle these two mechanisms.

Taken together, the results are consistent with the model's central prediction that the effect of contact on AA attitudes depends on students' prior beliefs and existing information. Attitude changes vary with baseline meritocratic beliefs, prior cross-caste exposure, and academic standing, suggesting that contact helps people with prior misbelieves update their beliefs stronger but update is stronger on the perceived benefits and costs of AA for the other group, consistent with model's key predictions.

9.1 Heterogeneity by Gender in Attitudes Toward Affirmative Action

I examine whether the effects of mixed-room assignment on support for affirmative action (AA) differ by gender in Table A10. Among upper-caste students, female students assigned to mixed-caste rooms exhibit substantially larger shifts away from support for AA. This pattern is notable in the institutional context: in 2018, the institution introduced a 20% supernumerary quota for women within each caste category to improve gender representation. These additional seats are allocated when women's representation in a specialization falls below 20%, and all women in my sample benefit from these quotas.

Although female students express greater baseline support for AA overall, exposure to mixed-caste roommates substantially attenuates this gender difference, eliminating the positive effect of mixing. I find no comparable gender heterogeneity among lower-caste students. One possible explanation is that upper-caste women perceive cross-caste integration as threatening the position of women within the institution, particularly if they anticipate a backlash against gender-based affirmative action. Alternatively, the intersection of caste and gender disadvantage may shape how students interpret integration differently across groups. These results need further inquiry to understand what channel is driving this reduction in support for AA among females.

9.2 Support for affirmative action: Medium run

In the control group, the caste gap in support for affirmative action narrows from 0.48 s.d. at entry to 0.32 s.d. For upper-caste students, the first-year effect of mixing does not persist: point estimates are negative, particularly for items on the costs and benefits of affirmative action to

their own group, though none is statistically significant. For lower-caste students, mixed-room assignment raises support for affirmative action, an effect significant at the 10% level and present on both the cost and the benefit dimensions. One possibility is that experiences in preparing for the labor market, including the stipend differences documented above, shape these attitudes in the later years of college.

I examine how support for affirmative action (AA) evolves over time for both groups in Figures A5 and A6. The narrowing of the gap in AA support is driven primarily by changes among upper-caste students, whose support increases across both dimensions of AA, beliefs related to disadvantaged groups and advantaged groups. Lower-caste students' support remains comparatively stable overall, following a slight decline in the first year and a subsequent recovery by the third year. There is, however, heterogeneity within this group: Scheduled Caste (SC) and Scheduled Tribe (ST) students reduce their support for AA over time. This decline may reflect the psychological costs or caste-related frictions they experience in mixed rooms, although the trends alone cannot establish these mechanisms.

9.3 Downstream Affirmative action attitudes: Medium run

Upper-caste students assigned to mixed rooms become less supportive of affirmative action (AA) in the medium run, particularly in employment. Their support for AA in jobs declines by 0.177 standard deviations, a statistically significant effect at the 5% level. Support for AA in politics also falls by 0.14 standard deviations, while the estimated effect on support for AA in elite higher education is smaller and negative. In contrast, I find no evidence of persistent changes in lower-caste students' support for AA in employment, politics, or education. The point estimates are small, positive, and statistically insignificant.

The decline in upper-caste students' support for affirmative action may partly reflect their decline in economic outcomes due to mixed-room assignment. The effect is largest for affirmative action in employment, where I also document a significant decline in internship stipends. This pattern is consistent with students becoming less supportive of policies perceived to disadvantage their own group when they experience adverse labor-market outcomes.

10 Discussion and Conclusion

This paper studies caste integration using randomly assigned first-year roommates among approximately 2,400 students at a highly selective public engineering institution in India. Combining administrative records with detailed surveys, I find that mixed-caste rooms improve disadvantaged students' academic performance but worsen their mental health, while advantaged students experience a decline in academic performance. Three years later, academic losses persist among upper-caste students, while gains among lower-caste students remain largely unchanged. Labor-market outcomes depend on the competitiveness of the opportunity: disadvantaged students have better outcomes in less competitive internships but are less likely to secure more com-

petitive internships, while advantaged students assigned mixed-caste roommates receive lower internship stipends. Integration also reshapes political attitudes. In the short run, students in both groups become more supportive of affirmative action policies benefiting the other group. In the medium run, however, advantaged students' support for employment reservations declines as they experience losses in more competitive internships.

An important limitation of this study is that, because students are surveyed while still enrolled at the institution, I cannot examine the longer-term consequences of inter-caste exposure for employment, particularly access to and the quality of full-time jobs. This is an important dimension of economic mobility that the internship outcomes examined here cannot fully capture.

Several questions remain for future research. How do the effects of integration vary across educational settings with different levels of competition, initial social distance, and performance inequality? Do cross-caste interactions during college shape graduates' subsequent hiring decisions, residential choices, marriage patterns, and political beliefs? These questions are particularly relevant in contexts where caste continues to shape access to economic opportunities, including hiring Shukla, 2025. Evidence from developed-country settings suggests that intergroup contact during childhood can influence later hiring decisions Wren-Lewis et al., 2025 and residential choices Merlino et al., 2024. Whether similar effects emerge from exposure during college, particularly in contexts of persistent caste inequality, remains an open question. Future work could also examine whether institutional arrangements such as structured peer groups, cross-cohort mentorship, and student clubs can reduce identity-based frictions and strengthen the benefits of cross-caste interactions.

Caste integration creates opportunities for connections across historically entrenched social boundaries, but the returns to these connections depend on the resources students bring into these relationships and the frictions they encounter. Students enter educational institutions with unequal academic preparation and social capital, shaping both the resources they gain from cross-caste connections and the costs they bear. Assessing the consequences of integration therefore requires looking beyond academic performance and attitudes to consider psychological well-being, the quality of social interactions, and access to economic and labor-market opportunities, as well as how these effects evolve over time. These findings highlight that the consequences of integration depend not simply on bringing students together, but also on the social and institutional conditions under which they interact.

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11 Figures

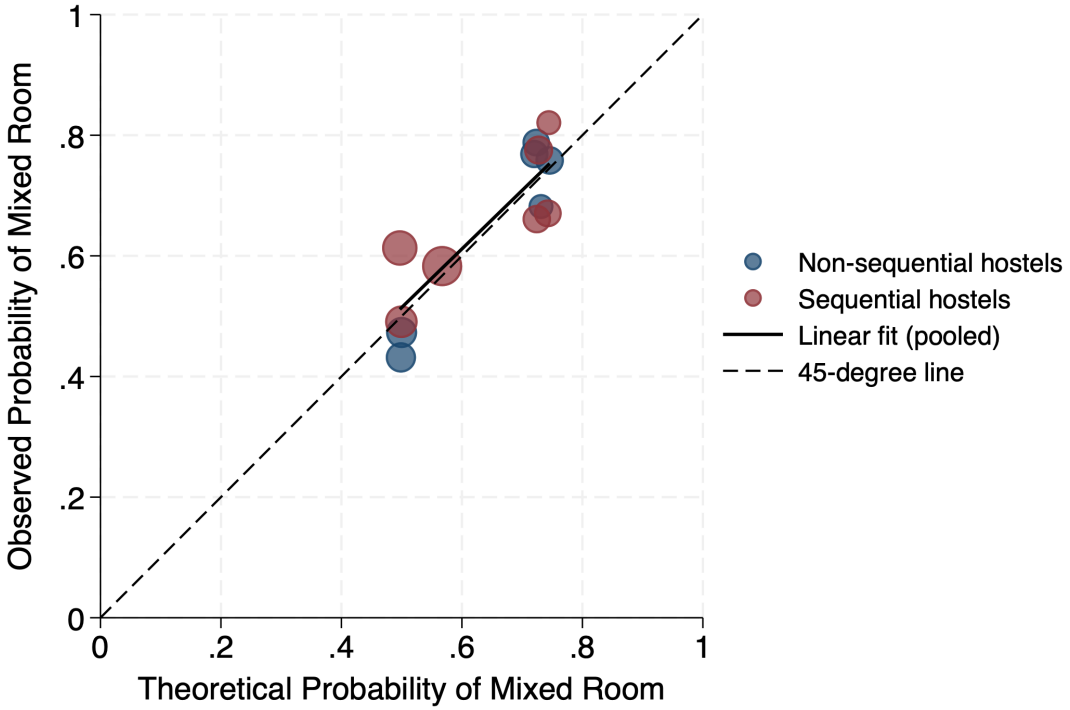


Figure 1: Observed and Theoretical Cross-Caste Mixing Across Hostels

Notes: Circle size represents hostel size (number of students). The fitted line is weighted by hostel size.

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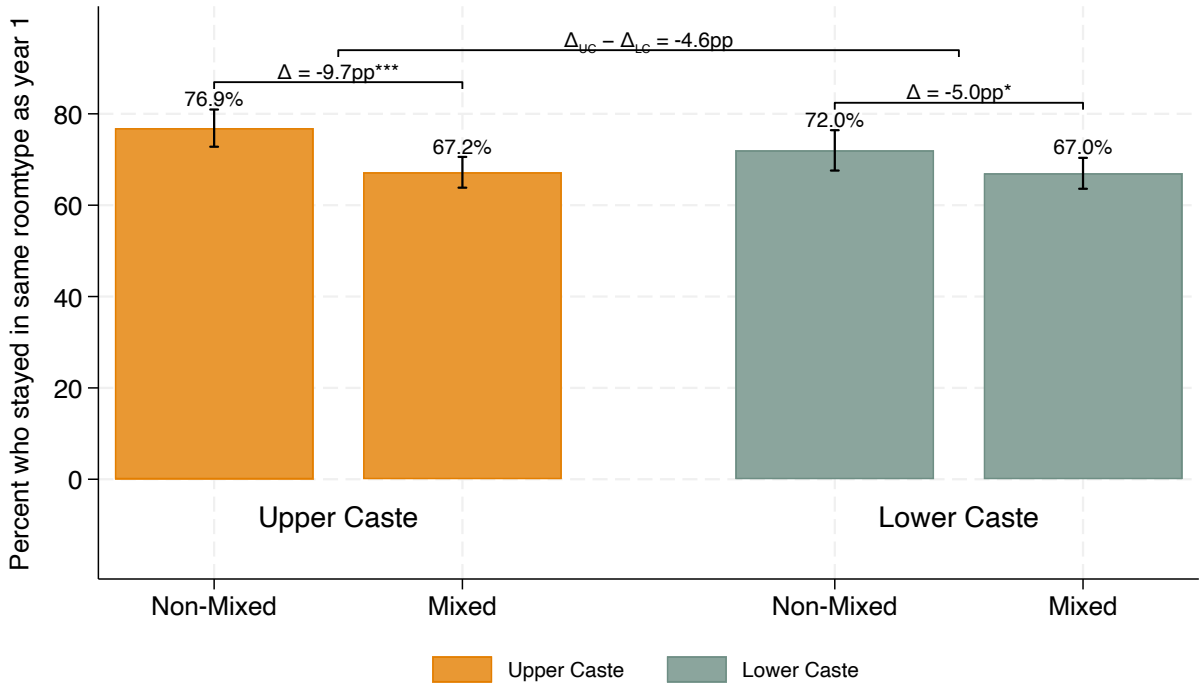


Figure 2: Room Transitions by caste group in Year 2

Notes: Bar represents share of each room type continues in the same room type in year 2. In the upper (lower) caste panel, First bars shows students assigned to non-mixed students continued in non-mixed rooms in year 2 and second bar shows share of students assigned to mixed rooms continued in mixed-rooms in year 2. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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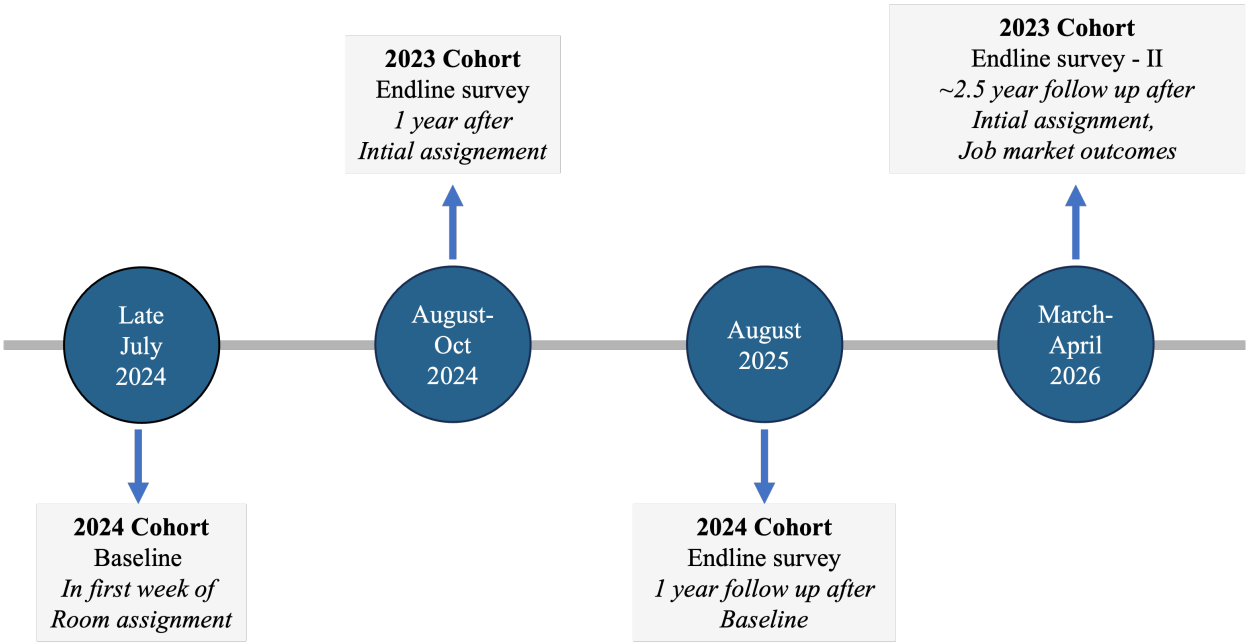


Figure 3: Timeline of Surveys: Baseline for 2024, Endline for 2023, 2024 cohorts and Endline-II for 2023 cohort

12 Tables

Table 1: Summary statistics on characteristics

	Upper Caste			Lower Caste			Upper Caste - Lower Caste	
	N	Mean	Sd	N	Mean	Sd	Difference	P-Val
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Socio Economic Indicators								
Mother's education (bachelors or higher)	902	0.855	0.353	881	0.571	0.495	0.284	0.000
Father's education (bachelors or higher)	896	0.756	0.430	876	0.614	0.487	0.141	0.000
Knows someone studied in IITs	912	0.265	0.442	907	0.184	0.388	0.081	0.000
Assets Index	824	0.150	0.547	809	-0.180	0.647	0.329	0.000
Taken a loan	885	0.148	0.355	890	0.230	0.421	-0.082	0.000
Family income (9L or above)	859	0.655	0.476	895	0.415	0.493	0.241	0.000
Owns a laptop/computer	824	0.900	0.300	809	0.759	0.428	0.142	0.000
Consumption Index	510	0.010	0.871	503	-0.097	0.740	0.107	0.036
B. Entrance Exam Preparation and Academic Background								
JEE Advanced score	1112	201.611	47.275	1091	141.166	47.111	60.445	0.000
Took more than 1 attempt	945	0.145	0.352	934	0.281	0.449	-0.136	0.000
Years: JEE preparation	964	2.473	1.386	964	2.175	1.365	0.298	0.000
Average daily study hours JEE	963	8.443	3.375	962	8.874	3.472	-0.431	0.006
JEE Prep Cost (in lakhs INR)	875	1.332	1.142	884	1.276	1.154	0.056	0.307
Score in Class XII	942	93.286	6.537	934	90.338	6.494	2.947	0.000
Score in Class X	939	95.044	7.282	935	92.877	8.109	2.167	0.000
C. Pre-college Friendship Networks								
Close friends count	488	11.004	10.604	484	9.531	9.869	1.473	0.025
Close friends in IITs	488	0.079	0.989	484	-0.034	1.067	0.113	0.087
Friends who did not clear entrance exam	488	4.973	8.160	484	5.983	9.111	-1.010	0.069
Share of friends from different caste	425	-0.340	1.025	418	0.485	0.759	-0.825	0.000
D. Attitudes towards Affirmative Action and Redistribution								
AA support: reserved group	455	-0.312	0.681	440	0.368	0.622	-0.679	0.000
AA support: non-reserved group	453	-0.349	0.730	442	0.476	0.699	-0.825	0.000
Quota for caste based AA (out of 100)	476	10.378	11.478	471	24.102	16.489	-13.724	0.000
Entrance exam: Merit only	452	0.103	1.042	429	-0.075	1.001	0.178	0.010
Opportunity inside college: Equal	453	0.008	1.092	429	0.013	0.937	-0.006	0.934
E. Mental Health at Baseline								
Mental health at Baseline	502	0.032	0.476	490	-0.045	0.489	0.077	0.012
PHQ4: anxiety and depression	455	0.017	0.813	438	-0.039	0.763	0.056	0.286
Relative wellbeing index	450	0.071	0.510	426	-0.049	0.498	0.120	0.000
Worry index	502	0.011	0.586	490	-0.064	0.687	0.075	0.065
Satisfied with Specialization/Department	474	0.538	0.499	461	0.638	0.481	-0.100	0.002

Notes: The table reports summary statistics for students from entry years 2023 and 2024 who had one or two roommates. Means and standard deviations are reported separately for upper-caste and lower-caste students. Mother's and father's education indicate whether the parent has a bachelor's degree or higher. Family income indicates annual household income above INR 9,00,000 (approximately USD 11,000). The asset ownership index combines ownership of a house, car, smart TV, agricultural land, non-agricultural land, air conditioner, and washing machine. The expenses index combines expenditure on food, movies/parties, education, and other miscellaneous items. Taken loan indicates whether the student's family has taken an education loan. Owns a laptop indicates whether the student owns a personal laptop. Entrance exam score is the score obtained on the entrance examination (out of 360). Took more than one attempt indicates whether the student took multiple attempts to pass the examination. Years of preparation and hours studied per day refer to preparation for the entrance examination. Cost of entrance preparation is the total preparation cost, reported in lakhs of INR. Class 10 and Class 12 scores are percentage scores on board examinations. Close friends measures the number of close friends reported before entering college; close friends in college measures the number of close friends at college; and close friends in other elite engineering institutes measures the number of close friends at other elite engineering institutes. Support for AA based on costs and benefits to disadvantaged (advantaged) groups measures support for affirmative action based on its perceived costs and benefits for disadvantaged (advantaged) groups. Support for Caste-Based AA measures the number of seats, out of 100, that students believe should be reserved for disadvantaged caste groups. Entrance Exam: Merit Only, Opportunity inside College: Equal, are measuring students' beliefs about merit in the entrance examination, equal opportunity in college, respectively. Mental health index is a composite index of PHQ-4, worry index, relative wellbeing index. All mental health variables are coded so that higher values indicate better mental health. Raw PHQ-4, relative wellbeing, and worry scale items are reverse-coded from their original direction, on which higher values indicate worse mental health. PHQ-4 measures anxiety and depression symptoms; the Relative Happiness Index measures happiness relative to students' pre-college level; the Worry Index measures concerns about both academic and social life of college. Department satisfaction is an indicator of whether the student is satisfied with their department.

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Table 2: Social Interaction and Substantial Help, Same-Caste Rooms

	Upper Caste			Lower Caste			Upper Caste - Lower Caste	
	N	Mean	Sd	N	Mean	Sd	Difference	P-Val
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Social interaction (once a week or more)								
Study together	286	0.462	0.499	296	0.574	0.495	-0.113	0.006
Eat together	288	0.597	0.491	298	0.651	0.477	-0.054	0.180
Go out together	286	0.304	0.461	297	0.343	0.476	-0.039	0.312
Play sports together	286	0.413	0.493	299	0.498	0.501	-0.086	0.037
B. Substantial help received								
Exam prep: help received	296	0.666	0.473	308	0.666	0.473	-0.000	0.999
Internships: help received	294	0.313	0.464	297	0.327	0.470	-0.014	0.722
Emotional support: help received	293	0.522	0.500	305	0.574	0.495	-0.052	0.206
Financial support: help received	292	0.298	0.458	296	0.463	0.499	-0.165	0.000
C. Substantial help provided								
Exam prep: help provided	277	0.643	0.480	288	0.701	0.458	-0.059	0.137
Internships: help provided	270	0.289	0.454	277	0.336	0.473	-0.047	0.238
Emotional support: help provided	275	0.524	0.500	283	0.618	0.487	-0.095	0.024
Financial support: help provided	275	0.342	0.475	285	0.495	0.501	-0.153	0.000

Notes: The table reports social interaction and substantial help exchanged between roommates, for students from entry years 2023 and 2024 with one or two roommates, restricted to same-caste rooms. In three-person rooms, roommate-reported outcomes take the maximum over the two roommates' reports, equal to one if the activity occurred with at least one roommate; in two-person rooms the single roommate report is used. Panel A reports indicators for social interaction – studying, eating, going out, and playing sports together – equal to one if the activity occurred once a week or more. Panel B reports indicators for substantial help received from a roommate with exam preparation, internships, emotional support, and financial support, equal to one if the roommate was reported as 'Sometimes helpful' or 'Mostly helpful' on that dimension. Panel C reports the analogous indicators for substantial help provided to a roommate, equal to one if the respondent reported being 'Sometimes' or 'Mostly' supportive. Mean, standard deviation, and sample size are reported separately for upper-caste and lower-caste students; the p-value is from an unequal-variance t-test of equal means.

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Table 3: Effect on GPA, Mental Health and Attitudes towards Affirmative Action

	CGPA : First Year (1)	Mental health index (2)	AA support: reserved group (3)	AA support: non-reserved group (4)
Panel A: Upper Caste Sample				
Mixed Room	-0.103**	-0.062	0.115**	0.048
	(0.050)	(0.044)	(0.057)	(0.058)
DV Mean in same caste rm	0.474	0.027	-0.251	-0.265
R-squared	0.473	0.149	0.172	0.184
N	947	935	820	817
Sharpened q-value	0.044	0.108	0.044	0.206
Panel B: Lower Caste Sample				
Mixed Room	0.091*	-0.082**	-0.022	0.110**
	(0.052)	(0.042)	(0.045)	(0.056)
DV Mean in same caste rm	-0.445	0.001	0.239	0.237
R-squared	0.433	0.146	0.110	0.131
N	896	928	829	820
Sharpened q-value	0.053	0.049	0.309	0.049
Panel C: Full Sample				
Mixed Room	-0.015	-0.073**	0.049	0.100**
	(0.038)	(0.032)	(0.040)	(0.047)
DV Mean in same caste rm	0.021	0.014	-0.002	-0.012
R-squared	0.506	0.109	0.138	0.135
N	1843	1863	1649	1637
Sharpened q-value	0.706	0.106	0.318	0.108
P val ($\beta_LC == \beta_UC$)	0.033	0.509	0.086	0.648

Notes: The table presents results for academic performance, mental health, and attitudes toward affirmative action. Column 1 reports estimates for cumulative GPA at the end of the first year. Column 2 reports estimates for a composite Mental Health Index, constructed using standardized scores of the PHQ-4 index, and the worry index capturing academic- and social-life concerns in college. Negative values of the index indicate worse mental health outcomes. Column 3 reports an index of support for AA based on its perceived costs and benefits for reserved-group students. Column 4 reports an index of support for AA based on its perceived costs and benefits for non-reserved-group students. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste, testing whether the lower- and upper-caste treatment effects differ. All regressions include hostel-by-cohort and specialization-year fixed effects. I use ANCOVA specification along with a control for the number of roommates. Standard errors are clustered at the room level. Sharpened q-values (Anderson 2008) correcting for multiple hypothesis testing within each panel are reported. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table 4: Heterogeneous Effects of Roommate's Entrance Exam Score on First-Year GPA

	Upper Caste			Lower Caste		
	(1) CGPA : First Year	(2) CGPA : First Year	(3) CGPA : First Year	(4) CGPA : First Year	(5) CGPA : First Year	(6) CGPA : First Year
Mixed Room	-0.103**	-0.118*	-0.100	0.091*	0.021	0.001
	(0.050)	(0.064)	(0.064)	(0.052)	(0.065)	(0.066)
Roommate's Entrance Score (SD)		-0.011	0.044		0.091**	0.136**
		(0.040)	(0.055)		(0.043)	(0.058)
Mixed Room × Roommate's Entrance Score			-0.084			-0.068
			(0.058)			(0.064)
DV Mean, Non-mixed	0.474	0.474	0.474	-0.445	-0.445	-0.445
R-squared	0.473	0.469	0.470	0.433	0.439	0.440
N	947	919	919	896	867	867

Notes: This table reports estimates of the effects of mixed-room assignment on GPA, controlling for roommates' entrance-exam scores and their interaction with mixed-room assignment. Columns 1–3 report results for upper-caste students, and Columns 4–6 report results for lower-caste students. Within each caste group, Columns 1 and 4 report the baseline specification without mediators; Columns 2 and 5 add roommates' entrance-exam scores; and Columns 3 and 6 additionally include the interaction between roommates' entrance-exam scores and mixed-room assignment. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors, clustered at the room level, are reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table 5: Effect of Mixed-Room Assignment on Roommate Interaction Intensity

	Upper Caste			Lower Caste		
	(1) Interaction Intensity (Composite Index)	(2) Study Interactions	(3) Social Interactions	(4) Interaction Intensity (Composite Index)	(5) Study Interactions	(6) Social Interactions
Mixed Room	-0.083	-0.016	-0.085	-0.159**	-0.042	-0.181***
	(0.073)	(0.060)	(0.077)	(0.067)	(0.057)	(0.069)
DV Mean, Non-mixed	-0.111	-0.080	-0.098	0.126	0.047	0.120
R-squared	0.141	0.173	0.150	0.116	0.116	0.126
N	865	906	864	883	938	879

Columns 1-3 restrict to upper-caste students, columns 4-6 to lower-caste students. Column 1(4) reports the Composite Interaction Intensity. Columns 2(5) report study-related interactions and 3(6) reports social interaction index. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates, Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table 6: Effect on GPA, Internships, Mental Health and Attitudes towards Affirmative Action: 3 Years after Admission

	CGPA : Third Year (1)	Got an internship offer (2)	Salary: 3rd year activity (in 1000 INR) (3)	Mental health: 3yr (4)	AA support: reserved group (5)	AA support: non-reserved group (6)
Panel A: Upper Caste Sample						
Mixed Room	-0.125*	-0.044	-32.542***	-0.044	-0.028	-0.074
	(0.075)	(0.038)	(11.716)	(0.049)	(0.065)	(0.073)
DV Mean in same caste rm	0.518	0.782	163.631	0.089	-0.138	-0.197
R-squared	0.327	0.260	0.418	0.068	0.070	0.079
N	471	469	446	472	452	454
Sharpened q-value	0.244	0.370	0.029	0.370	0.556	0.370
Panel B: Lower Caste Sample						
Mixed Room	0.085	-0.073*	-9.535	-0.016	0.112*	0.086
	(0.077)	(0.042)	(7.774)	(0.043)	(0.061)	(0.066)
DV Mean in same caste rm	-0.459	0.487	66.921	-0.072	0.118	0.165
R-squared	0.207	0.263	0.234	0.059	0.080	0.072
N	500	474	485	494	468	467
Sharpened q-value	0.321	0.259	0.321	0.710	0.259	0.321
Panel C: Full Sample						
Mixed Room	-0.016	-0.046	-21.309***	-0.028	0.042	-0.001
	(0.055)	(0.029)	(7.213)	(0.033)	(0.046)	(0.053)
DV Mean in same caste rm	-0.000	0.631	112.238	0.004	-0.004	-0.009
R-squared	0.365	0.270	0.365	0.049	0.042	0.049
N	971	943	931	966	920	921
Sharpened q-value	0.745	0.261	0.018	0.527	0.527	0.820
P val ($\beta_LC == \beta_UC$)	0.064	0.621	0.077	0.814	0.072	0.073

Notes: The table presents effects of mixed-room assignment on medium-run academic, labor-market, mental health, and attitudinal outcomes. Column 1 reports estimates for cumulative GPA through the fifth semester, the first semester of the third year. Column 2 reports results for third-year summer internship completion, an indicator for whether a student secured an internship, a direct input into final job offers. Column 3 reports results for internship stipend. Column 4 reports estimates for the composite Mental Health Index at the three-year follow-up, constructed using standardized scores of the PHQ-4 index, and the worry index capturing academic- and social-life concerns in college. Negative values of the index indicate worse mental health outcomes. Column 5 reports an index of support for AA based on its perceived costs and benefits for reserved-group students. Column 6 reports an index of support for AA based on its perceived costs and benefits for non-reserved-group students. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste, testing whether the lower- and upper-caste treatment effects differ. All regressions include hostel-by-cohort and specialization-year fixed effects. Columns 3 through 6 do not include a baseline control for the respective outcome, since baseline mental health and AA measures exist only for the 2024 cohort and follow up sample is from 2023 cohort. Standard errors are clustered at the room level. Sharpened q-values (Anderson 2008) correcting for multiple hypothesis testing within each panel are reported. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively.

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Table 7: Effect on Leadership Positions, Job Preferences, and Expected Salary: Pooled One- and Three-Year Sample

	Upper Caste			Lower Caste		
	(1) Clubs Leadership	(2) Preferred job sector: Finance/Consulting	(3) Expected salary (INR 1000s)	(4) Clubs Leadership	(5) Preferred job sector: Finance/Consulting	(6) Expected salary (INR 1000s)
Mixed Room	-0.091**	-0.078**	-80.385	0.005	-0.029	-34.579
	(0.039)	(0.033)	(358.694)	(0.036)	(0.029)	(320.247)
DV Mean, Non-mixed	0.079	0.479	5332.267	-0.083	0.234	4301.939
R-squared	0.113	0.131	0.142	0.087	0.095	0.066
N	968	961	1062	953	950	1061

Columns 1-3 restrict to upper-caste students, columns 4-6 to lower-caste students. Column 1 reports effects on holding a position of responsibility in student clubs (Clubs Leadership index). Column 2 reports effects on preferring a job in finance or consulting. Column 3 reports effects on expected first-job salary in thousands of INR, top-coded at the 95th percentile to reduce the influence of outliers. No additional controls are included. Outcomes pool each student's most recent available measurement: the one-year follow-up for the 2024 cohort and the three-year follow-up for the 2023 cohort. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table 8: Heterogeneous Effects on Affirmative Action Attitudes by Baseline Belief that the Entrance Exam Reflects Merit

	Upper Caste		Lower Caste	
	(1) AA support: reserved group	(2) AA support: non-reserved group	(3) AA support: reserved group	(4) AA support: non-reserved group
Mixed Room	0.074	0.010	-0.039	0.098
	(0.083)	(0.091)	(0.065)	(0.071)
Baseline Merit Belief(SD)	-0.182***	-0.186***	-0.142***	-0.164***
	(0.060)	(0.061)	(0.053)	(0.057)
Mixed Room × Baseline Merit Belief	0.142**	0.135*	0.051	0.143**
	(0.071)	(0.077)	(0.065)	(0.069)
DV Mean, Non-mixed	-0.251	-0.265	0.239	0.237
R-squared	0.128	0.159	0.108	0.118
N	374	376	354	354

Columns 1-2 restrict to upper-caste students, columns 3-4 to lower-caste students. Column 1 (3) the reserved-group index indicating support for AA based on its perceived costs and benefits for reserved-group students. Column 2 (4) the non-reserved-group index indicating support for AA based on its perceived costs and benefits for non-reserved-group students. Each regression interacts mixed-room assignment with the student's baseline belief that only merit is required to clear the entrance exam, to test whether the attitude effect varies with prior meritocratic belief. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects, control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table 9: Heterogeneous Effects on Affirmative Action Attitudes by Baseline Cross-Caste Friendship

	Upper Caste		Lower Caste	
	(1) AA support: reserved group	(2) AA support: non-reserved group	(3) AA support: reserved group	(4) AA support: non-reserved group
Mixed Room	0.007	-0.071	0.000	0.158
	(0.093)	(0.106)	(0.073)	(0.099)
Baseline Social Segregation	-0.119**	-0.128**	-0.107	-0.206**
	(0.057)	(0.062)	(0.073)	(0.104)
Mixed Room × Baseline Social Segregation	0.131*	0.123	0.068	0.108
	(0.070)	(0.076)	(0.088)	(0.110)
DV Mean, Non-mixed	-0.251	-0.265	0.239	0.237
R-squared	0.129	0.158	0.077	0.120
N	351	353	344	344

Columns 1-2 restrict to upper-caste students, columns 3-4 to lower-caste students. Column 1 (3) the reserved-group index indicating support for AA based on its perceived costs and benefits for reserved-group students. Column 2 (4) the non-reserved-group index indicating support for AA based on its perceived costs and benefits for non-reserved-group students. Each regression interacts mixed-room assignment with the student's pre-college share of other-caste friends, testing whether the attitude effect varies with prior cross-caste exposure. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects, control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table 10: Effect on Academic and Social Help Received from/Provided to Roommate

	Upper Caste				Lower Caste			
	(1) Got exam prep	(2) Got Emotional /Financial Help	(3) Provided exam prep	(4) Provided Emotional /Financial Help	(5) Got exam prep	(6) Got Emotional /Financial Help	(7) Provided exam prep	(8) Provided Emotional /Financial Help
Mixed Room	-0.170*	-0.040	0.144*	0.106	0.120	-0.130*	-0.142**	-0.167**
	(0.090)	(0.075)	(0.082)	(0.073)	(0.075)	(0.072)	(0.071)	(0.069)
DV Mean, Non-mixed	-0.040	-0.113	-0.081	-0.154	0.048	0.103	0.089	0.150
R-squared	0.097	0.092	0.092	0.107	0.081	0.089	0.078	0.093
N	880	878	835	836	887	882	841	834

Notes: This table reports the effects of mixed-room assignment on the types of help received from roommates. Columns 1–4 report results for upper-caste students, and Columns 5–8 report results for lower-caste students. Columns 1 and 5 report effects on academic help got from roommates; Columns 2 and 6 report effects on emotional or financial help received; Columns 3 and 7 report effects on academic help provided to roommates; and Columns 4 and 8 effects on emotional or financial help provided. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

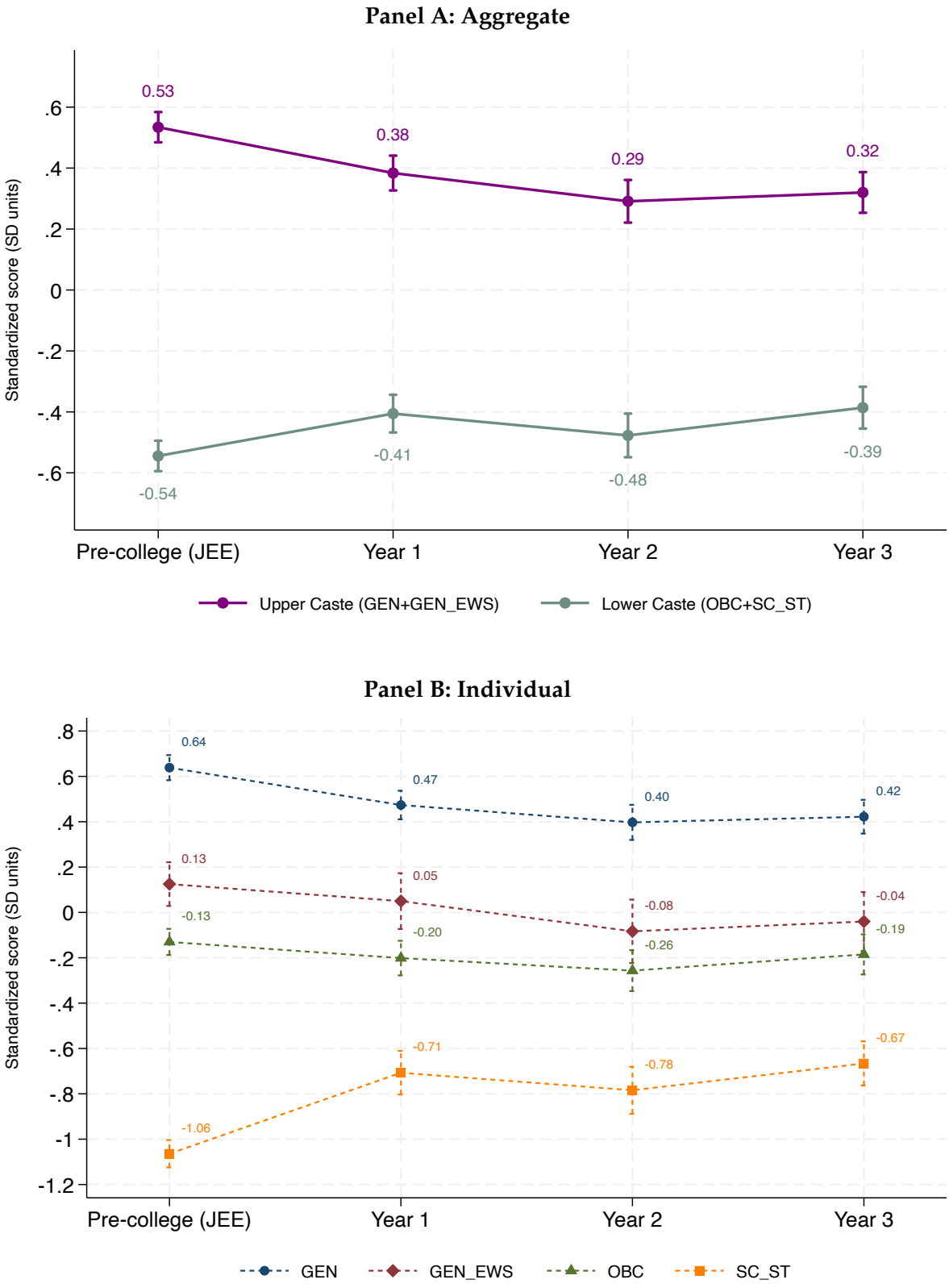
Table 11: Effect on Support for Affirmative Action in Education, Jobs, and Politics: Three Years after Admission

	Upper Caste			Lower Caste		
	(1) Education	(2) Jobs	(3) Politics	(4) Education	(5) Jobs	(6) Politics
Mixed Room	-0.077	-0.170**	-0.145	0.033	0.084	-0.036
	(0.095)	(0.085)	(0.094)	(0.081)	(0.077)	(0.088)
DV Mean, Non-mixed	-0.232	-0.189	-0.184	0.196	0.174	0.152
R-squared	0.035	0.032	0.027	0.032	0.046	0.037
N	459	461	456	465	464	456

Columns 1-3 restrict to upper-caste students, columns 4-6 to lower-caste students. Column 1 reports effects on support for caste-based reservations in elite educational institutions. Column 2 reports effects on support for reservations in public- and private-sector jobs. Column 3 reports effects on support for reservations in politics and local elections. No additional controls are included. Sample is restricted to the 2023 cohort's three-year follow-up. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

13 Online Appendix

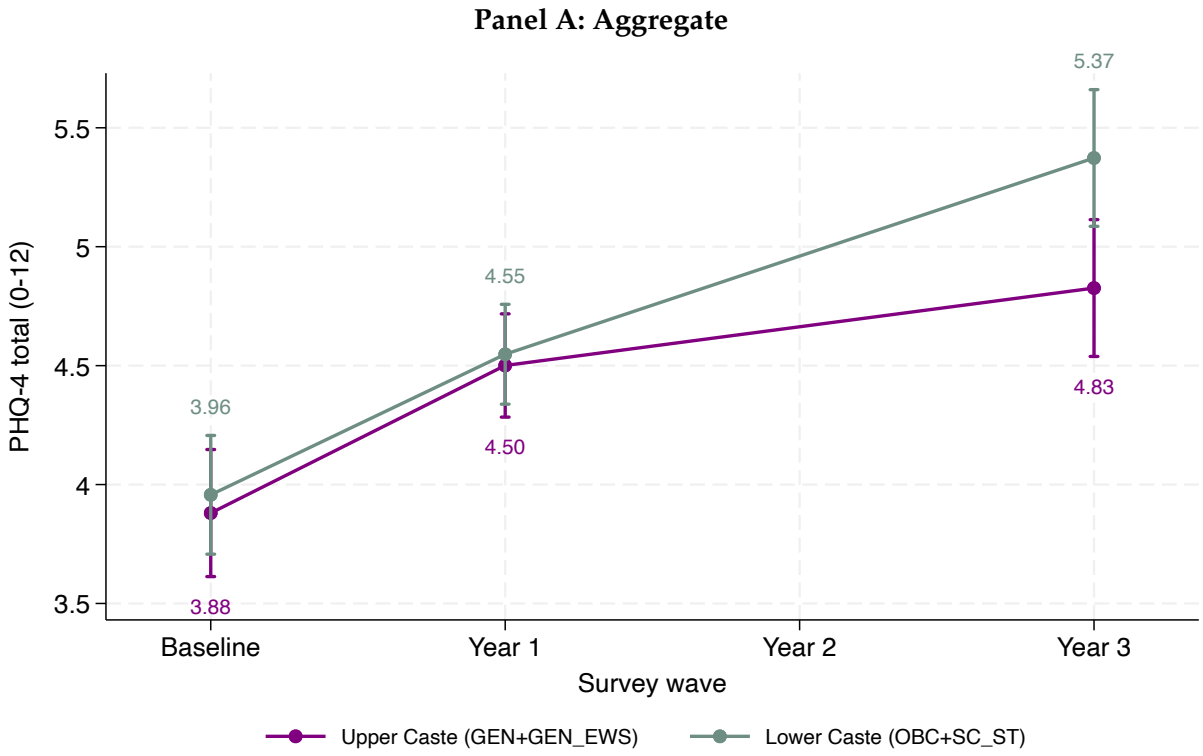
Figure A1: Academic Trajectories by caste from Baseline to year 3

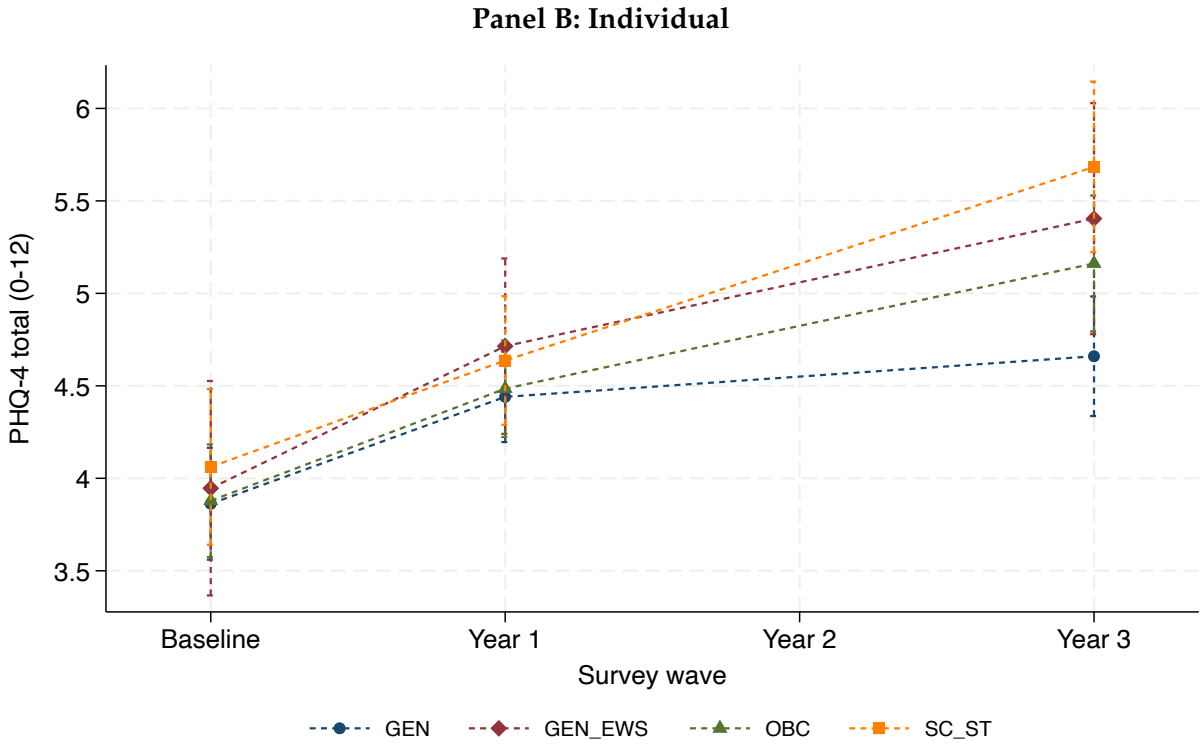


Notes: The figure shows trends in standardized GPA and baseline entrance-exam scores. Baseline exam scores are standardized using their own distribution, while GPA in Years 1, 2, and 3 is standardized relative to the Year 1 GPA distribution. Baseline and Year 1 data include both the 2023 and 2024 cohorts, whereas Years 2 and 3 include only the 2023 cohort. Panel A shows trends for upper- and lower-caste students. Panel B disaggregates these groups further to show trends for each subgroup.

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Figure A2: Mental Health (PHQ4) Trajectories by caste from Baseline to year 3





Notes: The figure shows trends in PHQ-4 scores. Baseline data include only the 2024 cohort, Year 1 data include both the 2023 and 2024 cohorts, and Year 3 data include only the 2023 cohort. Panel A shows trends for upper- and lower-caste students. Panel B further disaggregates these groups to show trends for each subgroup.

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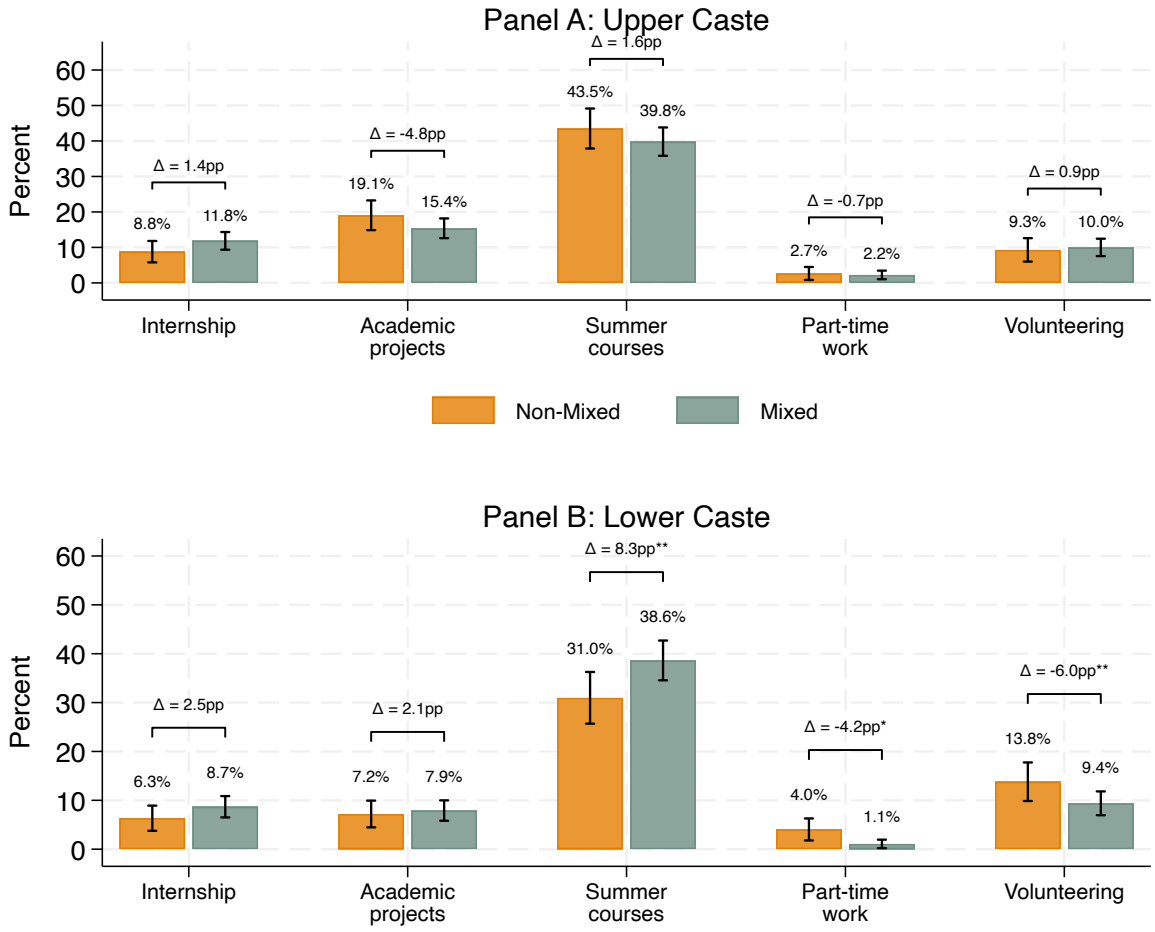


Figure A3: Summer Activities at the end of year 1

Notes: Each bar show mean of the activity by room type. Delta and significance is calculated by running baseline specification with hostel-year, specialization-year and roommate fixed effects and I also control for baseline entrance exam scores in all regressions and standard errors are clustered at room level ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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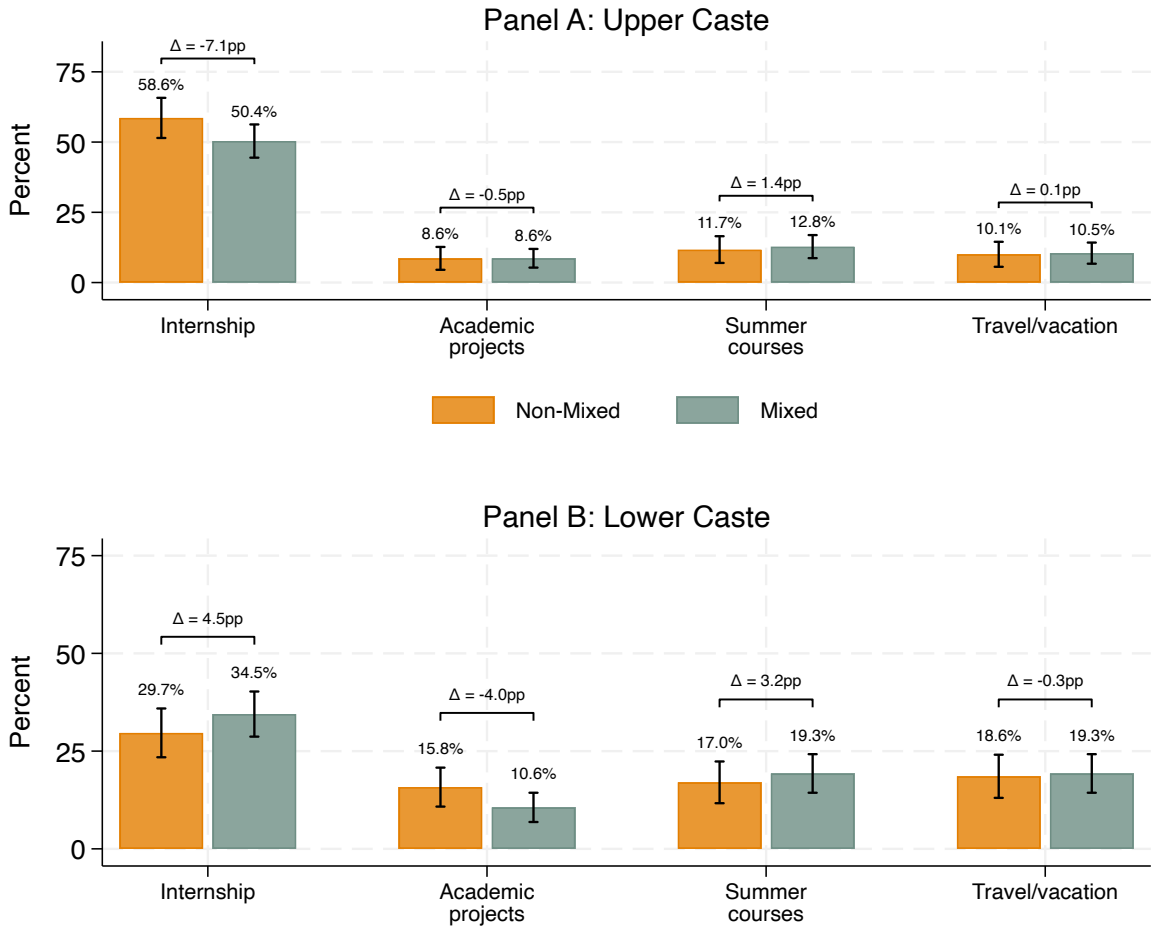
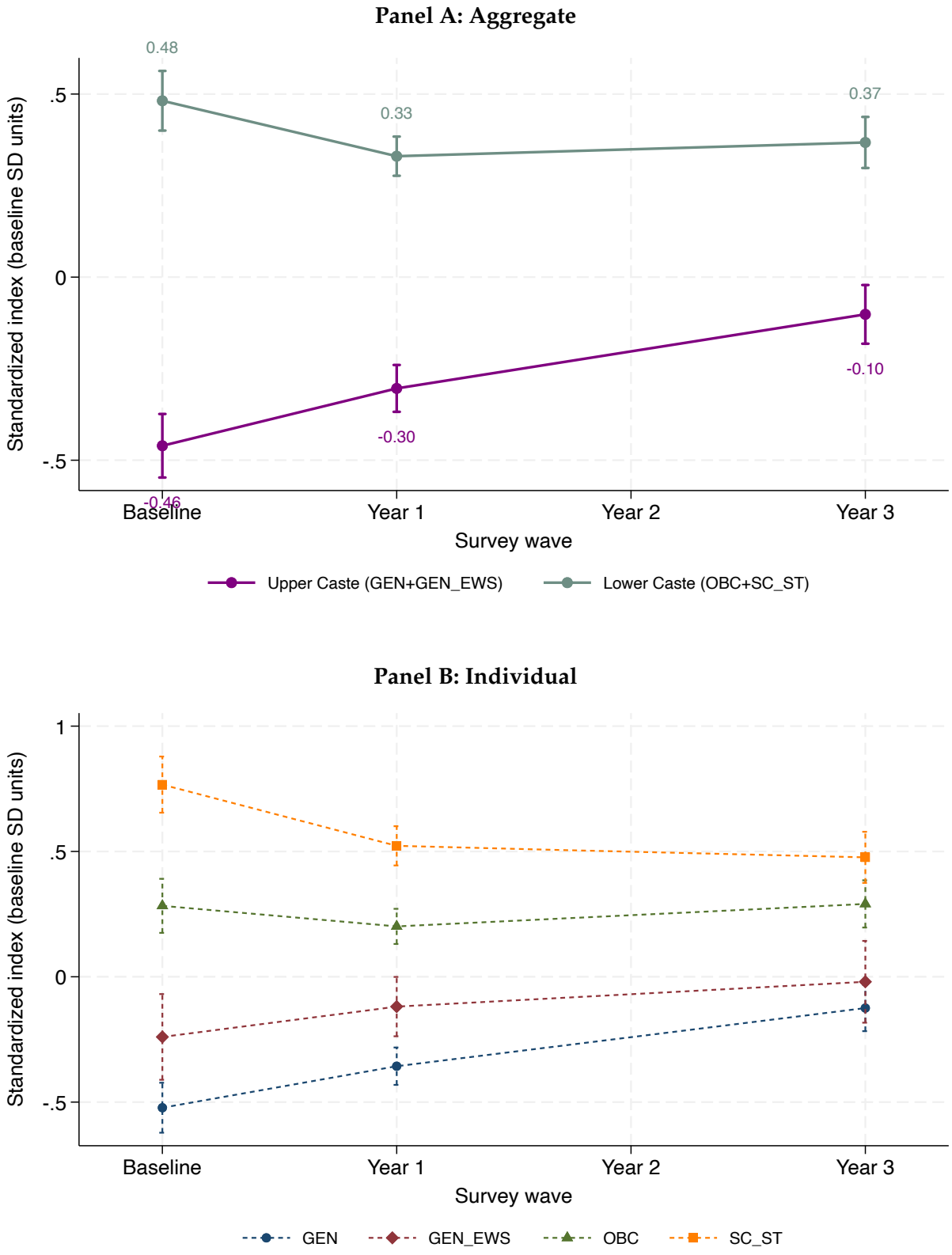


Figure A4: Summer Activities at the end of year 2

Notes: Each bar show mean of the activity by room type. Delta and significance is calculated by running baseline specification with hostel-year, specialization-year and roommate fixed effects and I also control for baseline entrance exam scores in all regressions and standard errors are clustered at room level ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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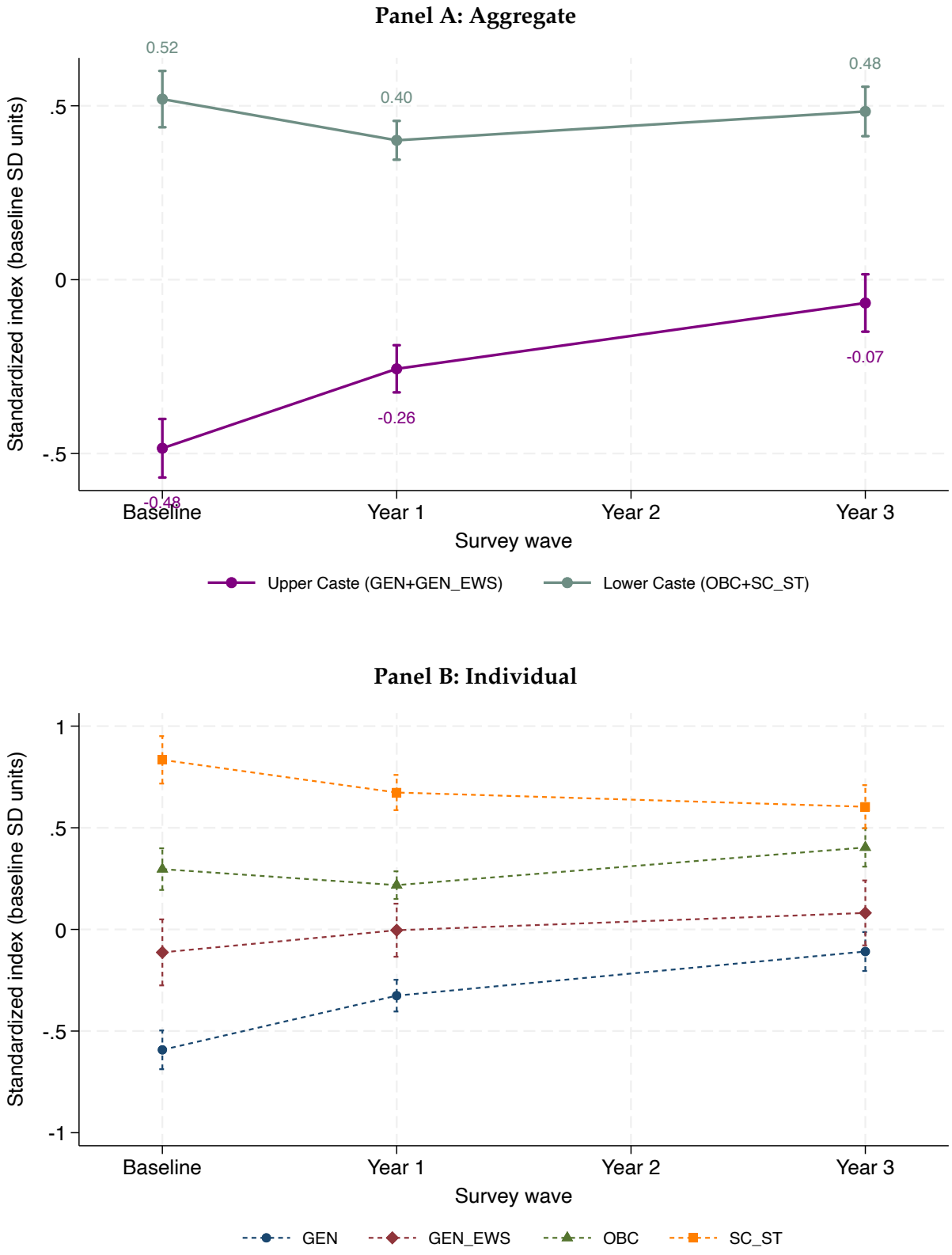
Figure A5: AA Supp: Costs and Benefits to reserved group from Baseline to year 3



Notes: The figure shows trends in support for affirmative action (AA), measured by an index of perceived costs and benefits to reserved groups. The indices are standardized relative to their baseline distributions. Baseline data include only the 2024 cohort, Year 1 data include both the 2023 and 2024 cohorts, and Year 3 data include only the 2023 cohort. Panel A shows trends for upper- and lower-caste students. Panel B further disaggregates these groups to show trends for each subgroup.

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Figure A6: AA Supp: Costs and Benefits to non-reserved group from Baseline to year 3



Notes: The figure shows trends in support for affirmative action (AA), measured by an index of perceived costs and benefits to non-reserved groups. The indices are standardized relative to their baseline distributions. Baseline data include only the 2024 cohort, Year 1 data include both the 2023 and 2024 cohorts, and Year 3 data include only the 2023 cohort. Panel A shows trends for upper- and lower-caste students. Panel B further disaggregates these groups to show trends for each subgroup.

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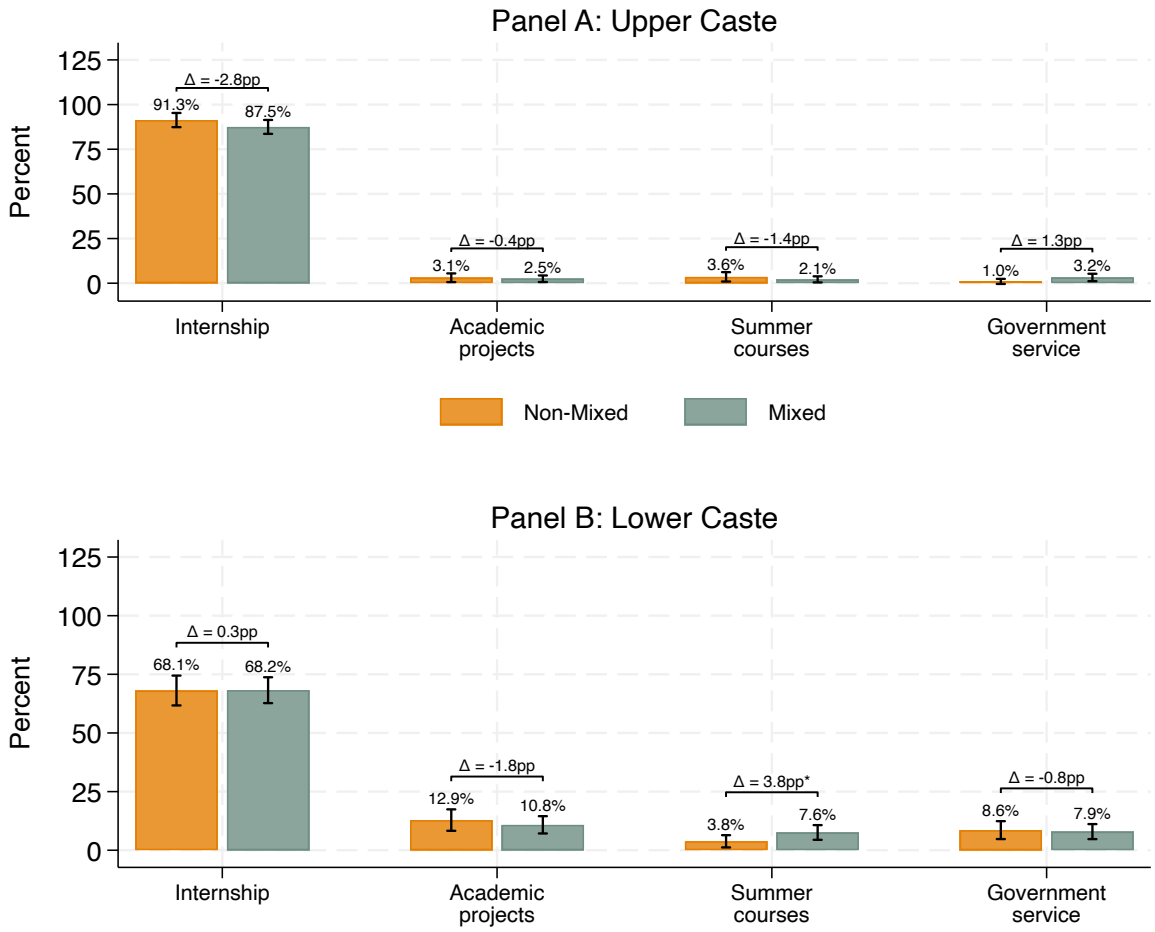


Figure A7: Summer Activities at the end of year 3

Notes: Each bar show mean of the activity by room type. Delta and significance is calculated by running baseline specification with hostel-year, specialization-year and roommate fixed effects and I also control for baseline entrance exam scores in all regressions and standard errors are clustered at room level ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

Table A1: Survey Response Rates by Wave

Wave	Sampled (N)	Opened Survey Link	Responded to ≥1 Outcome	Completed Survey
2024 Baseline	1,147	90.8%	86.4%	82.0%
2024 Endline	1,147	93.4%	92.1%	90.3%
2023 Endline (1 Year), Raw	1,120	85.6%	81.0%	77.6%
2023 Endline (1 Year), Effective			92.8%	90.2%
2023 Endline (3 Years)	1,120	90.7%	89.4%	87.1%

Notes: *Notes:* Sample is 2023 and 2024 admission cohorts and two surveys waves for each cohort. *Opened survey link* indicates that the student provided consent, entered their name and college ID, and completed at least one page of the survey instrument. *Responded* indicates that the student progressed beyond the demographic and academic-outcome pages. *Completed* indicates that the student finished the survey in full. The 2023 Endline (1 Year) survey was conducted in two stages. In the initial stage, the response rate was 71.7

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Table A2: Extensive and Intensive Margins of Help Between Roommates, Same-Caste Rooms

	Upper Caste			Lower Caste			Upper Caste - Lower Caste	
	N	Mean	Sd	N	Mean	Sd	Difference	P-Val
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Never asked for help: received								
Exam prep: never asked (received)	296	0.206	0.405	308	0.153	0.360	0.053	0.087
Internships: never asked (received)	294	0.544	0.499	297	0.431	0.496	0.113	0.006
Emotional support: never asked (received)	293	0.365	0.482	305	0.259	0.439	0.106	0.005
Financial support: never asked (received)	292	0.565	0.497	296	0.385	0.487	0.180	0.000
B. Never asked for help: provided								
Exam prep: roommate never asked	277	0.238	0.427	288	0.153	0.360	0.085	0.011
Internships: roommate never asked	270	0.570	0.496	277	0.422	0.495	0.148	0.001
Emotional support: roommate never asked	275	0.349	0.478	283	0.219	0.414	0.130	0.001
Financial support: roommate never asked	275	0.556	0.498	285	0.358	0.480	0.198	0.000
C. Substantial help, conditional on having asked: received								
Exam prep: substantial help received, if asked	235	0.838	0.369	261	0.785	0.411	0.053	0.132
Internships: substantial help received, if asked	134	0.687	0.466	169	0.574	0.496	0.113	0.043
Emotional support: substantial help received, if asked	186	0.823	0.383	226	0.774	0.419	0.048	0.223
Financial support: substantial help received, if asked	127	0.685	0.466	182	0.753	0.433	-0.068	0.197
D. Substantial help, conditional on having asked: provided								
Exam prep: substantial help provided, if asked	211	0.844	0.364	244	0.828	0.378	0.016	0.652
Internships: substantial help provided, if asked	116	0.672	0.471	160	0.581	0.495	0.091	0.122
Emotional support: substantial help provided, if asked	179	0.804	0.398	221	0.792	0.407	0.013	0.755
Financial support: substantial help provided, if asked	122	0.770	0.422	183	0.770	0.422	0.000	1.000

Notes: The table reports the extensive and intensive margins of help exchanged between roommates, for the same sample as Table 1 (same-caste rooms only), using the same roommate-report combination rule, except that in three-person rooms the never-asked indicators in Panels A-B take the minimum over the two roommates' reports, so that the pair is coded as having asked for help if either roommate reports asking; in two-person rooms the single roommate report is used. Panels A and B report the extensive margin: whether help was ever asked for, equal to one if the respondent reports never having asked a roommate for help (Panel A, received) or never having been asked for help by a roommate (Panel B, provided), across exam preparation, internships, emotional support, and financial support. Panels C and D report the intensive margin – substantial help, conditional on having asked – restricting to the subsample for whom some help was asked for or given (Panels A-B equal to zero), and reporting Table 1's substantial-help indicator ('Sometimes' or 'Mostly' helpful/supportive) within that subsample. Mean, standard deviation, and sample size are reported separately for upper-caste and lower-caste students; the p-value is from an unequal-variance t-test of equal means.

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Table A3: Time Use, Active Membership, and Positions in Clubs and Activities, Same-Caste Rooms

	Upper Caste			Lower Caste			Upper Caste - Lower Caste	
	N	Mean	Sd	N	Mean	Sd	Difference	P-Val
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Time use								
Study hours : normal week	334	10.987	12.079	344	13.907	15.307	-2.920	0.006
Study hours: exam week	334	32.710	22.183	344	29.846	22.796	2.864	0.098
Hours in student clubs	334	9.970	13.626	343	8.714	12.733	1.256	0.216
Hours in internships/jobs	333	5.111	9.940	343	6.216	11.704	-1.105	0.186
Hours in part-time jobs	333	1.988	7.963	342	3.085	10.155	-1.097	0.118
Hours in other tasks	332	9.816	15.028	342	9.557	14.329	0.259	0.819
B. Indicator for active membership								
Participation in cultural clubs	304	0.401	0.491	300	0.277	0.448	0.125	0.001
Participation in hostel clubs	305	0.410	0.493	300	0.327	0.470	0.083	0.034
Participation in co-curricular clubs	305	0.295	0.457	299	0.184	0.388	0.111	0.001
Participation in volunteering clubs	305	0.239	0.427	301	0.292	0.456	-0.053	0.140
Participation in sports	303	0.370	0.484	300	0.427	0.495	-0.057	0.153
C. Positions held in clubs and activities								
Cultural clubs	336	0.497	0.501	332	0.383	0.487	0.114	0.003
Technical clubs	336	0.286	0.452	332	0.157	0.364	0.129	0.000
Institute board	336	0.274	0.447	332	0.166	0.372	0.108	0.001
Hostel positions	336	0.304	0.460	332	0.217	0.413	0.087	0.011
Volunteering clubs	336	0.077	0.268	332	0.123	0.329	-0.046	0.048
Cultural fest	336	0.304	0.460	332	0.407	0.492	-0.103	0.005

Notes: The table reports time use and club/activity involvement, for the same sample as Table 1 (same-caste rooms only). Panel A reports the respondent's own weekly time use (not a roommate report), in hours, across studying in a normal week, studying during exam week, student clubs, internships/job search, part-time jobs, and other activities, as recorded in the survey; no roommate-report combination rule is applied since these are individual self-reports. Panel B reports indicators for active membership in sports, cultural clubs, hostel activities, co-curricular activities, and volunteering, equal to one if the respondent reports being an active member of that club or activity type. Panel C reports indicators for holding a position in cultural clubs, technical clubs, the institute board, hostel activities, volunteering clubs, or the cultural fest, equal to one if the respondent reports holding such a position. Mean, standard deviation, and sample size are reported separately for upper-caste and lower-caste students; the p-value is from an unequal-variance t-test of equal means.

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Table A4: Balance table on Attrition, Socio Economic Indicators and College Preparation

	Upper Caste				Lower Caste			
	N	Non-mixed mean	Diff	P-Val	N	Non-mixed mean	Diff	P-Val
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Survey attrition indicators								
Response	1145	0.838	-0.005	0.848	1121	0.891	-0.015	0.527
Survey completed	1145	0.800	-0.006	0.828	1121	0.872	0.005	0.859
B. Socio Economic Indicators								
Mother's education (bachelors or higher)	902	0.876	-0.043	0.148	881	0.566	-0.008	0.834
Father's education (bachelors or higher)	896	0.786	-0.023	0.447	876	0.641	-0.039	0.302
Assets Index	824	0.166	-0.016	0.728	809	-0.171	-0.009	0.856
Taken a loan	885	0.148	0.006	0.829	890	0.194	0.034	0.241
Family income (9L or above)	859	0.699	-0.079	0.047	895	0.425	-0.039	0.287
Speaks English with family	917	0.494	-0.027	0.462	938	0.246	0.048	0.133
Speaks English with friends	917	0.705	-0.003	0.934	938	0.558	0.027	0.452
Knows someone who did engineering	912	0.682	-0.060	0.088	907	0.490	-0.005	0.888
Knows someone studied in IITs	912	0.259	-0.005	0.885	907	0.168	0.013	0.642
C. JEE Preparation and Academic Background								
Score in Class XII	942	93.089	0.985	0.135	934	90.175	0.271	0.576
Score in Class X	939	94.995	0.424	0.540	935	92.942	-0.374	0.569
JEE Advanced score	1112	204.125	-1.279	0.354	1091	139.368	1.354	0.527
Years: JEE preparation	964	2.520	-0.130	0.216	964	2.251	-0.020	0.842
Average daily study hours JEE	963	8.225	0.167	0.582	962	8.588	0.467	0.080
JEE Prep Cost (in lakhs INR)	875	1.384	-0.022	0.812	884	1.394	-0.088	0.320
Took more than 1 attempt	945	0.147	-0.004	0.862	934	0.317	-0.063	0.066
Doesn't Know programming	912	0.455	0.041	0.291	907	0.671	-0.028	0.427
Joint Test p-value				0.181				0.296

Notes: The table presents balance on socioeconomic indicators, JEE preparation, and pre-college academic outcomes. Panel A reports balance on attrition indicators, including whether students responded to the survey and whether they completed the survey. Panels B and C report balance on socioeconomic characteristics and pre-college academic outcomes, respectively. The sample includes students from entry years 2023 and 2024 who had one or two roommates. Columns 2 and 6 report means for same-caste rooms for upper-caste and lower-caste students, respectively. Columns 3–4 and 7–8 report differences in means and corresponding p-values for upper-caste and lower-caste students. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level. The last row reports the p-value from the joint F-test of orthogonality.

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Table A5: Balance table on Pre-college Networks

	Upper Caste				Lower Caste			
	N	Non-mixed mean	Diff	P-Val	N	Non-mixed mean	Diff	P-Val
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Pre-college Networks								
Close friends count	488	10.929	-0.057	0.957	484	9.173	0.595	0.568
Close friends wrote jee	488	10.489	-0.550	0.604	484	9.624	0.103	0.929
Close friends in college	488	2.979	-0.104	0.865	484	2.549	0.407	0.455
Close friends in NITs	488	1.851	0.819	0.188	484	2.692	0.784	0.189
Friends who did not clear entrance exam	488	4.574	0.911	0.290	484	5.684	1.080	0.237
Prog: Learned from friends	477	0.036	0.018	0.419	464	0.031	-0.012	0.451
Share of friends from different caste	425	-0.475	0.208	0.080	418	0.467	0.039	0.668
B. Consumption Expenditure at Baseline								
Food expenses	510	1763.514	-34.175	0.856	503	1500.000	-63.167	0.676
Movies and partying expenses	510	1283.784	-36.779	0.837	503	1021.429	-82.599	0.461
Education expenses	510	2354.730	-262.736	0.311	503	2567.857	-436.217	0.137
Other miscellaneous expenses	510	2111.486	-321.836	0.191	503	2128.571	-353.494	0.203
Joint Test p-value				0.268				0.899

Notes: The table presents balance on pre-college networks and self-reported skills at baseline. Panel A reports balance on friendship networks pre college and panel B represents balance on consumption expenditure at baseline. The sample includes students only from entry year 2024 who had one or two roommates. Columns 2 and 6 report means for same-caste rooms for upper-caste and lower-caste students, respectively. Columns 3–4 and 7–8 report differences in means and corresponding p-values for upper-caste and lower-caste students. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level. The last row of each panel reports the p-value from the joint F-test of orthogonality.

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Table A6: Early exposure results on Mental Health, Attitudes towards Affirmative Action and Skills

	Upper Caste				Lower Caste			
	N	Non-mixed mean	Diff	P-Val	N	Non-mixed mean	Diff	P-Val
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mental Health, AA support & Skill indices at Baseline								
PHQ4: anxiety and depression	455	0.013	0.044	0.639	438	-0.006	0.009	0.916
Relative wellbeing index	450	0.037	0.073	0.167	426	-0.037	0.030	0.605
Worry index	502	0.006	0.054	0.408	490	-0.015	-0.106	0.147
AA support: reserved group	455	-0.287	-0.089	0.306	440	0.314	0.053	0.416
AA support: non-reserved group	453	-0.369	0.021	0.807	442	0.408	0.020	0.780
Self rating of skills	490	0.251	-0.110	0.086	484	-0.303	0.112	0.151
Joint Test p-value				0.073				0.328

Notes: The table presents balance on mental health, attitudes towards Affirmative Action and self-reported skills at baseline. It reports early exposure effects on PHQ4 index, relative happiness index, worry index, academic merit index, social justice index, and skills index. The sample includes students only from entry year 2024 who had one or two roommates. Columns 2 and 6 report means for same-caste rooms for upper-caste and lower-caste students, respectively. Columns 3–4 and 7–8 report differences in means and corresponding p-values for upper-caste and lower-caste students. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level. The last row of each panel reports the p-value from the joint F-test of orthogonality.

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Table A7: Attrition on Socio Economic Indicators and College Preparation

	Upper Caste				Lower Caste			
	N	Non-mixed mean	Diff	P-Val	N	Non-mixed mean	Diff	P-Val
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Socio Economic Indicators								
Mother's education (bachelors or higher) Missing	1059	0.169	-0.031	0.206	1055	0.146	0.034	0.148
Father's education (bachelors or higher) Missing	1059	0.169	-0.025	0.344	1055	0.173	-0.000	0.998
Assets Index Missing	1059	0.203	0.059	0.038	1055	0.253	-0.009	0.761
Taken a loan Missing	1059	0.194	-0.026	0.310	1055	0.192	-0.028	0.283
Family income (9L or above) Missing	1059	0.217	-0.013	0.636	1055	0.159	0.003	0.897
Speaks English with family Missing	1059	0.144	-0.017	0.447	1055	0.118	-0.006	0.785
Speaks English with friends Missing	1059	0.144	-0.015	0.507	1055	0.118	-0.007	0.764
Knows someone in engg Missing	1059	0.153	-0.010	0.661	1055	0.165	-0.021	0.402
Knows someone with IIT degree Missing	1059	0.153	-0.010	0.661	1055	0.165	-0.021	0.402
B. JEE Preparation and Academic Background								
Score in Class XII Missing	1059	0.122	-0.015	0.482	1055	0.102	0.014	0.519
Score in Class X Missing	1059	0.125	-0.014	0.525	1055	0.096	0.020	0.345
JEE Advanced score Missing	1059	0.028	0.007	0.576	1055	0.027	-0.005	0.678
Years: JEE preparation Missing	1059	0.092	-0.010	0.623	1055	0.069	0.018	0.308
Average daily study hours JEE Missing	1059	0.097	-0.016	0.431	1055	0.074	0.014	0.432
JEE Prep Cost (in lakhs INR) Missing	1059	0.181	0.003	0.895	1055	0.151	0.021	0.404
Took more than 1 attempt Missing	1059	0.114	-0.015	0.498	1055	0.099	0.020	0.361
Doesn't Know programming Missing	1059	0.144	-0.016	0.519	1055	0.124	0.024	0.290

Notes: The table presents attrition on socioeconomic indicators, JEE preparation, and pre-college academic outcomes. Panels A and B report attrition on socioeconomic characteristics and pre-college academic outcomes, respectively. The sample includes students from entry years 2023 and 2024 who had one or two roommates. Columns 2 and 6 report means for same-caste rooms for upper-caste and lower-caste students, respectively. Columns 3–4 and 7–8 report differences in means and corresponding p-values for upper-caste and lower-caste students. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level.

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Table A8: Attrition on Mental Health and Academic Outcomes

	Upper Caste				Lower Caste			
	N	Non-mixed mean	Diff	P-Val	N	Non-mixed mean	Diff	P-Val
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Refused to Answer								
Nervous or anxious - Refused	953	0.086	0.006	0.756	969	0.096	0.024	0.303
Can't stop worrying - Refused	953	0.086	0.005	0.807	969	0.099	0.021	0.385
Feeling down - Refused	953	0.086	0.006	0.785	969	0.104	0.007	0.750
Little interest in activities - Refused	953	0.086	0.005	0.823	969	0.107	0.009	0.695
CGPA : First Year - Refused	982	0.021	0.023	0.066	987	0.073	-0.002	0.885
CGPA : Second Year - Refused	422	0.031	-0.015	0.370	461	0.015	0.013	0.400
CGPA : Third Year - Refused	422	0.031	-0.015	0.370	461	0.015	0.013	0.400
B. Did Not Respond to Survey								
Nervous or anxious - Not responded	977	0.108	0.004	0.876	949	0.087	-0.002	0.931
Can't stop worrying - Not responded	977	0.108	0.003	0.885	944	0.088	-0.004	0.866
Feeling down - Not responded	976	0.108	0.004	0.854	948	0.088	-0.006	0.777
Little interest in activities - Not responded	977	0.108	0.004	0.874	941	0.088	-0.004	0.855
CGPA : First Year - Not responded	1024	0.071	0.011	0.543	964	0.059	0.012	0.518
CGPA : Second Year - Not responded	470	0.148	-0.040	0.225	478	0.034	0.031	0.181
CGPA : Third Year - Not responded	470	0.148	-0.040	0.225	478	0.034	0.031	0.181

Notes: The table presents attrition on the mental health (PHQ-4) and academic (CGPA) variables. Panel A reports the share who were asked but explicitly refused to answer; Panel B reports the share who did not respond to the survey at all and so never reached the item. Students from the 2023 cohort who haven't been sampled in intensive tracking i.e., those who didn't responsd in the first phase but also not sampled in tracking) were left out of the analysis, since their non-response reflects sampling, not refusal or survey attrition. The sample includes students from entry years 2023 and 2024 who had one or two roommates. Columns 2 and 6 report means for same-caste rooms for upper-caste and lower-caste students, respectively. Columns 3–4 and 7–8 report differences in means and corresponding p-values for upper-caste and lower-caste students. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates.Standard errors are clustered at the room level.

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Table A9: Attrition on Affirmative Action Attitudes

	Upper Caste				Lower Caste			
	N	Non-mixed mean	Diff	P-Val	N	Non-mixed mean	Diff	P-Val
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Refused to Answer								
Promote social justice - Refused	942	0.146	0.014	0.577	967	0.167	0.007	0.816
Foster diversity - Refused	942	0.146	0.017	0.511	967	0.176	0.016	0.590
Hurt reserved groups - Refused	942	0.156	0.005	0.845	967	0.170	-0.017	0.536
Reduce incentive to work (rev) - Refused	942	0.150	0.007	0.795	967	0.170	-0.008	0.776
Unfair to merit - Refused	906	0.156	0.008	0.778	941	0.175	0.004	0.881
Compromise academic quality (rev) - Refused	942	0.146	0.012	0.631	967	0.182	-0.017	0.549
Benefit to non-reserved groups - Refused	906	0.160	0.027	0.310	941	0.200	-0.004	0.895
B. Did Not Respond to Survey								
Promote social justice - Not responded	921	0.125	0.009	0.715	889	0.094	-0.008	0.744
Foster diversity - Not responded	918	0.125	0.008	0.751	875	0.095	-0.004	0.850
Hurt reserved groups - Not responded	919	0.126	0.007	0.791	899	0.094	-0.009	0.712
Reduce incentive to work (rev) - Not responded	923	0.125	0.007	0.772	892	0.094	-0.006	0.790
Unfair to merit - Not responded	921	0.170	0.001	0.981	885	0.143	-0.024	0.392
Compromise academic quality (rev) - Not responded	923	0.125	0.009	0.720	887	0.096	-0.008	0.723
Benefit to non-reserved groups - Not responded	907	0.170	0.005	0.860	867	0.147	-0.023	0.423

Notes: The table presents attrition on the affirmative-action attitude variables. Panel A reports the share who were asked but explicitly refused to answer; Panel B reports the share who did not respond to the survey at all and so never reached the item. Students from the 2023 cohort who haven't been sampled in intensive tracking i.e., those who didn't responsd in the first phase but also not sampled in tracking) were left out of the analysis, since their non-response reflects sampling, not refusal or survey attrition. The sample includes students from entry years 2023 and 2024 who had one or two roommates. Columns 2 and 6 report means for same-caste rooms for upper-caste and lower-caste students, respectively. Columns 3–4 and 7–8 report differences in means and corresponding p-values for upper-caste and lower-caste students. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates.Standard errors are clustered at the room level

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Table A10: Heterogeneous Effects on Main Outcomes by Gender

	Upper Caste				Lower Caste			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	CGPA : First Year	Mental health index	AA support: reserved group	AA support: non-reserved group	CGPA : First Year	Mental health index	AA support: reserved group	AA support: non-reserved group
Mixed Room	-0.108*	-0.051	0.168***	0.109	0.102*	-0.052	-0.037	0.084
	(0.056)	(0.053)	(0.060)	(0.070)	(0.061)	(0.047)	(0.051)	(0.064)
Female	-0.056	-0.215***	0.279***	0.276***	-0.103	-0.137	0.053	-0.016
	(0.103)	(0.071)	(0.080)	(0.089)	(0.110)	(0.112)	(0.085)	(0.109)
Mixed Room × Female	0.079	-0.026	-0.416***	-0.386***	0.028	-0.107	-0.041	0.053
	(0.120)	(0.096)	(0.114)	(0.132)	(0.133)	(0.129)	(0.108)	(0.132)
DV Mean, Non-mixed	0.474	0.027	-0.251	-0.265	-0.445	0.001	0.239	0.237
R-squared	0.459	0.112	0.134	0.150	0.409	0.107	0.081	0.090
N	947	935	820	817	896	928	829	820

Columns 1-4 restrict to upper-caste students, columns 5-8 to lower-caste students. Within each caste block, Column 1 reports cumulative GPA at the end of the first year; Column 2 the composite Mental Health Index; Column 3 the reserved-group Support for Affirmative Action sub-index; Column 4 the non-reserved-group sub-index – the same four outcomes as the main results table. Each regression interacts mixed-room assignment with a Female indicator to test whether the treatment effect varies by gender within each caste group. Regressions include specialization-by- cohort fixed effects and control for the number of roommates, but exclude hostel-by-cohort fixed effects, since hostel assignment is itself gendered and would absorb the Female main effect. Regressions are. Baseline controls for the respective outcome are included, with a missingness dummy where the baseline measure is missing. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A11: Effect on PHQ4 index and its components

	PHQ4 index (1)	Nervous or anxious (rev) (2)	Can't stop worrying (rev) (3)	Feeling down (rev) (4)	Little interest in activities (rev) (5)
Panel A: Upper Caste Sample					
Mixed Room	-0.093	-0.110	-0.036	-0.099	-0.143*
	(0.068)	(0.079)	(0.082)	(0.078)	(0.083)
DV Mean in same caste rm	0.016	-0.037	-0.009	0.065	0.061
R-squared	0.153	0.150	0.139	0.152	0.113
N	875	871	871	870	871
Sharpened q-value	0.255	0.255	0.662	0.255	0.255
Panel B: Lower Caste Sample					
Mixed Room	-0.084	-0.160**	-0.083	-0.095	-0.031
	(0.064)	(0.075)	(0.079)	(0.079)	(0.080)
DV Mean in same caste rm	-0.010	0.048	0.025	-0.054	-0.054
R-squared	0.125	0.126	0.140	0.113	0.113
N	877	863	858	862	855
Sharpened q-value	0.364	0.167	0.364	0.364	0.695
Panel C: Full Sample					
Mixed Room	-0.099**	-0.139**	-0.058	-0.108*	-0.111*
	(0.047)	(0.057)	(0.059)	(0.056)	(0.057)
DV Mean in same caste rm	0.003	0.006	0.008	0.005	0.003
R-squared	0.103	0.095	0.102	0.096	0.076
N	1752	1734	1729	1732	1726
Sharpened q-value	0.013	0.013	0.059	0.013	0.013
P val ($\beta_LC == \beta_UC$)	0.911	0.443	0.677	0.784	0.560

Notes: The table presents results for PHQ-4 index and its components, including anxiety and depression symptoms. Column 1 reports estimates for a composite PHQ-4 Index constructed using standardized scores of the individual PHQ-4 components. Column 2 reports results for feeling nervous, anxious, or on edge. Column 3 reports results for being unable to stop or control worrying. Column 4 reports results for feeling down, depressed, or hopeless. Column 5 reports results for little interest or pleasure in doing things. In all the columns symptoms are reverse coded so that higher values indicate better mental health outcomes. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste testing whether the lower- and upper-caste treatment effects differ. I use ANCOVA specification along with hostel-by-cohort and specialization-year fixed effects and a control for the number of roommates. Standard errors are clustered at the room level. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A12: Effect on Worry Index and its components

	Worry index (1)	Academics (rev) (2)	Getting an Internship (rev) (3)	Time Management (rev) (4)	Cultural Adaptation (rev) (5)	Staying Away from Home (rev) (6)	Discrimination (rev) (7)
Panel A: Upper Caste Sample							
Mixed Room	-0.024	-0.022	0.017	-0.122	0.021	-0.118	0.050
	(0.045)	(0.080)	(0.070)	(0.076)	(0.071)	(0.074)	(0.072)
DV Mean in same caste rm	0.052	0.043	0.014	0.023	0.142	0.079	0.026
R-squared	0.135	0.110	0.133	0.095	0.101	0.071	0.097
N	890	881	877	883	884	879	878
Sharpened q-value	0.804	0.804	0.804	0.392	0.804	0.392	0.804
Panel B: Lower Caste Sample							
Mixed Room	-0.048	-0.073	-0.034	-0.014	-0.085	0.025	-0.116
	(0.045)	(0.072)	(0.072)	(0.072)	(0.075)	(0.076)	(0.073)
DV Mean in same caste rm	-0.046	-0.038	-0.003	-0.011	-0.130	-0.077	-0.016
R-squared	0.124	0.129	0.102	0.102	0.080	0.101	0.071
N	892	882	873	884	881	880	871
Sharpened q-value	0.541	0.541	0.850	0.850	0.541	0.850	0.541
Panel C: Full Sample							
Mixed Room	-0.038	-0.035	-0.004	-0.057	-0.045	-0.059	-0.033
	(0.033)	(0.054)	(0.050)	(0.053)	(0.052)	(0.054)	(0.051)
DV Mean in same caste rm	0.003	0.003	0.005	0.006	0.005	0.001	0.005
R-squared	0.091	0.086	0.085	0.053	0.041	0.058	0.049
N	1782	1763	1750	1767	1765	1759	1749
Sharpened q-value	0.615	0.627	0.938	0.615	0.615	0.615	0.627
P val ($\beta_LC == \beta_UC$)	0.693	0.970	0.912	0.294	0.098	0.176	0.041

Notes: The table presents results for worry index and its components, capturing worries about various aspects of college life. Column 1 reports estimates for a composite of worry index about college aspects. Column 2 reports results for how worried about academic performance. Column 3 reports results for how worried about internship opportunities. Column 4 reports results for how worried about time management. Column 5 reports results for how worried about language and cultural adjustment. Column 6 reports results for how worried about being far away from home. Column 7 reports results for how worried about discrimination. I reverse code all the columns so that higher values of these indices indicate better mental health outcomes. All the components are measured on a scale of 1-5 and standardized with respect to same caste group sample. The final row of the full-sample panel (Panel C) reports the p-value on the interaction between mixed-room assignment and a lower-caste indicator from a pooled specification, testing whether the lower- and upper-caste treatment effects differ. All regressions control for the baseline PHQ-4 index and relative wellbeing index (missing values imputed as zero, with an indicator for missingness included) and for the number of roommates. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A13: Effect on Relative well-being index and its components

	Relative well being index (1)	More happy now (2)	Less control in life (rev) (3)	More stressed (rev) (4)	Better support system (5)
Panel A: Upper Caste Sample					
Mixed Room	-0.007	0.065	0.018	-0.090	-0.023
	(0.042)	(0.082)	(0.080)	(0.086)	(0.079)
DV Mean in same caste rm	-0.006	-0.080	0.010	0.075	-0.018
R-squared	0.076	0.093	0.067	0.112	0.109
N	839	828	832	829	825
Sharpened q-value	0.873	0.873	0.873	0.873	0.873
Panel B: Lower Caste Sample					
Mixed Room	0.005	-0.001	-0.027	0.015	0.072
	(0.042)	(0.073)	(0.078)	(0.074)	(0.074)
DV Mean in same caste rm	0.012	0.061	0.003	-0.050	0.005
R-squared	0.082	0.111	0.084	0.109	0.111
N	850	838	830	841	832
Sharpened q-value	0.994	0.994	0.994	0.994	0.994
Panel C: Full Sample					
Mixed Room	0.006	0.051	-0.017	-0.029	0.034
	(0.030)	(0.054)	(0.054)	(0.057)	(0.054)
DV Mean in same caste rm	0.003	-0.008	0.006	0.011	-0.006
R-squared	0.042	0.062	0.042	0.071	0.057
N	1689	1666	1662	1670	1657
Sharpened q-value	0.840	0.840	0.840	0.840	0.840
P val ($\beta_LC == \beta_UC$)	0.877	0.435	0.918	0.776	0.524

Notes: The table presents results for relative happiness and its components, capturing subjective well-being relative to the pre-college period. Column 1 reports estimates for a composite of relative happiness compared to pre-college period. Column 2 reports results for feeling more happy now. Column 3 reports results for feeling less in control. Column 4 reports results for feeling more stressed. Column 5 reports results for feeling better about social support. I reverse code columns 3 and 4 so that higher values of these indices indicate better mental health outcomes. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste testing whether the lower- and upper-caste treatment effects differ. I use ANCOVA specification along with hostel-by-cohort and specialization-year fixed effects and a control for the number of roommates. Standard errors are clustered at the room level. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A14: Effect on Support for Affirmative Action: Benefits of AA Policies to Reserved Groups

	AA support: reserved group (1)	Promote social justice (2)	Foster diversity (3)	Hurt reserved groups (rev) (4)	Reduce incentive to work (rev) (5)
Panel A: Upper Caste Sample					
Mixed Room	0.115**	0.130	0.110	0.109	0.045
	(0.057)	(0.085)	(0.092)	(0.085)	(0.082)
DV Mean in same caste rm	-0.251	-0.294	-0.148	-0.251	-0.303
R-squared	0.172	0.126	0.120	0.122	0.125
N	820	804	801	802	806
Sharpened q-value	0.218	0.291	0.291	0.291	0.587
Panel B: Lower Caste Sample					
Mixed Room	-0.022	-0.057	-0.064	-0.080	0.084
	(0.045)	(0.070)	(0.076)	(0.083)	(0.079)
DV Mean in same caste rm	0.239	0.283	0.149	0.251	0.275
R-squared	0.110	0.093	0.071	0.078	0.115
N	829	801	787	811	804
Sharpened q-value	0.618	0.528	0.528	0.528	0.528
Panel C: Full Sample					
Mixed Room	0.049	0.035	0.023	0.032	0.056
	(0.040)	(0.061)	(0.063)	(0.062)	(0.059)
DV Mean in same caste rm	-0.002	-0.003	0.001	0.003	-0.011
R-squared	0.138	0.088	0.065	0.063	0.090
N	1649	1605	1588	1613	1610
Sharpened q-value	0.749	0.749	0.749	0.749	0.749
P val ($\beta_LC == \beta_UC$)	0.086	0.249	0.257	0.078	0.929

Notes: The table presents results for attitudes towards Affirmative Action specific to the reserved groups. Column 1 reports estimates for an index of attitudes that are related to benefits and costs (reverse coded) of AA policies for groups that receive reservation. It is composed of components in columns 2-5. Column 2 reports results for attitudes on if AA is justified to balance historical disadvantage. Column 3 reports results for attitudes on if AA is justified to foster diversity. Column 4 reports results for attitudes on if AA hurts reserved students by placing them in challenging academic environments. Column 5 reports results for if AA policies reduce incentives for reserved students to work hard. Columns 4 and 5 are reverse coded, and higher estimated values indicate more support for the benefits to the reserved groups. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level. All the regressions include baseline controls for the respective dependent variables and a missing indicator for the baseline controls and number of roommates.

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Table A15: Effect on Support for Affirmative Action: Costs of AA Policies to Non-Reserved Groups

	AA support: non-reserved group (1)	Unfair to merit (rev) (2)	Compromise academic quality (rev) (3)	Benefit to non-reserved groups (4)
Panel A: Upper Caste Sample				
Mixed Room	0.048	-0.016	0.142*	-0.019
	(0.058)	(0.078)	(0.077)	(0.083)
DV Mean in same caste rm	-0.265	-0.295	-0.311	-0.178
R-squared	0.184	0.137	0.128	0.121
N	817	768	806	754
Sharpened q-value	0.822	0.838	0.262	0.838
Panel B: Lower Caste Sample				
Mixed Room	0.110**	0.114	0.047	0.008
	(0.056)	(0.072)	(0.076)	(0.072)
DV Mean in same caste rm	0.237	0.280	0.304	0.178
R-squared	0.131	0.146	0.108	0.084
N	820	771	799	753
Sharpened q-value	0.192	0.227	0.709	0.910
Panel C: Full Sample				
Mixed Room	0.100**	0.051	0.124**	0.010
	(0.047)	(0.055)	(0.060)	(0.056)
DV Mean in same caste rm	-0.012	-0.005	-0.003	-0.001
R-squared	0.135	0.126	0.088	0.072
N	1637	1539	1605	1507
Sharpened q-value	0.108	0.534	0.108	0.868
P val ($\beta_LC == \beta_UC$)	0.648	0.296	0.319	0.584

Notes: The table presents results for attitudes towards Affirmative Action specific to the non reserved groups. Column 1 reports estimates for an index of attitudes that are related to benefits (reverse coded) and costs of AA policies for groups that don't benefit from AA. It is composed of components in columns 2-4. Column 2 reports results for if AA policies are unfair compared to meritocratic policies. Column 3 reports results for attitudes on if AA policies compromise academic standards of the university. Column 4 reports results for whether non-reserved-group students believe they benefit from AA policies. Columns 2 and 3 are reverse coded, and higher estimated values indicate greater belief in higher costs of AA to the non-reserved groups. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level. All the regressions include baseline controls for the respective dependent variables and a missing indicator for the baseline controls and number of roommates.

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Table A16: Effect on GPA, Mental Health and Attitudes towards Affirmative Action: Demographic Controls

	CGPA : First Year (1)	Mental health index (2)	AA support: reserved group (3)	AA support: non-reserved group (4)
Panel A: Upper Caste Sample				
Mixed Room	-0.114**	-0.055	0.112**	0.044
	(0.049)	(0.046)	(0.056)	(0.057)
DV Mean in same caste rm	0.474	0.027	-0.251	-0.265
R-squared	0.496	0.157	0.179	0.201
N	947	935	820	817
Sharpened q-value	0.041	0.153	0.047	0.223
Panel B: Lower Caste Sample				
Mixed Room	0.091*	-0.084**	-0.006	0.108*
	(0.052)	(0.042)	(0.044)	(0.056)
DV Mean in same caste rm	-0.445	0.001	0.239	0.237
R-squared	0.466	0.154	0.126	0.136
N	896	928	829	820
Sharpened q-value	0.108	0.108	0.900	0.108
Panel C: Full Sample				
Mixed Room	-0.025	-0.069**	0.046	0.094**
	(0.037)	(0.032)	(0.040)	(0.046)
DV Mean in same caste rm	0.021	0.014	-0.002	-0.012
R-squared	0.528	0.111	0.154	0.158
N	1843	1863	1649	1637
Sharpened q-value	0.514	0.125	0.342	0.125
P val ($\beta_LC == \beta_UC$)	0.012	0.540	0.107	0.656

Notes: Robustness check on the main results table 2: identical specification, with pre-college family background added as controls to every regression – family income, mother's and father's education, and Class XII score (all mean-centered, with missingness dummies). Column 1 reports estimates for cumulative GPA at the end of the first year. Column 2 reports estimates for a composite Mental Health Index, constructed using standardized scores of the PHQ-4 index, and the worry index capturing academic- and social-life concerns in college. Negative values of the index indicate worse mental health outcomes. Column 3 reports an index of support for AA based on its perceived costs and benefits for reserved-group students. Column 4 reports an index of support for AA based on its perceived costs and benefits for non-reserved-group students. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste, testing whether the lower- and upper-caste treatment effects differ. All regressions include hostel-by-cohort and specialization-year fixed effects. I use ANCOVA specification along with controls for the number of roommates, family income, mother's and father's education, and Class XII score. Standard errors are clustered at the room level. Sharpened q-values (Anderson 2008) correcting for multiple hypothesis testing within each panel are reported. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A17: Effect on GPA, Mental Health and Attitudes towards Affirmative Action: Caste Fixed Effects

	CGPA : First Year (1)	Mental health index (2)	AA support: reserved group (3)	AA support: non-reserved group (4)
Panel A: Upper Caste Sample				
Mixed Room	-0.092*	-0.058	0.107*	0.036
	(0.050)	(0.045)	(0.057)	(0.058)
DV Mean in same caste rm	0.474	0.027	-0.251	-0.265
R-squared	0.482	0.151	0.178	0.194
N	947	935	820	817
Sharpened q-value	0.132	0.267	0.132	0.544
Panel B: Lower Caste Sample				
Mixed Room	0.083	-0.083**	-0.019	0.113**
	(0.052)	(0.042)	(0.044)	(0.055)
DV Mean in same caste rm	-0.445	0.001	0.239	0.237
R-squared	0.436	0.146	0.137	0.176
N	896	928	829	820
Sharpened q-value	0.074	0.048	0.333	0.048
Panel C: Full Sample				
Mixed Room	-0.019	-0.070**	0.043	0.092**
	(0.038)	(0.032)	(0.037)	(0.045)
DV Mean in same caste rm	0.021	0.014	-0.002	-0.012
R-squared	0.518	0.111	0.218	0.219
N	1843	1863	1649	1637
Sharpened q-value	0.638	0.139	0.405	0.158
P val ($\beta_LC == \beta_UC$)	0.097	0.461	0.114	0.510

Notes: Robustness check on the main results table 2: identical specification, with caste added as an additional fixed effect alongside hostel-by-cohort and specialization-by-cohort, on top of the existing lower-/upper-caste panel split. Column 1 reports estimates for cumulative GPA at the end of the first year. Column 2 reports estimates for a composite Mental Health Index, constructed using standardized scores of the PHQ-4 index, and the worry index capturing academic- and social-life concerns in college. Negative values of the index indicate worse mental health outcomes. Column 3 reports an index of support for AA based on its perceived costs and benefits for reserved-group students. Column 4 reports an index of support for AA based on its perceived costs and benefits for non-reserved-group students. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste, testing whether the lower- and upper-caste treatment effects differ. All regressions include hostel-by-cohort, specialization-year, and caste category fixed effects. I use ANCOVA specification along with a control for the number of roommates. Standard errors are clustered at the room level. Sharpened q-values (Anderson 2008) correcting for multiple hypothesis testing within each panel are reported. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A18: Effect on Help Received from Roommate

	Upper Caste				Lower Caste			
	(1) Got Help (Composite Ind)	(2) Got exam prep	(3) Got Internship Help	(4) Got Emotional /Financial Help	(5) Got Help (Composite Ind)	(6) Got exam prep	(7) Got Internship Help	(8) Got Emotional /Financial Help
Mixed Room	-0.102	-0.170*	-0.120	-0.040	-0.028	0.120	-0.009	-0.130*
	(0.068)	(0.090)	(0.079)	(0.075)	(0.063)	(0.075)	(0.078)	(0.072)
DV Mean, Non-mixed	-0.065	-0.040	-0.038	-0.113	0.085	0.048	0.052	0.103
R-squared	0.100	0.097	0.099	0.092	0.084	0.081	0.074	0.089
N	886	880	875	878	895	887	864	882

Notes: This table reports the effects of mixed-room assignment on the types of help received from roommates. Columns 1–4 report results for upper-caste students, and Columns 5–8 report results for lower-caste students. Within each caste group, Columns 1 and 5 report effects on a composite index of all types of help received from roommates; Columns 2 and 6 report effects on academic help received; Columns 3 and 7 report effects on internship-related help received; and Columns 4 and 8 report effects on emotional and financial help received. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A19: Effect on Help Received from Other Hostel Mates

	Upper Caste				Lower Caste			
	(1) Got Help (Composite Index)	(2) Got Academic Help	(3) Got Internship Help	(4) Got Emotional /Financial Help	(5) Got Help (Composite Index)	(6) Got Academic Help	(7) Got Internship Help	(8) Got Emotional /Financial Help
Mixed Room	-0.045	-0.076	-0.060	-0.003	0.051	0.072	0.046	0.052
	(0.058)	(0.071)	(0.073)	(0.071)	(0.057)	(0.072)	(0.068)	(0.071)
DV Mean, Non-mixed	0.047	0.083	0.047	0.017	-0.054	-0.091	-0.043	-0.032
R-squared	0.093	0.090	0.096	0.088	0.081	0.082	0.087	0.066
N	846	844	836	837	844	839	821	832

Notes: This table reports the effects of mixed-room assignment on the types of help received from other hostel mates (OHM). Columns 1–4 report results for upper-caste students, and Columns 5–8 report results for lower-caste students. Within each caste group, Columns 1 and 5 report effects on a composite index of all types of help received from OHM; Columns 2 and 6 report effects on academic help received; Columns 3 and 7 report effects on internship-related help received; and Columns 4 and 8 report effects on emotional and financial help received. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A20: Heterogeneous Effects on Academic Outcomes by Sequence Match with Roommate

	Upper Caste				Lower Caste			
	(1) CGPA : First Year	(2) Group work	(3) Study Interactions	(4) Effort in academics	(5) CGPA : First Year	(6) Group work	(7) Study Interactions	(8) Effort in academics
Mixed Room	-0.078	0.166*	-0.002	-0.049	0.133	0.123	0.044	-0.027
	(0.082)	(0.092)	(0.082)	(0.085)	(0.090)	(0.095)	(0.091)	(0.102)
Same Course Sequence as Roommate	0.094	0.456***	0.310***	0.038	0.047	0.385***	0.297***	0.010
	(0.081)	(0.126)	(0.108)	(0.097)	(0.095)	(0.126)	(0.106)	(0.115)
Mixed Room × Same Course seq	-0.060	-0.243	-0.059	0.188	-0.069	-0.132	-0.132	-0.013
	(0.113)	(0.158)	(0.133)	(0.129)	(0.124)	(0.154)	(0.130)	(0.141)
DV Mean, Non-mixed	0.474	-0.002	-0.080	-0.051	-0.445	-0.003	0.047	0.057
R-squared	0.475	0.186	0.187	0.099	0.432	0.154	0.128	0.109
N	939	901	901	975	895	937	937	985

Notes: This table reports the effects of academic overlap on GPA, group work, and academic effort and study interactions. I look at heterogeneity by academic overlap with roommate on these outcomes. Columns 1–4 report results for upper-caste students, and Columns 5–8 report results for lower-caste students. Within each caste group, Columns 1 and 5 report effects on GPA; Columns 2 and 6 report effects on group work, shared assignments, and lab work; Columns 3 and 7 report effects on study interactions; and Columns 4 and 8 report effects on self-reported overall academic effort. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A21: Effect on Academic Engagement with Roommates

	Upper Caste			Lower Caste		
	(1) Group work	(2) Study Interactions	(3) Effort in academics	(4) Group work	(5) Study Interactions	(6) Effort in academics
Mixed Room	0.057	-0.020	0.040	0.030	-0.042	-0.031
	(0.073)	(0.060)	(0.060)	(0.070)	(0.058)	(0.060)
DV Mean, Non-mixed	-0.002	-0.080	-0.051	-0.003	0.047	0.057
R-squared	0.173	0.174	0.087	0.141	0.116	0.110
N	906	906	983	938	938	986

Notes: This table reports the effects of academic overlap on GPA, group work, and academic effort and study interactions. Columns 1–3 report results for upper-caste students, and Columns 4–6 report results for lower-caste students. Columns 1 and 4 report effects on group work, shared assignments, and lab work; Columns 2 and 5 report effects on study interactions; and Columns 3 and 6 report effects on self-reported overall academic effort. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A22: Early exposure results on Mental Health and Attitudes towards Affirmative Action and Skills by Components

	Upper Caste				Lower Caste			
	N	Non-mixed mean	Diff	P-Val	N	Non-mixed mean	Diff	P-Val
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. PHQ4: Anxiety and Depression Symptoms at Baseline (Reverse coded)								
Nervous or anxious (rev)	453	-0.017	0.062	0.623	434	0.050	0.004	0.974
Can't stop worrying (rev)	450	0.047	-0.019	0.873	429	-0.032	0.026	0.821
Feeling down (rev)	453	0.026	-0.025	0.828	431	-0.017	0.085	0.458
Little interest in activities (rev)	448	-0.003	0.135	0.210	423	-0.018	-0.003	0.979
B. Relative Wellbeing components at Baseline								
More happy now	448	-0.004	0.038	0.759	422	0.021	-0.132	0.232
Less control in life (rev)	443	0.079	0.018	0.880	415	-0.087	0.213	0.084
More stressed (rev)	447	-0.076	0.159	0.166	423	0.052	0.034	0.747
Better support system	446	0.148	0.059	0.604	420	-0.153	0.038	0.735
C. Worry Index Components at Baseline (Reverse coded)								
Academics (rev)	502	-0.113	0.178	0.146	490	0.150	-0.155	0.133
Getting an Internship (rev)	502	-0.042	0.093	0.394	490	0.048	-0.110	0.335
Cultural Adaptation (rev)	501	0.269	-0.147	0.107	490	-0.309	-0.123	0.342
Staying Away from Home (rev)	502	-0.004	-0.009	0.925	490	-0.034	0.048	0.666
Discrimination (rev)	501	0.072	0.065	0.462	490	-0.088	-0.261	0.047
Time Management (rev)	502	-0.122	0.123	0.215	490	0.142	-0.036	0.727
D. Attitudes towards AA at Baseline								
Promote social justice	446	-0.383	-0.117	0.346	433	0.397	0.030	0.772
Foster diversity	445	-0.170	-0.178	0.219	423	0.180	0.154	0.131
Hurt reserved groups (rev)	444	-0.233	-0.054	0.626	429	0.236	0.071	0.531
Reduce incentive to work (rev)	451	-0.394	-0.018	0.857	434	0.396	-0.037	0.752
Benefit to non-reserved groups	441	-0.259	-0.047	0.715	418	0.268	0.039	0.705
Compromise academic quality (rev)	451	-0.392	0.128	0.208	429	0.444	0.093	0.430
Unfair to merit (rev)	447	-0.446	-0.022	0.824	432	0.462	-0.033	0.762
E. Self-reported Skills at Baseline and Friendship with Roommate								
Rating of math & prog	490	0.286	-0.117	0.117	484	-0.334	0.120	0.142
Rating: English	490	0.336	-0.064	0.493	484	-0.400	0.127	0.305
Rating of soft skills	490	0.242	-0.108	0.131	484	-0.295	0.111	0.194
Expected Friendship with roommate	473	2.372	-0.197	0.039	452	2.242	0.004	0.964
Joint Test p-value				0.030				0.734

Notes: The table presents balance on pre-college networks and self-reported skills at baseline. Panel A reports early exposure effects on PHQ4 index components. Panels B and C report effects on components of relative happiness and worry indices. Panel E reports effects on attitudes towards Affirmative Action: Social Justice and Meritocratic concerns indices. Panel F reports effects on self-reported skills both soft skills and hard skills and expected friendship with roommate. The sample includes students only from entry year 2024 who had one or two roommates. Columns 2 and 6 report means for same-caste rooms for upper-caste and lower-caste students, respectively. Columns 3–4 and 7–8 report differences in means and corresponding p-values for upper-caste and lower-caste students. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level. The last row of each panel reports the p-value from the joint F-test of orthogonality.

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Table A23: Effect on Help Provided to Roommate

	Upper Caste				Lower Caste			
	(1) Provided Help (Composite Ind)	(2) Provided exam prep	(3) Provided Internship Help	(4) Provided Emotional /Financial Help	(5) Provided Help (Composite Ind)	(6) Provided exam prep	(7) Provided Internship Help	(8) Provided Emotional /Financial Help
Mixed Room	0.093	0.144*	0.065	0.106	-0.115**	-0.142**	-0.054	-0.167**
	(0.064)	(0.082)	(0.074)	(0.073)	(0.058)	(0.071)	(0.070)	(0.069)
DV Mean, Non-mixed	-0.113	-0.081	-0.094	-0.154	0.116	0.089	0.105	0.150
R-squared	0.115	0.092	0.116	0.107	0.089	0.078	0.085	0.093
N	838	835	829	836	847	841	828	834

Notes: This table reports the effects of mixed-room assignment on the types of help provided to roommates. Columns 1–4 report results for upper-caste students, and Columns 5–8 report results for lower-caste students. Within each caste group, Columns 1 and 5 report effects on a composite index of all types of help provided to roommates; Columns 2 and 6 report effects on academic help provided; Columns 3 and 7 report effects on internship-related help provided; and Columns 4 and 8 report effects on emotional and financial help provided. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A24: Effect on Student Club and Sports Active Participation

	Upper Caste			Lower Caste		
	(1) Participation in clubs and sports	(2) Participation in student clubs	(3) Participation in sports and volunteering clubs	(4) Participation in clubs and sports	(5) Participation in student clubs	(6) Participation in sports and volunteering clubs
Mixed Room	0.019	0.003	0.050	0.017	0.066	-0.054
	(0.047)	(0.057)	(0.053)	(0.047)	(0.053)	(0.059)
DV Mean, Non-mixed	0.034	0.105	-0.077	-0.033	-0.103	0.065
R-squared	0.098	0.086	0.098	0.090	0.093	0.075
N	893	893	893	885	882	885

Columns 1-3 restrict to upper-caste, columns 4-6 to lower-caste students. Column 1 reports the composite Participation in Clubs and Sports index. Column 2 reports participation in student clubs, combining cultural, co-curricular, and hostel activities. Column 3 reports participation in sports and volunteering clubs. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects, control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses.***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A25: Mental Health, Controlling for Roommate Interaction

	Upper Caste			Lower Caste		
	(1) Mental health index	(2) Mental health index	(3) Mental health index	(4) Mental health index	(5) Mental health index	(6) Mental health index
Mixed Room	-0.062	-0.034	-0.034	-0.080*	-0.024	-0.026
	(0.045)	(0.035)	(0.035)	(0.041)	(0.032)	(0.032)
Social Interactions		0.019	0.031		0.067***	0.046**
		(0.017)	(0.020)		(0.019)	(0.023)
Study Interactions			-0.023			0.039
			(0.026)			(0.025)
DV Mean, Non-mixed	0.027	0.027	0.027	0.001	0.001	0.001
R-squared	0.151	0.158	0.159	0.152	0.148	0.151
N	935	856	856	928	872	872

Columns 1-3 restrict to upper-caste students, columns 4-6 to lower-caste students. Column 1 (4) reports the total effect of mixed-room assignment on Mental Health on baseline specification. Column 2 (5) adds Social Interaction; column 3 (6) adds Study-Related Interaction along with as well. This is suggestive of a mediating channel but not a clean decomposition. All regressions control for the baseline PHQ4 index and relative well-being index. I add a include hostel-by-cohort and specialization-by-cohort fixed effects, control for the number of roommates, Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A26: Effect on GPA, Mental Health and Attitudes towards Affirmative Action: Two-Person Rooms Only

	CGPA : First Year (1)	Mental health index (2)	AA support: reserved group (3)	AA support: non-reserved group (4)
Panel A: Upper Caste Sample				
Mixed Room	-0.160*	-0.152**	-0.008	0.005
	(0.086)	(0.063)	(0.077)	(0.086)
DV Mean in same caste rm	0.437	0.049	-0.250	-0.236
R-squared	0.501	0.249	0.216	0.266
N	371	358	304	303
Sharpened q-value	0.096	0.052	0.712	0.712
Panel B: Lower Caste Sample				
Mixed Room	0.077	-0.094	0.027	0.116
	(0.085)	(0.066)	(0.076)	(0.093)
DV Mean in same caste rm	-0.457	-0.015	0.228	0.249
R-squared	0.488	0.168	0.162	0.112
N	350	348	316	312
Sharpened q-value	0.487	0.423	0.725	0.423
Panel C: Full Sample				
Mixed Room	-0.029	-0.112**	-0.005	0.051
	(0.062)	(0.047)	(0.059)	(0.067)
DV Mean in same caste rm	0.010	0.019	-0.011	0.006
R-squared	0.528	0.159	0.159	0.151
N	732	723	627	622
Sharpened q-value	0.671	0.034	0.699	0.671
P val ($\beta_LC == \beta_UC$)	0.069	0.397	0.917	0.222

Notes: Robustness check on the main results table 2: identical specification, restricted to two-person rooms only. Column 1 reports estimates for cumulative GPA at the end of the first year. Column 2 reports estimates for a composite Mental Health Index, constructed using standardized scores of the PHQ-4 index, and the worry index capturing academic- and social-life concerns in college. Negative values of the index indicate worse mental health outcomes. Column 3 reports an index of support for AA based on its perceived costs and benefits for reserved-group students. Column 4 reports an index of support for AA based on its perceived costs and benefits for non-reserved-group students. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste, testing whether the lower- and upper-caste treatment effects differ. Sample restricted to two-person rooms only; the number-of-roommates control is constant by construction under this restriction and drops out of the fixed effects. All regressions include hostel-by-cohort and specialization-year fixed effects. I use ANCOVA specification. Standard errors are clustered at the room level. Sharpened q-values (Anderson 2008) correcting for multiple hypothesis testing within each panel are reported. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A27: Effect on GPA, Mental Health and Attitudes towards Affirmative Action: Partial vs. Full Mixed-Room Assignment, Three-Person Rooms Only

	CGPA : First Year	Mental health index	AA support: reserved group	AA support: non-reserved group
	(1)	(2)	(3)	(4)
Panel A: Upper Caste Sample				
Partial Mixed Room	-0.071	0.027	0.194**	0.094
	(0.068)	(0.070)	(0.085)	(0.097)
Full Mixed Room	-0.071	-0.018	0.198**	0.057
	(0.082)	(0.074)	(0.095)	(0.107)
P-val (Partial = Full)	0.996	0.437	0.960	0.716
DV Mean in same caste rm	0.526	-0.003	-0.251	-0.301
R-squared	0.494	0.183	0.206	0.198
N	569	567	509	507
Panel B: Lower Caste Sample				
Partial Mixed Room	0.078	-0.098*	-0.016	0.124*
	(0.078)	(0.057)	(0.055)	(0.069)
Full Mixed Room	0.070	-0.113*	-0.059	0.030
	(0.090)	(0.063)	(0.069)	(0.087)
P-val (Partial = Full)	0.922	0.777	0.435	0.183
DV Mean in same caste rm	-0.431	0.019	0.253	0.223
R-squared	0.429	0.207	0.149	0.229
N	539	570	505	501

Notes: Robustness check on the main results table's partial-vs-full mixed-room specification: identical specification, restricted to three-person rooms only – the only room size where partial mixing (one of two roommates from the other caste group) and full mixing (both roommates) are separately identified; in two-person rooms every mixed room is fully mixed by construction. Column 1 reports estimates for cumulative GPA at the end of the first year. Column 2 reports estimates for a composite Mental Health Index, constructed using standardized scores of the PHQ-4 index, and the worry index capturing academic- and social-life concerns in college. Negative values of the index indicate worse mental health outcomes. Column 3 reports an index of support for AA based on its perceived costs and benefits for reserved-group students. Column 4 reports an index of support for AA based on its perceived costs and benefits for non-reserved-group students. The reported p-value in each column tests equality of the partial- and full-mixing coefficients. Sample restricted to three-person rooms only; the number-of-roommates control is constant by construction under this restriction and drops out of the fixed effects. All regressions include hostel-by-cohort and specialization-year fixed effects. I use ANCOVA specification. Standard errors are clustered at the room level. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A28: Effect on GPA, Internships, Mental Health and Attitudes towards Affirmative Action

	CGPA : First Year (1)	Mental health index (2)	AA support: reserved group (3)	AA support: non-reserved group (4)
Panel A: Upper Caste Sample				
Caste distance with roommates	-0.090**	-0.036	0.059	0.039
	(0.040)	(0.028)	(0.037)	(0.041)
DV Mean in same caste rm	0.480	0.030	-0.256	-0.273
R-squared	0.475	0.104	0.127	0.183
N	945	933	818	815
Sharpened q-value	0.106	0.259	0.221	0.336
Panel B: Lower Caste Sample				
Caste distance with roommates	0.019	-0.086**	0.047	0.120***
	(0.049)	(0.036)	(0.036)	(0.040)
DV Mean in same caste rm	-0.453	0.020	0.251	0.246
R-squared	0.431	0.116	0.068	0.137
N	896	928	829	820
Sharpened q-value	0.351	0.018	0.130	0.005
Panel C: Full Sample				
Caste distance with roommates	-0.045	-0.046**	0.123***	0.137***
	(0.033)	(0.023)	(0.028)	(0.032)
DV Mean in same caste rm	0.233	0.027	-0.123	-0.139
R-squared	0.507	0.071	0.067	0.150
N	1841	1861	1647	1635
Sharpened q-value	0.052	0.015	0.000	0.000
P val ($\beta_LC == \beta_UC$)	0.777	0.699	0.829	0.323

Notes: The table presents results for academic performance, mental health, and attitudes toward affirmative action. Caste distance is defined as an average absolute caste distance a student has with their roommates. Column 1 reports estimates for cumulative GPA at the end of the first year. Column 2 reports estimates for a composite Mental Health Index, constructed using standardized scores of the PHQ-4 index, and the worry index capturing academic- and social-life concerns in college. Negative values of the index indicate worse mental health outcomes. Column 3 reports an index of support for AA based on its perceived costs and benefits for reserved-group students. Column 4 reports an index of support for AA based on its perceived costs and benefits for non-reserved-group students. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste, testing whether the lower- and upper-caste treatment effects differ. All regressions include hostel-by-cohort and specialization-year fixed effects. I use ANCOVA specification along with a control for the number of roommates. Standard errors are clustered at the room level. Sharpened q-values (Anderson 2008) correcting for multiple hypothesis testing within each panel are reported. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A29: Room Composition Transitions: Year 1 to Year 2

	Upper Caste				Lower Caste			
Year 1 Room Type	Non-Mixed	Partial-Mixed	Full-Mixed	N	Non-Mixed	Partial-Mixed	Full-Mixed	N
A. 2-Person Rooms (Year 1)								
Non-Mixed	78.8	2.1	19.1	236	81.7	3.0	15.3	202
Full-Mixed	32.3	5.8	61.9	223	30.4	6.2	63.4	224
B. 3-Person Rooms (Year 1)								
Non-Mixed	75.8	12.4	11.8	161	63.7	22.9	13.4	179
Partial-Mixed	34.9	42.7	22.5	347	36.9	45.0	18.1	331
Full-Mixed	28.5	23.6	47.9	165	27.4	21.7	50.9	175

Notes: Sample restricted to the 2023 and 2024 admission cohorts, 2- and 3-person rooms in Year 1 (single rooms and 4-person rooms excluded). Cell entries are row percentages among students with an observed Year 2 room: among students in a given Year 1 room type and caste group who have a Year 2 room on record, the percentage who are in each Year 2 composition category. Students with no observed Year 2 room (1.0% of this sample, e.g. due to attrition) are excluded from both the percentages and the N column. *Partial-Mixed* is achievable only in 3-person rooms by construction (a 2-person room's caste mix can only be 0, 50, or 100 percent upper-caste, and 50 percent falls outside the partial-mixed range), so Panel A has only two Year 1 room types. Composition state is defined from each student's own caste perspective.

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Table A30: Effect on Experiences of Discrimination and Exclusion: Three Years after Admission

	Upper Caste			Lower Caste		
	(1) Experience of Discrimination (SD)	(2) Exclusion in Study Groups (SD)	(3) Exclusion in Social Events (SD)	(4) Experience of Discrimination (SD)	(5) Exclusion in Study Groups (SD)	(6) Exclusion in Social Events (SD)
Mixed Room	0.085	0.059	-0.003	0.344**	0.016	0.047
	(0.099)	(0.083)	(0.096)	(0.137)	(0.103)	(0.099)
DV Mean, Non-mixed	-0.077	-0.124	-0.030	0.068	0.110	0.026
R-squared	0.029	0.034	0.041	0.035	0.023	0.013
N	431	431	431	450	450	450

Columns 1-3 restrict to upper-caste students, columns 4-6 to lower-caste students. Column 1 reports effects on experience of discrimination. Column 2 reports effects on exclusion in study groups. Column 3 reports effects on exclusion in social events. Higher values indicate more frequently reported negative experiences. No additional controls are included. Sample is restricted to the 2023 cohort's three-year follow-up. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A31: Effect on Experiences of Discrimination and Exclusion

	Upper Caste			Lower Caste		
	(1) Experience of Discrimination (SD)	(2) Exclusion in Study Groups (SD)	(3) Exclusion in Social Events (SD)	(4) Experience of Discrimination (SD)	(5) Exclusion in Study Groups (SD)	(6) Exclusion in Social Events (SD)
Mixed Room	0.010	0.117*	0.077	0.081	-0.024	0.032
	(0.074)	(0.067)	(0.072)	(0.073)	(0.076)	(0.069)
DV Mean, Non-mixed	-0.005	-0.086	0.006	0.005	0.086	-0.006
R-squared	0.054	0.059	0.055	0.074	0.064	0.063
N	966	966	966	978	978	978

Columns 1-3 restrict to upper-caste students, columns 4-6 to lower-caste students. Column 1 reports effects on experience of discrimination. Column 2 reports effects on exclusion in study groups. Column 3 reports effects on exclusion in social events. Higher values indicate more frequently reported negative experiences. No additional controls are included. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects, control for the number of roommates, and are weighted using survey weights correcting for the two-tier sampling design. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A32: Effect on GPA, Internships: 2 Years after Admission

	CGPA : Second Year (1)	Internship: 2nd Summer (2)	Salary: 2nd year activity (in 1000 INR) (3)	Plan to do internship: 3rd year (4)
Panel A: Upper Caste Sample				
Mixed Room	-0.119*	-0.070	-10.287	-0.022
	(0.070)	(0.045)	(6.403)	(0.027)
DV Mean in same caste rm	0.539	0.584	37.809	0.910
R-squared	0.367	0.077	0.104	0.172
N	471	455	458	468
Sharpened q-value	0.156	0.156	0.156	0.417
Panel B: Lower Caste Sample				
Mixed Room	0.119*	0.036	5.985*	-0.004
	(0.072)	(0.044)	(3.542)	(0.045)
DV Mean in same caste rm	-0.478	0.303	9.938	0.678
R-squared	0.266	0.085	0.077	0.122
N	500	462	468	476
Sharpened q-value	0.196	0.556	0.196	0.933
Panel C: Full Sample				
Mixed Room	0.003	-0.019	-3.465	-0.001
	(0.051)	(0.031)	(3.712)	(0.030)
DV Mean in same caste rm	-0.001	0.436	23.062	0.791
R-squared	0.415	0.084	0.059	0.094
N	971	917	926	944
Sharpened q-value	0.971	0.971	0.971	0.971
P val ($\beta_LC == \beta_UC$)	0.037	0.071	0.017	0.605

Notes: The table presents effects of mixed-room assignment on medium-run academic, labor-market, 2 years after entering college. Column 1 reports estimates for cumulative GPA at the end of the 2nd year, Column 2 reports summer internship completion at the end of 2st year summer, the Column 3 reports results for internship stipend. Column 4 reports results for plan to do 3rd-year summer internship, this includes students who both received offers and who intend to do a summer internship. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste, testing whether the lower- and upper-caste treatment effects differ. All regressions include hostel-by-cohort and specialization-year fixed effects. I use ANCOVA specification along with a control for the number of roommates. Standard errors are clustered at the room level. Sharpened q-values (Anderson 2008) correcting for multiple hypothesis testing within each panel are reported. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

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Table A33: Heterogeneous Effects on Affirmative Action Attitudes by Own Entrance Exam Score

	Upper Caste		Lower Caste	
	(1) AA support: reserved group	(2) AA support: non-reserved group	(3) AA support: reserved group	(4) AA support: non-reserved group
Mixed Room	0.037	-0.063	0.036	0.172***
	(0.060)	(0.063)	(0.052)	(0.067)
Own Entrance Score (SD)	-0.235***	-0.315***	-0.276***	-0.360***
	(0.068)	(0.085)	(0.049)	(0.063)
Mixed Room × Own Entrance Score	0.118*	0.185**	0.142***	0.147**
	(0.068)	(0.079)	(0.054)	(0.068)
DV Mean, Non-mixed	-0.251	-0.265	0.239	0.237
R-squared	0.184	0.196	0.150	0.184
N	803	800	814	806

Columns 1-2 restrict to upper-caste students, columns 3-4 to lower-caste students. Column 1 (3) the reserved-group index indicating support for AA based on its perceived costs and benefits for reserved-group students. Column 2 (4) the non-reserved-group index indicating support for AA based on its perceived costs and benefits for non-reserved-group students. Each regression interacts mixed-room assignment with the student's own standardized entrance exam score, testing whether the attitude effect varies with own academic preparation. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects, control for the number of roommates. Baseline controls for the respective outcome are included, with a missingness dummy where the baseline measure is missing. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

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Table A34: Summary statistics: Mental Health and Attitudes on Affirmative Action at Baseline

	Upper Caste			Lower Caste			Upper Caste - Lower Caste	
	N	Mean	Sd	N	Mean	Sd	Difference	P-Val
Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. PHQ4: Anxiety and Depression Symptoms at Baseline								
Nervous or anxious (rev)	453	0.002	1.005	434	0.024	0.925	-0.023	0.726
Can't stop worrying (rev)	450	0.012	1.009	429	-0.058	1.021	0.070	0.308
Feeling down (rev)	453	-0.036	1.019	431	-0.022	0.947	-0.014	0.832
Little interest in activities (rev)	448	0.080	1.018	423	-0.040	0.993	0.119	0.080
B. Relative unhappiness components at Baseline								
More happy now	448	0.026	1.028	422	-0.067	0.974	0.093	0.173
Less control in life (rev)	443	0.041	1.035	415	0.013	0.943	0.029	0.671
More stressed (rev)	447	0.015	1.067	423	0.005	0.932	0.010	0.881
Better support system	446	0.186	1.107	420	-0.151	0.999	0.337	0.000
C. Worry Index Components at Baseline								
Academics (rev)	502	-0.076	1.027	490	0.058	1.033	-0.134	0.041
Getting an Internship (rev)	502	-0.036	1.023	490	0.001	1.001	-0.037	0.569
Cultural Adaptation (rev)	501	0.183	0.864	490	-0.383	1.217	0.566	0.000
Staying Away from Home (rev)	502	-0.029	0.998	490	0.044	0.957	-0.074	0.236
Discrimination (rev)	501	0.085	0.860	490	-0.230	1.135	0.316	0.000
Time Management (rev)	502	-0.057	1.010	490	0.124	1.005	-0.180	0.005
D. Costs and Benefits of Affirmative Action to the Reserved Groups at Baseline								
Promote social justice	446	-0.414	0.940	433	0.445	0.858	-0.859	0.000
Foster diversity	445	-0.218	1.112	423	0.302	0.928	-0.519	0.000
Hurt reserved groups (rev)	444	-0.250	1.013	429	0.308	0.973	-0.558	0.000
Reduce incentive to work (rev)	451	-0.388	0.883	434	0.380	1.067	-0.768	0.000
E. Costs and Benefits of AA to the Non-Reserved Groups at Baseline								
Benefit to non-reserved groups	441	-0.288	1.051	418	0.367	0.809	-0.655	0.000
Compromise academic quality (rev)	451	-0.318	0.904	429	0.533	0.972	-0.850	0.000
Unfair to merit (rev)	447	-0.430	0.854	432	0.509	0.967	-0.938	0.000

Notes: The table presents summary statistics for socioeconomic indicators. The sample includes students from entry year of 2024 who had one or two roommates. I report the sample size, mean, and standard deviation separately for upper-caste and lower-caste students. PHQ4 is a standard measure of anxiety and depression symptoms, with higher values indicating worse mental health. Relative Unhappiness Index measures students' happiness relative to their pre-college happiness levels, with higher values indicating worse happiness. Worry Index measures the extent to which students are worried about college life, with higher values indicating more worry. Social Justice and Diversity benefits of Affirmative Action index measures students' attitudes toward the social justice aspects of affirmative action policies, with higher values indicating more support. Meritocratic concerns of Affirmative Action index measures students' attitudes toward whether affirmative action policies undermine meritocracy, with higher values indicating more concern/less supported. Students responded to a series of statements using a five-point Likert scale ranging from strongly disagree to strongly agree, and I standardized the responses to create z-scores for each variable with respect to full sample.

Table A35: Balance table on Attrition, Socio Economic Indicators and College Preparation: Partial and Full Mixed Specification

	Upper Caste						Lower Caste					
	N	Non mixed mean	Partial Mixed diff	P-Val	Fully Mixed diff	P-Val	N	Non mixed mean	Partial Mixed diff	P-Val	Fully Mixed diff	P-Val
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
A. Survey attrition indicators												
Response	1145	0.838	0.005	0.883	-0.010	0.739	1121	0.891	-0.005	0.863	-0.022	0.422
Survey completed	1145	0.800	-0.017	0.630	-0.001	0.977	1121	0.872	0.015	0.660	-0.001	0.967
B. Socio Economic Indicators												
Mother's education (bachelors or higher)	902	0.876	0.006	0.876	-0.066	0.044	881	0.566	-0.030	0.549	0.005	0.913
Father's education (bachelors or higher)	896	0.786	0.023	0.549	-0.046	0.181	876	0.641	-0.053	0.254	-0.030	0.481
Assets Index	824	0.166	0.016	0.789	-0.033	0.502	809	-0.171	-0.043	0.500	0.013	0.817
Taken a loan	885	0.148	-0.019	0.573	0.019	0.529	890	0.194	0.036	0.368	0.033	0.306
Family income (9L or above)	859	0.699	-0.074	0.150	-0.081	0.059	895	0.425	-0.058	0.226	-0.027	0.513
Speaks English with family	917	0.494	0.024	0.633	-0.053	0.193	938	0.246	0.069	0.086	0.036	0.329
Speaks English with friends	917	0.705	0.039	0.363	-0.024	0.535	938	0.558	0.037	0.416	0.021	0.601
Knows someone who did engineering	912	0.682	-0.007	0.884	-0.088	0.029	907	0.490	-0.018	0.681	0.003	0.932
Knows someone studied in IITs	912	0.259	0.003	0.938	-0.008	0.809	907	0.168	0.021	0.554	0.008	0.807
C. JEE Preparation and Academic Background												
Score in Class XII	942	93.089	1.356	0.048	0.796	0.285	934	90.175	-0.150	0.774	0.536	0.346
Score in Class X	939	94.995	0.943	0.175	0.157	0.844	935	92.942	0.183	0.782	-0.724	0.352
JEE Advanced score	1112	204.125	0.034	0.984	-1.927	0.208	1091	139.368	4.373	0.106	-0.445	0.849
Years: JEE preparation	964	2.520	-0.133	0.329	-0.129	0.270	964	2.251	0.025	0.837	-0.048	0.679
Average daily study hours JEE	963	8.225	-0.343	0.409	0.421	0.171	962	8.588	0.623	0.059	0.371	0.218
JEE Prep Cost (in lakhs INR)	875	1.384	0.026	0.819	-0.045	0.650	884	1.394	-0.116	0.305	-0.070	0.481
Took more than 1 attempt	945	0.147	-0.043	0.174	0.015	0.604	934	0.317	-0.050	0.238	-0.070	0.069
Doesn't Know programming	912	0.455	0.023	0.660	0.050	0.225	907	0.671	-0.025	0.586	-0.030	0.452
Joint Test p-value				0.439		0.449				0.651		0.316

Notes: The table presents balance on attrition, socioeconomic indicators, and pre-college academic outcomes across partially and fully mixed rooms. Panel A reports balance on attrition indicators, including whether students responded to and completed the survey. Panels B and C report balance on socioeconomic characteristics and pre-college academic outcomes, respectively. The sample includes students from entry years 2023 and 2024 assigned to three-person rooms, the only room size in which a partially mixed assignment is possible. Same-caste rooms are the omitted category. Columns 2 and 8 report same-caste room means for upper-caste and lower-caste students, respectively. For upper-caste students, columns 3 and 5 report differences in means for partially and fully mixed rooms relative to same-caste rooms, with corresponding p-values in columns 4 and 6 ; columns 9 and 11 report the analogous differences for lower-caste students, with p-values in columns 10 and 12. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level. The last row reports the p-value from the joint F-test of orthogonality for each treatment.

Table A36: Heterogeneous Effects on Main Outcomes by Family Income

	Upper Caste				Lower Caste			
	(1) CGPA : First Year	(2) Mental health index	(3) AA support: reserved group	(4) AA support: non-reserved group	(5) CGPA : First Year	(6) Mental health index	(7) AA support: reserved group	(8) AA support: non-reserved group
Mixed Room	0.006	-0.027	0.020	-0.157	0.060	-0.094	-0.015	0.085
	(0.086)	(0.075)	(0.088)	(0.103)	(0.074)	(0.064)	(0.058)	(0.083)
Family income (9L or above)	0.173**	0.092	-0.164*	-0.292***	0.002	0.005	-0.023	-0.057
	(0.080)	(0.077)	(0.088)	(0.106)	(0.095)	(0.080)	(0.078)	(0.112)
Mixed Room × Family Income (¿ Rs. 9L)	-0.163	-0.057	0.130	0.237*	0.109	-0.047	0.085	0.104
	(0.102)	(0.091)	(0.112)	(0.131)	(0.116)	(0.093)	(0.094)	(0.125)
DV Mean, Non-mixed	0.474	0.027	-0.251	-0.265	-0.445	0.001	0.239	0.237
R-squared	0.506	0.192	0.190	0.212	0.454	0.161	0.120	0.142
N	785	786	701	699	778	820	739	734

Columns 1-4 restrict to upper-caste students, columns 5-8 to lower-caste students. Within each caste block, Column 1 reports cumulative GPA at the end of the first year; Column 2 the composite Mental Health Index; Column 3 the reserved-group Support for Affirmative Action sub-index; Column 4 the non-reserved-group sub-index – the same four outcomes as the main results table. Each regression interacts mixed-room assignment with an indicator for self-reported family income above Rs. 9 lakh/year, to test whether the treatment effect varies by pre-college family income within each caste group. All regressions include hostel-by-cohort and specialization-by- cohort fixed effects and control for the number of roommates. Baseline controls for the respective outcome are included, with a missingness dummy where the baseline measure is missing. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A37: Effect on GPA, Mental Health and Attitudes towards Affirmative Action

	CGPA : First Year	Mental health index	AA support: reserved group	AA support: non-reserved group
	(1)	(2)	(3)	(4)
Panel A: Upper Caste Sample				
Partial Mixed Room	-0.085	-0.011	0.136*	0.081
	(0.062)	(0.061)	(0.077)	(0.086)
Full Mixed Room	-0.112*	-0.088*	0.104*	0.031
	(0.057)	(0.045)	(0.059)	(0.064)
P-val (Partial = Full)	0.668	0.158	0.649	0.575
DV Mean in same caste rm	0.474	0.027	-0.251	-0.265
R-squared	0.473	0.152	0.173	0.185
N	947	935	820	817
Panel B: Lower Caste Sample				
Partial Mixed Room	0.098	-0.084	-0.011	0.153**
	(0.070)	(0.055)	(0.052)	(0.067)
Full Mixed Room	0.087	-0.081*	-0.029	0.083
	(0.058)	(0.045)	(0.051)	(0.063)
P-val (Partial = Full)	0.881	0.960	0.723	0.296
DV Mean in same caste rm	-0.445	0.001	0.239	0.237
R-squared	0.433	0.146	0.110	0.132
N	896	928	829	820

Notes: The table presents results for academic performance, mental health, and attitudes toward affirmative action. Column 1 reports estimates for cumulative GPA at the end of the first year. Column 2 reports estimates for a composite Mental Health Index, constructed using standardized scores of the PHQ-4 index, and the worry index capturing academic- and social-life concerns in college. Negative values of the index indicate worse mental health outcomes. Column 3 reports an index of support for AA based on its perceived costs and benefits for reserved-group students. Column 4 reports an index of support for AA based on its perceived costs and benefits for non-reserved-group students. The final row of the full-sample panel (Panel C) reports the p-value from a pooled specification interacting mixed-room assignment with lower-caste, testing whether the lower- and upper-caste treatment effects differ. All regressions include hostel-by-cohort and specialization-year fixed effects. I use ANCOVA specification along with a control for the number of roommates. Standard errors are clustered at the room level. Sharpened q-values (Anderson 2008) correcting for multiple hypothesis testing within each panel are reported. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

Table A38: Effect on Mental Health: PHQ4, Relative Happiness, Worry Index

	Mental health index	PHQ4 index	Relative well being index	Worry index
	(1)	(2)	(3)	(4)
Panel A: Upper Caste Sample				
Mixed Room	-0.062	-0.093	-0.007	-0.024
	(0.044)	(0.068)	(0.042)	(0.045)
DV Mean in same caste rm	0.027	0.016	-0.006	0.052
R-squared	0.149	0.153	0.076	0.135
N	935	875	839	890
Sharpened q-value	0.342	0.342	0.873	0.784
Panel B: Lower Caste Sample				
Mixed Room	-0.082**	-0.084	0.005	-0.048
	(0.042)	(0.064)	(0.042)	(0.045)
DV Mean in same caste rm	0.001	-0.010	0.012	-0.046
R-squared	0.146	0.125	0.082	0.124
N	928	877	850	892
Sharpened q-value	0.197	0.377	0.904	0.381
Panel C: Full Sample				
Mixed Room	-0.073**	-0.099**	0.006	-0.038
	(0.032)	(0.047)	(0.030)	(0.033)
DV Mean in same caste rm	0.014	0.003	0.003	0.003
R-squared	0.109	0.103	0.042	0.091
N	1863	1752	1689	1782
Sharpened q-value	0.035	0.035	0.420	0.163
P val ($\beta_LC == \beta_UC$)	0.553	0.911	0.877	0.693

Notes: The table presents results for mental health outcomes: the Mental Health Index, the PHQ-4 index, the relative wellbeing index, and the worry index. In all columns, higher values indicate better mental health outcomes. Column 1 reports estimates for a composite Mental Health Index, constructed using standardized scores of the individual mental health measures. Column 2 reports results for the PHQ-4 index, the standardized PHQ-4 score measuring anxiety and depression symptoms, reverse-coded so that higher values indicate fewer symptoms. Column 3 reports results for the relative wellbeing index, an index of four components, with higher values indicating better relative wellbeing. Column 4 reports results for the standardized worry index, an index of self-reported worries about academic and social aspects of college life. The final row of the full-sample panel (Panel C) reports the p-value on the interaction between mixed-room assignment and a lower-caste indicator from a pooled specification, testing whether the lower- and upper-caste treatment effects differ. All regressions control for the baseline PHQ-4 index and relative wellbeing index (missing values imputed as zero, with an indicator for missingness included) and for the number of roommates. ***, **, * indicates significance at the 1%, 5%, and 10% level respectively

Table A39: Effect on Support for Affirmative Action: Benefits and Costs to Different Groups

	Supports AA index (1)	AA support: reserved group (2)	AA support: non-reserved group (3)	Opportunities inside IIT: Equal (4)	Clearing JEE: Merit only (5)
Panel A: Upper Caste Sample					
Mixed Room	0.088*	0.115**	0.048	0.191*	0.087
	(0.053)	(0.057)	(0.058)	(0.105)	(0.117)
DV Mean in same caste rm	-0.256	-0.251	-0.265	-0.021	0.088
R-squared	0.196	0.172	0.184	0.099	0.124
N	824	820	817	449	443
Sharpened q-value	0.163	0.163	0.456	0.163	0.456
Panel B: Lower Caste Sample					
Mixed Room	0.046	-0.022	0.110**	0.174	0.012
	(0.044)	(0.045)	(0.056)	(0.109)	(0.101)
DV Mean in same caste rm	0.240	0.239	0.237	0.011	-0.079
R-squared	0.117	0.110	0.131	0.076	0.100
N	834	829	820	432	433
Sharpened q-value	0.495	0.772	0.240	0.280	0.906
Panel C: Full Sample					
Mixed Room	0.077*	0.049	0.100**	0.178**	0.042
	(0.040)	(0.040)	(0.047)	(0.076)	(0.079)
DV Mean in same caste rm	-0.005	-0.002	-0.012	-0.006	0.009
R-squared	0.158	0.138	0.135	0.061	0.086
N	1658	1649	1637	881	876
Sharpened q-value	0.125	0.298	0.125	0.125	0.622
P val ($\beta_LC == \beta_UC$)	0.465	0.086	0.648	0.825	0.696

Notes: The table presents results for attitudes towards Affirmative Action, including benefits to the group who benefit from AA policy and perceptions on costs to those who don't benefit from AA policy. Column 1 reports estimates for a composite Affirmative Action support Index constructed as the mean of dependent variables in columns 2 and 3. Column 2 reports estimates for an index of attitudes that are related to benefits and costs (reverse coded) of AA policies for groups that receive reservation. Column 3 reports estimates for an index of attitudes that are related to benefits and costs (reverse coded) of AA policies for groups that don't benefit from AA. Column 4 reports if students believe opportunities inside university are equal for all. Column 5 reports if students believe that only merit is required to clear entrance exam. Higher values in columns 1, 2, and 3 indicate stronger support for Affirmative Action. All regressions include hostel-by-cohort and specialization fixed effects and control for the number of roommates. Standard errors are clustered at the room level. All the regressions include baseline controls for the respective dependent variables and a missing indicator for the baseline controls and number of roommates.

Table A40: Effect on Friendship Networks and Cross-Caste Friendships

	Upper Caste					Lower Caste				
	(1) Friends network index	(2) Close friends in college	(3) Close Friends Outside IIT	(4) Different caste friends within IIT	(5) Different caste friends outside IIT	(6) Friends network index	(7) Close friends in college	(8) Close Friends Outside IIT	(9) Different caste friends within IIT	(10) Different caste friends outside IIT
Mixed Room	-0.004	0.022	-0.030	0.181**	0.058	-0.075	-0.005	-0.144**	-0.022	0.027
	(0.072)	(0.080)	(0.078)	(0.087)	(0.088)	(0.064)	(0.077)	(0.067)	(0.073)	(0.072)
DV Mean, Non-mixed	-0.016	0.031	-0.068	-0.345	-0.383	0.020	-0.030	0.065	0.360	0.390
R-squared	0.098	0.094	0.091	0.074	0.074	0.096	0.072	0.117	0.093	0.068
N	922	922	922	748	738	944	944	944	710	714

Columns 1-5 restrict to upper-caste students, columns 6-10 to lower-caste students. Within each caste block, Column 1 (6) reports the Friendship Index, a composite of close friends at college and close friends outside college; Column 2 (7) reports the number of close friends at college; Column 3 (8) reports the number of close friends outside college; Column 4 (9) reports the number of other-caste friends within college; Column 5 (10) reports the number of other-caste friends outside college. All outcomes are standardized (z-scores) with respect to the full sample. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Columns 1-3 (6-8) include baseline controls for the respective outcome, with a missingness dummy where the baseline measure is unavailable; no baseline measure is available for columns 4-5 (9-10). Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A41: Heterogeneous Effects on Main Outcomes by Gender: Three Years after Admission

	Upper Caste				Lower Caste			
	(1) Third-Year GPA (SD)	(2) Mental health: 3yr	(3) AA support: reserved group	(4) AA support: non-reserved group	(5) Third-Year GPA (SD)	(6) Mental health: 3yr	(7) AA support: reserved group	(8) AA support: non-reserved group
Mixed Room	-0.127	-0.067	-0.053	-0.080	0.082	-0.042	0.068	0.073
	(0.087)	(0.063)	(0.076)	(0.089)	(0.085)	(0.052)	(0.064)	(0.073)
Female	0.190	-0.193**	0.030	0.210*	0.180	-0.146**	0.063	0.180*
	(0.148)	(0.081)	(0.101)	(0.112)	(0.132)	(0.072)	(0.111)	(0.093)
Mixed Room × Female	-0.016	0.091	0.022	-0.005	-0.119	0.080	0.109	-0.008
	(0.183)	(0.108)	(0.140)	(0.156)	(0.192)	(0.096)	(0.173)	(0.159)
DV Mean, Non-mixed	0.518	0.089	-0.138	-0.197	-0.459	-0.072	0.118	0.165
R-squared	0.284	0.017	0.003	0.014	0.145	0.008	0.012	0.017
N	471	472	452	454	500	494	468	467

Columns 1-4 restrict to upper-caste students, columns 5-8 to lower-caste students. Within each caste block, Column 1 reports cumulative GPA through the third year; Column 2 the composite Mental Health Index at the three-year follow-up; Column 3 the reserved-group Support for Affirmative Action sub-index; Column 4 the non-reserved-group sub-index. Each regression interacts mixed-room assignment with a Female indicator to test whether the treatment effect varies by gender within each caste group. Regressions control for the number of roommates, but exclude both hostel-by-cohort fixed effects – since hostel assignment is itself gendered – and cohort fixed effects, since the three-year follow-up is a single cohort. Baseline controls for the respective outcome are included, with a missingness dummy where the baseline measure is unavailable. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A42: Effect on Leadership Positions, Job Preferences, and Expected Salary

	Upper Caste			Lower Caste		
	(1) Clubs Leadership	(2) Preferred job sector: Finance/Consulting	(3) Expected salary (INR 1000s)	(4) Clubs Leadership	(5) Preferred job sector: Finance/Consulting	(6) Expected salary (INR 1000s)
Mixed Room	-0.126**	-0.106**	1.882	0.001	-0.019	-115.705
	(0.060)	(0.049)	(375.368)	(0.056)	(0.049)	(350.906)
DV Mean, Non-mixed	0.134	0.520	5271.515	-0.164	0.308	4348.665
R-squared	0.097	0.128	0.140	0.073	0.083	0.066
N	505	496	970	475	483	972

Columns 1-3 restrict to upper-caste students, columns 4-6 to lower-caste students. Column 1 reports effects on holding a position of responsibility in student clubs (Clubs Leadership index). Column 2 reports effects on preferring a job in finance or consulting. Column 3 reports effects on expected first-job salary in thousands of INR, top-coded at the 95th percentile to reduce the influence of outliers. No additional controls are included. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects, control for the number of roommates, and are weighted using survey weights correcting for the two-tier sampling design. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A43: Effect on Leadership Positions, Job Preferences, and Expected Salary: Three Years after Admission

	Upper Caste			Lower Caste		
	(1)	(2)	(3)	(4)	(5)	(6)
	Clubs Leadership	Preferred job sector: Finance/Consulting	Expected salary (INR 1000s)	Clubs Leadership	Preferred job sector: Finance/Consulting	Expected salary (INR 1000s)
Mixed Room	-0.034	-0.087**	59.539	0.019	-0.035	-248.799
	(0.053)	(0.044)	(562.060)	(0.048)	(0.036)	(381.740)
DV Mean, Non-mixed	0.031	0.446	5440.314	-0.029	0.184	4014.218
R-squared	0.046	0.061	0.075	0.010	0.039	0.025
N	463	465	482	478	467	512

Columns 1-3 restrict to upper-caste students, columns 4-6 to lower-caste students. Column 1 reports effects on holding a position of responsibility in student clubs (Clubs Leadership index). Column 2 reports effects on preferring a job in finance or consulting. Column 3 reports effects on expected first-job salary in thousands of INR, top-coded at the 95th percentile to reduce the influence of outliers. No additional controls are included. Sample is restricted to the 2023 cohort's three-year follow-up. All regressions include hostel-by-cohort and specialization-by-cohort fixed effects and control for the number of roommates. Standard errors are clustered at the room level and reported in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

A Survey Protocol and Data cleaning

I recruited four student survey managers who served as the primary points of contact for survey administration. For each survey wave, the survey managers recruited two to three student ambassadors in each hostel. Hostel ambassadors assisted with communicating information about the survey, organizing survey sessions, and following up with students within their respective hostels. All questionnaires were self-administered. I used a hybrid survey administration protocol that combined online questionnaires with in-person survey sessions and hostel-level tracking and follow-up.

A.1 Recruitment and Tracking

2024 cohort: For the 2024 cohort baseline survey, data collection took place during the first week of college, coinciding with the hostel orientation and induction period for first-year students. These orientation activities are typically attended by nearly all students in the incoming cohort. I, together with the survey managers, visited the hostels during these sessions. Students were gathered in a common hall within the hostel, where we explained the purpose and procedures of the study and invited them to participate. Students were informed that the survey concerned general aspects of college life, including their expectations about college and how their experiences and trajectories evolve over time. Students were provided with a survey link and completed the questionnaire individually. Students who were unable to attend the group sessions were contacted separately and encouraged to complete the questionnaire. Hostel ambassadors conducted individual follow-up with these students and provided them with the survey link. Approximately 90% of all baseline responses were obtained within 2–7 days of students' entry into the college, while the remaining 10% were obtained during the following week through individual follow-up. Thus, the baseline survey was administered within approximately two weeks of college entry for the full sample.

The endline survey for the 2024 cohort followed the same basic recruitment structure. Hostel ambassadors organized survey sessions in which students were gathered in common areas and invited to complete the questionnaire. Attendance at these sessions was approximately 60 percent of students in a hostel, on average. To reach students who did not attend the group sessions, hostel ambassadors conducted additional door-to-door recruitment using posters and pamphlets to publicize the survey and encourage participation. During this phase, I provided tracking sheets to hostel ambassadors to monitor participation and identify students who had not yet completed the questionnaire for follow-up.

2023 cohort: For both survey waves of the 2023 cohort, recruitment relied primarily on hostel-based communication and individual follow-up. Hostel ambassadors publicized the survey through posters, direct door-to-door outreach within the hostels, and messages distributed through hostel WhatsApp groups. These channels were used to inform students about the

survey, encourage participation, and follow up with students who had not yet completed the questionnaire.

For the first wave of the 2023 cohort survey, the initial recruitment round yielded responses from 71.8% of students in the cohort. After this initial round, I fixed the set of initial nonrespondents and drew a simple random sample of 25% of these students. This resulted in a subsample of 85 students from the 338 initial nonrespondents. I then conducted intensive individual follow-up with these selected students and asked them to complete the survey. In analyses, I account for this two stage sampling design using sampling weights. The 71.8% of students who responded during the initial round receive a weight of 1, while the 25% random subsample of initial nonrespondents receives a weight of 4, corresponding to the inverse of the probability of selection for intensive follow-up. Regressions using outcomes collected through this sampling procedure are estimated using these sampling weights.

The follow-up survey of the 2023 cohort was conducted separately as a single-stage survey. Two-stage sampling and sampling weights are not applied to this wave. Response rates were high across both cohorts and survey waves. The response rates for each survey wave are reported in Table A1.

A.2 Hostel Allocation and Student Lists

Student and hostel lists: I obtained cohort-level lists containing students' names, college identification numbers, and hostel assignments through student survey managers who had access to these lists through college sources. These lists were used to identify students belonging to each cohort, organize recruitment by hostel, and establish the survey contact frame.

I obtained hostel allocation information from multiple sources, depending on the cohort and year. For the 2024 cohort, I obtained first- and second-year hostel allocation lists for all hostels from the student heads of the respective hostels. These lists were obtained during the hostel orientation and recruitment period. For the 2023 cohort, I similarly obtained the second-year hostel allocation lists from the student heads of the respective hostels.

For the 2023 cohort's first-year allocation, I obtained administrative hostel rosters for 9 of the 13 hostels by visiting the respective hostel administration offices. The rosters were available as electronic Excel files or hard copies, which I manually transcribed into Excel during my visits. For the remaining four hostels, I reconstructed room allocations using students' self-reported hostel, room number, and roommate names collected in the survey. I cross-checked self-reported information against the available administrative records and conducted additional verification when discrepancies or missing information were identified. For hostels without administrative records, hostel ambassadors assisted in recontacting students to resolve discrepancies. I manually reviewed the resulting room and roommate records for consistency within and across hostels.

Respondent identification and data validation. Students reported their college identification number, institutional email address, room number, and roommate names in the survey. I used the

hostel allocation lists and cohort contact lists to verify these information and to resolve discrepancies as part of the data-cleaning process. College identification numbers and institutional email addresses are linked and provide a unique identifier for each student. I used these identifiers to verify respondent identity and to identify duplicate submissions within each survey wave.

A.3 JEE rank and score construction:

I constructed students' JEE rank and score measures by combining survey responses with JEE examination records published by the institute. I first scraped the institute's published JEE reports, which contain student-level information on JEE score, JEE rank, admission category, specialization, and institute. The examination ID also contains information on the examination centre or location. The published records additionally report gender- and category-specific opening and closing ranks for each specialization, which provide additional information for validating student-level matches.

I first collected students' self-reported JEE rank and score in the survey. I then matched students to the published JEE records using their reported rank, restricting matches to the relevant cohort and specialization. Because JEE rank is unique within these cells, exact matches provide a direct way to identify students in the published records. Any missing information due to non-response or mismatch due to typos or misreporting, I used the combination of examination location, specialization, gender, and category-specific opening and closing ranks to identify the most plausible record. City, gender- and category-specific cutoffs were particularly useful for resolving cases for missing ranks. I also used publicly available information, including rank information published by coaching institutes and information available through students' public profiles or school records, to cross-check and resolve remaining unmatched observations.

This procedure resulted in identified JEE records for 99 percent of students in the 2024 cohort and 95 percent of students in the 2023 cohort. As a validation of the resulting classification, the distribution of admission categories in the cleaned survey data closely matches the corresponding category shares in the institute's published records. The final JEE rank, score, and category measures are constructed from these matched records rather than relying solely on students' self-reported responses.

A.4 Survey Questions

Table A44: Survey wording: mental health

Item	Survey Question	Scale / Type
GAD-2 anxiety items (PHQ4)	"Over the last two weeks, how often have you been bothered by the following problems:" (1) "Feeling nervous, anxious or on edge"; (2) "Not being able to stop or control worrying"	4-pt frequency: Not at all / Some days / More than half the days / Almost Everyday / DNWA
PHQ-2 depression items (PHQ4)	"Over the last two weeks, how often have you been bothered by the following problems:" (1) "Feeling down, depressed or hopeless"; (2) "Little interest or pleasure in doing things"	4-pt frequency: Not at all / Some days / More than half the days / Almost Everyday / DNWA
Relative well-being vs. pre-college period	"Examine each of the five statements below and indicate your opinion": (1) "I am more happy now compared to my JEE preparation days"; (2) "I find little joy in my daily activities compared to when I was preparing for the JEE"; (3) "I have a better support system now than I did during my JEE preparation"; (4) "I feel less in control of my life now than I did when I was preparing for the JEE"; (5) "I am more stressed on life now than when I was preparing for the JEE"	5-pt agreement: Strongly Agree / Agree / Neutral / Disagree / Strongly Disagree / DNWA
College worries	"How worried are you about the following things (your day to day college life):" (1) "Academics"; (2) "Getting internships/jobs"; (3) "Language/cultural differences"; (4) "Staying away from home"; (5) "Discrimination/Exclusion in study groups"; (6) "Time management (Acads and clubs)"	5-pt worry scale: Not at all(1) / A Little(2) / Mildly worried(3) / Worried(4) / Extremely worried(5) / DNWA
Loneliness	"Examine each of the three statements below and answer them": (1) "How often do you feel you do not have friends?"; (2) "How often do you feel left out?"; (3) "How often do you feel isolated from others?"	4-pt frequency: Not at all / Sometimes / Most times / Yes, always / DNWA

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Table A44 continued

Item	Survey Question	Scale / Type
Self-efficacy	"Examine each of the six statements below and answer them": (1) "If someone opposes you, can you find some way to get what you want?"; (2) "Is it easy for you to stick to and accomplish your goals?"; (3) "Are you confident that you could deal efficiently with unexpected events?"; (4) "Do you know how to handle unforeseen situations by using your resourcefulness?"; (5) "Can you remain calm when facing difficulties by relying on your coping abilities?"; (6) "Can you usually handle whatever comes your way?"	4-pt frequency: Not at all / Sometimes / Most times / Yes, always / DNWA

Table A45: Survey wording: affirmative action attitudes

Item	Survey Question	Scale / Type
Benefit: social justice (reserved group)	"Reservations promote social justice by compensating for historical or systemic inequalities"	5-pt agreement: Strongly Agree / Agree / Neutral / Disagree / Strongly Disagree / DNWA
Benefit (reserved group): diversity and inclusion	"Reservations can foster diversity and inclusion within colleges and universities"	5-pt agreement (as above)
Cost (reserved group): reduces incentive to work hard	"Reservations may reduce the incentive for students to work hard"	5-pt agreement (as above)
Cost: harder academic placement (reserved group)	"Reservations ultimately hurt students belonging to reserved groups when they are placed into a more challenging academic placement"	5-pt agreement (as above)
Benefit: non-reserved-group students	"Reservations ultimately benefit students without reservation groups by promoting diversity in student demographics"	5-pt agreement (as above)
Cost (non-reserved group): fairness vs. merit	"Reservations is an unfair system compared to that based on pure academic merit"	5-pt agreement (as above)

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Table A45 continued

Item	Survey Question	Scale / Type
Cost (non-reserved group): academic quality	"Reservations might compromise the academic quality of colleges and universities"	5-pt agreement (as above)
General redistribution belief	"Governments should aim to reduce existing inequalities using more comprehensive social policies"	5-pt agreement (as above)
Merit belief (JEE-Advanced)	"For clearing JEE, only merit and effort matters not socio-economic conditions of the students"	5-pt agreement (as above)
Equality of opportunity (post-entry)	"Opportunities for jobs, participation in student clubs, and academic success is equal for everyone once you enter college"	5-pt agreement (as above)
Support for reservations by domain	"Overall, how much do you support reservations in the following areas?": Elite public universities like IITs, IIMs, AIIMS etc.; Other national institutes like NITs, IIITs etc.; Public sector jobs like IAS, IPS, Engineering service; Other public sector jobs like industries (PSU), state public services etc.; Private sector jobs; Politics (State/Central level); Local elections	5-pt support: Strongly support / Moderate support / Neutral / Moderate Oppose / Strongly Oppose / DNWA
Effect of reservations on own opportunities	"In your view, how have reservation policies in IIT admissions affected your opportunities?"	5-pt scale: Improved a lot / Somewhat improved / No difference / Reduced somewhat / Reduced a lot / DNWA
Hypothetical seat allocation	"Hypothetically, if there are 100 seats to be allocated in a public university with reservations similar to IITs, how would you allocate the seats for different groups below?" (with a follow-up asking how respondents would change the current allocation)	Numeric allocation across groups (sums to 100) / free-text comments

Table A46: Survey wording: internships

Item	Survey Question	Scale / Type
Summer activity, year 2	"What did you do in the summer of your 2nd year (May - July, 2025)"	Multiple choice, 12 categories (incl. "Other" free text)
Summer plans, year 3 (prospective)	"What are your plans in the summer of your 3rd year (May - July, 2026)"	Multiple choice, 12 categories (same list as above)
Summer activity, year 1	"What did you do in the summer after your 1st year?" Instruction: "Select upto 3 activities"	Multiple choice, up to 3 of 11 categories
Internship offer obtained	"Did you get an internship offer?"	Single select: No offer / Yes / DNWA
Stipend, year 3 internship	"What was your compensation for your third year internship/other activity?"	Single select, 12 categorical R bins (Unpaid to R4 lakhs+)
Stipend, year 2 internship	"What was your compensation for your second year summer internship/other activity?"	Single select, 13 categorical R bins (Unpaid to R3 lakhs+)
Internship/project counts	"In total how many internships/projects you have done since starting college?" Items: "In person/hybrid internships"; "Fully remote internships"; "Projects in college"; "Projects Outside college"	Numeric entry per item
Stipend, year 1 internship	"What was your monthly compensation for the internship/partime job?"	Single select, 11 categorical R bins (Unpaid to R2 lakh+)
Stipend, years 1–3 (pooled)	"What was your monthly compensation for the activity?"	Numeric entry; -100 = DNWA
Pay (activity type ambiguous)	"What was your monthly compensation for the activity?" Instruction: "Any stipend/Scholarship you received for internships/projects"	Numeric entry

B A Self-Image Model of Deservedness

B.1 Setup

Let $\psi \in [0,1]$ be the true share of the caste preparation gap attributable to circumstance rather than ability, and let α_i be student i's support for affirmative action (AA). Each student holds an ego-protective prior b_i, the baseline belief that admission reflects merit, which maps into a prior belief $\psi(b_i)$ about ψ with $\psi'(b_i) < 0$. In the spirit of Bénabou and Tirole (2002, 2006), the student does not simply adopt the Bayesian posterior ψ_i^B but chooses, in the style of Alesina and Angeletos (2005),

$$\hat{\psi}_i = \arg\max_{\psi} \Big\{ -e(r_i)\,\psi - \frac{1}{2\kappa}\big(\psi - \psi_i^B\big)^2 \Big\}, \tag{8}$$

where r_i is own academic standing, $e(r_i) \geq 0$ with $e' > 0$ is the ego stake in a low-ψ belief ("I deserve my place"), and $\kappa > 0$ scales malleability. The objective is strictly concave, so the first-order condition gives the equilibrium distortion

$$\hat{\psi}_i = \psi_i^B - \kappa\, e(r_i), \qquad \alpha_i = \alpha_0 + \varphi\, \hat{\psi}_i + \zeta_i, \tag{9}$$

with $\varphi > 0$ and $\mathbb{E}[\zeta_i \mid \cdot] = 0$. The prior and the co-residence signal z_i are normally distributed with precisions ξ_0 and $\xi(d_i)$, so the posterior mean is the precision-weighted average

$$\psi_i^B = \big(1 - \ell(d_i)\big)\,\psi(b_i) + \ell(d_i)\, z_i, \qquad \ell(d_i) = \frac{\xi(d_i)}{\xi(d_i) + \xi_0}. \tag{10}$$

Signal precision ξ is increasing in caste distance d_i with $\xi(0) = 0$: a same-caste roommate reveals nothing about the caste gap, so $\ell(0) = 0$ and $\ell' > 0$.

Assumption 7 (Beliefs are not payoff-relevant)**.** *ψ_i and α_i enter none of the payoff objects in §6.3: neither wellbeing, network capital, internships, nor utility. Support for AA is something a student comes to believe, not something traded off against social time. This severs the beliefs→behavior link: no belief feeds back into chosen social time, so no fixed point arises between what a student believes and how they behave.*

Assumption 8 (Beliefs update off exposure, not contact)**.** *The signal precision $\xi(d_i)$ and the prior-precision objects $\xi_0^k(c_i)$ depend on the identity of the roommate (caste distance d_i) and the student's own caste, not on realized interaction $\tilde{s}_{ij}$. Informativeness is a property of* who *the roommate is, not of* how the pair spend their time*.*

B.2 Correction of mis-anchored beliefs

Proposition 4 (Correction of mis-anchored beliefs)**.**

(i) *If standing r_i is pre-assignment ($r_i = \theta_i$), then under (8)–(10) the mixed- versus same-caste contrast in support is*

$$\alpha_i\big|_{d_i>0} - \alpha_i\big|_{d_i=0} = \varphi\, \ell(d_i)\big(z_i - \psi(b_i)\big). \tag{11}$$

The ego wedge $-\varphi\kappa\, e(r_i)$ is common to both room types and cancels; the contrast is identified purely off the correction channel, and is larger the more strongly anchored the prior (larger b_i) and the more caste-distant the roommate.

(ii) *If standing responds to assignment (r_i includes g_i, with e evaluated at expected standing), the contrast acquires a residual,*

$$\Delta\alpha_i = \varphi\,\ell(d_i)\big(z_i - \psi(b_i)\big) - \varphi\kappa\big[e(r_i\big|_{mixed}) - e(r_i\big|_{same})\big]. \tag{12}$$

For lower-caste students whose expected grade change is positive (the condition in Corollary 1(i)), the residual is negative and the correction is attenuated; for upper-caste students, whose expected grade change is negative (Corollary 1(ii)), the residual is positive and the correction is amplified. The clean contrast (11) therefore understates the upper-caste attitude shift and overstates the lower-caste one.

Part (ii) implies that the identifying assumption behind (11) is conservative for the paper's main identified result (the upper-caste benefits correction): relaxing it strengthens the predicted shift. For the lower-caste fairness correction, which is already flagged as consistent but not separately identified, (11) is an upper bound.

Proof of Proposition 4. *(i)* With $r_i = \theta_i$, standing is fixed by the entrance score and does not depend on room type. Differencing (9) between $d_i > 0$ and $d_i = 0$, the ego wedge $-\varphi\kappa\, e(r_i)$ cancels. By (10) and $\ell(0) = 0$, $\psi_i^B\big|_{d_i=0} = \psi(b_i)$, and the difference of posteriors is $\big[(1-\ell)\psi(b_i) + \ell z_i\big] - \psi(b_i) = \ell(d_i)(z_i - \psi(b_i))$, giving (11). Since $\psi'(b_i) < 0$, larger b_i lowers $\psi(b_i)$ and so raises $z_i - \psi(b_i)$; since $\ell' > 0$, the contrast rises in d_i.

(ii) From (9), $\alpha_i = \alpha_0 + \varphi\,\psi_i^B - \varphi\kappa\, e(r_i) + \zeta_i$. Differencing across room types,

$$\Delta\alpha_i = \varphi\big[\psi_i^B\big|_{\text{mixed}} - \psi_i^B\big|_{\text{same}}\big] - \varphi\kappa\big[e(r_i\big|_{\text{mixed}}) - e(r_i\big|_{\text{same}})\big].$$

The first term is $\varphi\,\ell(d_i)(z_i - \psi(b_i))$ as in part (i). For the second, $e' > 0$, so the sign of $e(r_i|_{\text{mixed}}) - e(r_i|_{\text{same}})$ is the sign of $r_i|_{\text{mixed}} - r_i|_{\text{same}}$. If r_i includes g_i, then for a lower-caste student with a positive expected net grade effect, $r_i|_{\text{mixed}} > r_i|_{\text{same}}$ and the residual $-\varphi\kappa[\cdot] < 0$: the correction is attenuated. For an upper-caste student the expected grade change is negative, the inequality reverses, and $-\varphi\kappa[\cdot] > 0$: the correction is amplified. □

B.3 Which dimension moves

ψ is a single parameter, but deservedness has two sides a student forms views about: whether the policy corrects a real disadvantage (benefits) and whether one's own place is deserved (fairness). Co-residence delivers a single signal z_i about ψ; the question is whether students with different starting positions update on different dimensions. Before co-residence, each student already holds relatively precise priors about the dimension they have lived. Co-residence reveals the roommate's circumstances A_j (family resources, schooling quality, daily life behind the score) which,

alongside the already-known θ_j, lets the student update the production function $\theta = f(\text{ability}, A)$ and hence ψ. But this update moves beliefs only on the dimension where the prior was imprecise.

Let $\xi_0^k(c_i)$ denote student i's prior precision on dimension $k \in \{ben, fair\}$, reflecting accumulated knowledge from living on side c_i of the gap. The posterior weight co-residence receives on dimension k is

$$\ell^k(d_i, c_i) = \frac{\xi(d_i)}{\xi(d_i) + \xi_0^k(c_i)}, \tag{13}$$

and the update on dimension k is

$$\Delta\alpha_i^k = \varphi^k \, \ell^k(d_i, c_i)\big(z_i - \psi(b_i)\big), \qquad \varphi^k > 0. \tag{14}$$

The prior-precision ordering follows from which side of the gap each group has lived:

$\xi_0^{ben}(L) > \xi_0^{ben}(U)$: lower-caste students have direct experience of disadvantage.

$\xi_0^{fair}(U) > \xi_0^{fair}(L)$: upper-caste students were not admitted through AA, so the fairness of their own place is settled by the admissions process itself.

These are informational endowments implied by the institutional structure and Assumption 1, not behavioral assumptions about motivated reasoning.

Proposition 5 (Dimension-specific updating)**.** *Under* (13)–(14) *and the prior-precision ordering above, holding $z_i - \psi(b_i) > 0$ and d_i fixed across castes:*

(i) $\Delta\alpha_U^{ben} > \Delta\alpha_L^{ben}$*: the benefits correction is larger for upper-caste students.*

(ii) $\Delta\alpha_L^{fair} > \Delta\alpha_U^{fair}$*: the fairness correction is larger for lower-caste students.*

(iii) *The two corrections move the between-caste support gap in opposite directions on the two dimensions. With $G^k \equiv \alpha_U^k - \alpha_L^k$,*

$$\Delta G^{ben} = \Delta\alpha_U^{ben} - \Delta\alpha_L^{ben} > 0, \qquad \Delta G^{fair} = \Delta\alpha_U^{fair} - \Delta\alpha_L^{fair} < 0.$$

If upper-caste students start less supportive on benefits and lower-caste students more supportive on fairness (an empirical premise, not implied by the model), the benefits gap closes and the fairness gap widens. The net effect on overall support is not signed and depends on $\varphi^{ben}[\ell^{ben}(d, U) - \ell^{ben}(d, L)]$ versus $\varphi^{fair}[\ell^{fair}(d, L) - \ell^{fair}(d, U)]$.

The opposite-sign prediction of part (iii) is the sharpest testable restriction the model produces on attitudes: an exposure-only specification predicts a common shift across dimensions and cannot generate $\Delta G^{ben} > 0 > \Delta G^{fair}$. The split vanishes when prior knowledge is symmetric ($\xi_0^k(U) = \xi_0^k(L)$), so students with prior cross-caste contact, who may have relatively symmetric priors, should show an attenuated dimensional asymmetry.

Remark (General ψ)**.** *The belief ψ is modeled as learning about the production function $\theta = f(\text{ability}, A)$, not just the caste gap per se. Any co-residence pairing with $d_{ij} > 0$ delivers information about f, including within caste.*

Proof of Proposition 5. Fix the correction size $z_i - \psi(b_i) > 0$ and caste distance d_i. By (14), $\Delta\alpha_i^k = \varphi^k \, \ell^k(d_i, c_i)(z_i - \psi(b_i))$, where ℓ^k is strictly decreasing in $\xi_0^k(c_i)$: $\partial \ell^k / \partial \xi_0^k = -\xi(d_i)/(\xi(d_i)+\xi_0^k)^2 < 0$.

(i) $\xi_0^{ben}(L) > \xi_0^{ben}(U)$ implies $\ell^{ben}(d, L) < \ell^{ben}(d, U)$. Since $\varphi^{ben} > 0$ and the correction size is common, $\Delta\alpha_U^{ben} > \Delta\alpha_L^{ben}$.

(ii) $\xi_0^{fair}(U) > \xi_0^{fair}(L)$ implies $\ell^{fair}(d, U) < \ell^{fair}(d, L)$, hence $\Delta\alpha_L^{fair} > \Delta\alpha_U^{fair}$.

(iii) The change induced by mixing is

$$\Delta G^k = \Delta\alpha_U^k - \Delta\alpha_L^k = \varphi^k \big[\ell^k(d, U) - \ell^k(d, L)\big]\big(z_i - \psi(b_i)\big).$$

For benefits the bracket is positive by (i), so $\Delta G^{ben} > 0$. For fairness the bracket is negative by (ii), so $\Delta G^{fair} < 0$. The opposite signs are a joint implication of the prior-precision ordering. An exposure-only specification, which delivers a common shift $\Delta\alpha$ across dimensions, predicts ΔG^{ben} and ΔG^{fair} of the same sign, so the opposite-sign pattern distinguishes the correction model from a pure dose-response. Whether the gap on a dimension closes or widens in levels depends on its baseline sign, which the model does not determine. The net effect $\Delta G = \Delta G^{ben} + \Delta G^{fair}$ is unsigned: it depends on the two magnitudes in the statement, which the model does not rank.

For the attenuation prediction, $\ell^{ben}(d, U) - \ell^{ben}(d, L) = \xi(d)\big[\xi_0^{ben}(L) - \xi_0^{ben}(U)\big] / \big[(\xi + \xi_0^{ben}(U))(\xi + \xi_0^{ben}(L))\big]$ is proportional to the precision gap and vanishes with it, and likewise for fairness. Prior cross-caste contact raises the precision of the otherwise imprecise dimension, narrowing the gap and compressing both brackets toward zero. □

C Proofs for Section 6

C.1 Notation and maintained conditions

For student i with roommate j, write $\chi_i \equiv \chi_{c_i}(d_{ij})$, $\tau_i \equiv \lambda_{ij} - \iota_i$ and $B_i(s) = \iota_i + \tau_i s$. Bridging has two sources: the room-sourced part, $(1 - \lambda_{ij})s$, is valued at the roommate's network return π_{ij}, and the club-sourced part, $(1 - \iota_i)(1 - s)$, at a constant $\pi_{c,i} \geq 0$, the value of the network reached outside the room. The constant $\pi_{c,i}$ does not depend on anyone's time allocation (price taking: clubs are large relative to any one student's participation, and random assignment moves club composition negligibly). It may differ across students through access, for example taking a lower value for club-excluded students. Network capital is

$$N_i = A_i + V_i, \qquad V_i \equiv \pi_{ij}(1 - \lambda_{ij})s + \pi_{c,i}(1 - \iota_i)(1 - s) \geq 0,$$

where V_i is the value of bridging; $V_i = \pi_{ij} R_i$ if $\pi_{c,i} = \pi_{ij}$. Network capital is affine in s with slope $\kappa_i \equiv \pi_{ij}(1 - \lambda_{ij}) - \pi_{c,i}(1 - \iota_i)$. Since belonging is also affine in s, for $\tau_i \neq 0$ utility (7) equals

$\Phi_i(B_i(s)) - \chi_i s$ plus terms that do not depend on contact s, where

$$\Phi_i(B) \equiv v(B) - \pi_{\text{eff},i}\, B - c(1-B;\Delta_i), \qquad \pi_{\text{eff},i} \equiv -\frac{\kappa_i}{\tau_i} = \frac{\pi_{c,i}(1-\iota_i) - \pi_{ij}(1-\lambda_{ij})}{\tau_i} = \pi_{ij} + \frac{(\pi_{c,i} - \pi_{ij})(1-\iota_i)}{\tau_i}.$$

The effective return $\pi_{\text{eff},i}$ is the network value gained per unit of belonging given up. It equals π_{ij} if $\pi_{c,i} = \pi_{ij}$, equals $\pi_{c,i}$ in a same-caste room ($\lambda_{ij} = 1$), and may be negative when room time carries more network value than club time. Hence $\Phi_i'(B) = v'(B) + c_R(1-B;\Delta_i) - \pi_{\text{eff},i}$, where c_R is the derivative of c in its first argument and $c_{R\Delta}$ the cross-partial. Own advantage A_i and grades g_i are fixed by the assignment. The knife-edge $\tau_i = 0$ is set aside. The main text sets $\pi_{c,i} = \pi_{ij}$, a single return to bridging whatever the venue; the proofs below allow $\pi_{c,i}$ to differ, and Remark C.3 lists what simplifies in the baseline. The proofs use the following conditions, all of which are stated in the assumptions of the main text.

(M1) v is strictly concave and increasing; c is twice continuously differentiable, increasing and convex in R, so $\Phi_i'' = v'' - c_{RR} < 0$ (Assumption 3).

(M2) $c(0;\Delta) = c_R(0;\Delta) = 0$ and $c_{R\Delta} \geq 0$ for all Δ; hence $c_\Delta(0;\Delta) = 0$ and c_R is increasing in both arguments (Assumption 3).

(M3) $\pi(\cdot) \geq 0$ is nondecreasing; the club network values satisfy $\pi_{c,i} \geq 0$ and are constants; $\chi_c(d) \geq 0$ with $\chi_c(0) = 0$; $A_i \geq 0$ (Assumptions 3 and 5). The ordering of club and roommate network values stated in the main text, $\bar{\pi}_L \leq \mathbb{E}[\pi_{c,i}] \leq \bar{\pi}_U$, is not used in any proof.

(M4) Club alignment satisfies $\iota_i \in [0,1)$, is nondecreasing in advantage within caste, and $\iota(U,x) > \iota(L,x)$ at every advantage level x (Assumption 2); club-excluded students have $\iota_i = 0$ and club-access thresholds satisfy $\underline{A}_L \geq \underline{A}_U$ (Assumption 6).

(M5) Within a room type, the roommate is drawn independently of the student's own characteristics, conditional on the pairing of castes (random assignment, §6.1).

(M6) $\pi(A) \leq v'(1)$ on the support of lower-caste advantage, and $\pi_{c,i} \leq v'(1)$ for lower-caste students (Assumption 4). No such restriction is imposed for upper-caste students.

With (M4), effective alignment is $\iota_i = \iota(c_i, A_i)\mathbb{1}[A_i \geq \underline{A}_{c_i}]$, nondecreasing in A_i within caste, and $\iota(U,x) \geq \iota(L,x)$ at every advantage level x (for $x \in [\underline{A}_U, \underline{A}_L)$ the upper-caste student is club-capable and the lower-caste student is excluded).

C.2 Grades

Proof of Proposition 1. Take expectations of (5) conditional on room type; the noise term integrates to zero.

Same-caste baseline. In a same-caste room, $\lambda_{ij} = 1$ and $d_{ij} = 0$. Two students drawn from the same caste distribution satisfy $\Pr(\theta_j > \theta_i \mid \text{both caste } c) = 1/2$ by symmetry, so

$$\mathbb{E}\big[(\theta_j - \theta_i)_+ \mid \text{both } c\big] = \tfrac{1}{2}\,\delta_c,$$

and $\mathbb{E}[g_i \mid \text{same-caste}] = \theta_i + \beta \cdot \frac{1}{2} \cdot \delta_c$.

Mixed room. For a lower-caste student i with an upper-caste roommate j, the event $\{\theta_j > \theta_i\}$ has probability p (defined in (3)) and conditional gap δ, so $\mathbb{E}[(\theta_j - \theta_i)_+ \mid i \text{ lower-caste, mixed}] = p\,\delta$. Hence $\mathbb{E}[g_i \mid \text{mixed}] = \theta_i + \beta\lambda_{ij} p\,\delta - \mu_L d_{ij}$, and the contrast against the same-caste baseline is

$$\mathbb{E}[g_i \mid \text{mixed}] - \mathbb{E}[g_i \mid \text{same-caste}] = \beta\big[\lambda_{ij}\,p\,\delta - \delta_L/2\big] - \mu_L d_{ij},$$

giving (i). For an upper-caste student i, $(\theta_j - \theta_i)_+ > 0$ only on the reversal event, with probability $1 - p$ and conditional gap δ':

$$\mathbb{E}[g_i \mid \text{mixed}] - \mathbb{E}[g_i \mid \text{same-caste}] = \beta\big[\lambda_{ij}(1-p)\delta' - \delta_U/2\big] - \mu_U d_{ij},$$

giving (ii). The transmission part of the upper-caste bracket is negative whenever $\lambda_{ij}(1-p)\delta' < \delta_U/2$: cross-caste transmission is weighted by the reversal probability $1 - p$ and discounted by $\lambda_{ij} < 1$, while the forgone within-caste transmission $\delta_U/2$ is subtracted in full. This holds in the limit $p \to 1$ (Corollary 1) and under the empirical magnitudes below; it is not implied by the model's assumptions alone.

Empirical magnitudes. The between-caste gap $\Delta\theta$ is slightly greater than one within-type standard deviation of the entrance-score distribution, so $p \in [0.85, 0.90]$ (see Table [*]). Reversals occur in the overlap region where scores are close, implying $\delta' \ll \delta$ and, a fortiori, $(1-p)\delta' \ll p\,\delta$. Within-caste upward gaps δ_c are bounded above by the within-type standard deviation, which is substantially smaller than the cross-caste gap δ; accordingly $\delta_L/2$ is small relative to $\lambda_{ij} p\,\delta$. The composites $\beta[\lambda_{ij} p\,\delta - \delta_L/2]$ and $\beta[\lambda_{ij}(1-p)\delta' - \delta_U/2]$ are what the reduced-form grade regressions estimate; the individual objects δ, δ', δ_c are not separately identified from β and λ_{ij}. □

Proof of Corollary 1. Set $p = 1$ in Proposition 1.

(i) The lower-caste contrast becomes $\beta[\lambda_{ij}\delta - \delta_L/2] - \mu_L d_{ij}$. Since $\beta > 0$ and $\delta > 0$, it is positive if and only if $\lambda_{ij} > \delta_L/(2\delta) + \mu_L d_{ij}/(\beta\delta)$. Equivalently, the discounted return to cross-caste transmission, $\beta\lambda_{ij}\delta$, exceeds the sum of the opportunity cost of forgone within-caste transmission, $\beta\delta_L/2$, and the social-mixing cost, $\mu_L d_{ij}$. Setting $\mu_L d_{ij} = 0$ gives the transmission-only threshold $\lambda_{ij} > \delta_L/(2\delta)$, and any positive mixing cost raises the threshold strictly. The right-hand side must be below 1 for the inequality to be satisfiable, i.e. $\delta_L/2 + \mu_L d_{ij}/\beta < \delta$.

(ii) With $1 - p = 0$, the upper-caste contrast becomes $-\beta\delta_U/2 - \mu_U d_{ij}$, which is negative for every $\lambda_{ij} \in (0, 1]$ whenever $\delta_U > 0$ or $\mu_U d_{ij} > 0$, since $\beta > 0$ and $\mu_U \geq 0$.

Illustrative magnitudes (back-of-envelope). With $\delta \approx 60$ and $\delta_L \approx 38$, $\delta_L/(2\delta) \approx 0.32$, so transmission alone is positive as long as roommate alignment preserves roughly a third of full same-caste

alignment. This is only the opportunity-cost term: μ_L, d_{ij} and β are not pinned down here, so these numbers do not establish that the net threshold is cleared. □

Remark (Heterogeneity of the transmission loss in own score). *Let $H_c(x) \equiv \mathbb{E}[(\theta - x)_+ \mid c] = \int_x^\infty \big(1 - F_c^\theta(t)\big)dt$ be the expected upward gap above score x in caste c, with F_c^θ the distribution of $\theta \mid c$. For an upper-caste student with score $\theta_i = x$, the transmission term of equation (5) averages $\beta H_U(x)$ in a same-caste room and $\beta\lambda_M H_L(x)$ in a mixed room, so the forgone transmission is*

$$T(x) \equiv \beta\big[H_U(x) - \lambda_M H_L(x)\big] \geq \beta(1-\lambda_M)H_U(x) \geq 0,$$

where the inequality uses $H_L \leq H_U$, which follows from first-order stochastic dominance of $\theta \mid U$ over $\theta \mid L$ (Assumption 1). Moreover T is weakly decreasing in x: $T'(x) = -\beta\big[(1 - F_U^\theta(x)) - \lambda_M(1 - F_L^\theta(x))\big] \leq -\beta(1-\lambda_M)(1 - F_U^\theta(x)) \leq 0$, using $F_L^\theta \geq F_U^\theta$. The mixing cost $\mu_U d_M$ does not depend on x. The model therefore predicts that the upper-caste grade loss from forgone transmission is largest for students with low own scores and falls with own score, on top of a constant $\mu_U d_M$. A supermodular complementarity between own and roommate scores, which is not in the model, would instead make the loss rise with own score.

C.3 The social life of integration

Lemma 1 (Allocation of time). *Fix student i with roommate j and $\tau_i \neq 0$, and let*

$$\bar{B}_i \equiv \max\{\lambda_{ij},\, \iota_i + \tau_i \underline{s}\}, \qquad B_i^{lo} \equiv \min\{\lambda_{ij},\, \iota_i + \tau_i \underline{s}\}.$$

(a) *Choosing $s \in [\underline{s}, 1]$ is equivalent to choosing belonging $B \in [B_i^{lo}, \bar{B}_i]$ to maximize $\Phi_i(B) - (\chi_i/\tau_i)B$, a strictly concave problem. Here $\bar{B}_i = \lambda_{ij}$ if $\tau_i > 0$ (reached at $s = 1$) and $\bar{B}_i = \iota_i(1 - \underline{s}) + \lambda_{ij}\underline{s}$ if $\tau_i < 0$ (reached at $s = \underline{s}$).*

(b) *Let B_i^χ maximize $\Phi_i(B) - (\chi_i/\tau_i)B$ over $[0,1]$: it solves $\Phi_i'(B_i^\chi) = \chi_i/\tau_i$ when interior, and equals the frictionless optimum $B_i^\dagger$ ($v' + c_R = \pi_{\text{eff},i}$) when $\chi_i = 0$. Then $B_i^* = \min\{\max\{B_i^\chi, B_i^{lo}\}, \bar{B}_i\}$.*

(c) *Define $\underline{\pi}_i \equiv v'(\bar{B}_i) + c_R(1 - \bar{B}_i; \Delta_i) - \chi_i/\tau_i$. Then $B_i^\chi \geq \bar{B}_i$ if and only if $\pi_{\text{eff},i} \leq \underline{\pi}_i$. In that case the student is at their ceiling: $s_i^* = 1$ if $\tau_i > 0$ and $s_i^* = \underline{s}$ if $\tau_i < 0$.*

Proof. (a) Utility equals $\Phi_i(B_i(s)) - \chi_i s$ plus terms independent of s. For $\tau_i \neq 0$, $B_i(s)$ is affine and strictly monotone, so $s = (B - \iota_i)/\tau_i$ and $-\chi_i s = -(\chi_i/\tau_i)B + \chi_i\iota_i/\tau_i$. The interval $s \in [\underline{s}, 1]$ maps to the interval between $B_i(\underline{s}) = \iota_i + \tau_i\underline{s}$ and $B_i(1) = \lambda_{ij}$. If $\tau_i > 0$, $\iota_i + \tau_i\underline{s} \leq \lambda_{ij}$; if $\tau_i < 0$ the inequality reverses. Strict concavity follows from $\Phi_i'' < 0$ (M1) and linearity of the friction term.

(b) A strictly concave function on an interval is maximized at the projection of its unconstrained maximizer on the interval.

(c) $\Phi_i'(B) - \chi_i/\tau_i$ is strictly decreasing, so the maximizer over $[0,1]$ satisfies $B_i^\chi \geq \bar{B}_i$ if and only if $\Phi_i'(\bar{B}_i) \geq \chi_i/\tau_i$. Substituting $\Phi_i'(\bar{B}_i) = v'(\bar{B}_i) + c_R(1 - \bar{B}_i; \Delta_i) - \pi_{\text{eff},i}$ gives $\pi_{\text{eff},i} \leq \underline{\pi}_i$. The ceiling is the point $B_i(1)$ if $\tau_i > 0$ and $B_i(\underline{s})$ if $\tau_i < 0$. □

Regimes. Student i is *belonging-seeking* if $\pi_{\text{eff},i} \le \underline{\pi}_i$, and *bridging-seeking* (interior) if $\underline{s} < s_i^* < 1$ at an interior solution of the first-order condition. By Lemma 1(c), the belonging-seeking regime is exactly the set of students who are bonding-constrained in the sense of §C.6. In a same-caste room $\lambda_{ij} = 1$, $\chi_i = 0$, $\tau_i = 1 - \iota_i > 0$, $\bar{B}_i = 1$ and $\pi_{\text{eff},i} = \pi_{c,i}$, so by (M2) the regime condition is $\pi_{c,i} \le v'(1)$, which Assumption 4 guarantees in lower-caste same-caste rooms. In upper-caste same-caste rooms the condition can fail, in which case the student bridges (Lemma 3). In a mixed room the condition is not automatic for any group. For a room-oriented student it reads $\pi_{c,i}(1-\iota_i) - \pi_{ij}(1-\lambda_M) \le \tau_i\big[v'(\lambda_M) + c_R(1-\lambda_M;\Delta_i)\big] - \chi_i$, and for a club-oriented student $\pi_{ij}(1-\lambda_M) - \pi_{c,i}(1-\iota_i) \le |\tau_i|\big[v'(\bar{B}_i) + c_R(1-\bar{B}_i;\Delta_i)\big] + \chi_i$: the network value of room time must not exceed the belonging the student gives up, net of friction.

Lemma 2 (Contact and belonging in the belonging-seeking regime)**.** *Suppose both members of a pair are belonging-seeking. Then desired contact is $s_i^* = 1$ if $\tau_i > 0$ and $s_i^* = \underline{s}$ if $\tau_i \le 0$, and realized contact is $\tilde{s}_{ij} = \min\{s_i^*, s_j^*\}$. In a mixed room, where $\tau_i > 0$ means i is room-oriented (Definition 4) and $\tau_i \le 0$ club-oriented:*

(a) *both room-oriented: $\tilde{s}_{ij} = 1$ and $B_i = \lambda_M$;*

(b) *i room-oriented, j club-oriented: $\tilde{s}_{ij} = \underline{s}$ and $B_i = \iota_i(1-\underline{s}) + \lambda_M\underline{s} < \lambda_M$, equal to $\lambda_M\underline{s}$ if i is club-excluded;*

(c) *i club-oriented: $\tilde{s}_{ij} = \underline{s}$ and $B_i = \iota_i(1-\underline{s}) + \lambda_M\underline{s} \ge \lambda_M$.*

In a same-caste room both students have $\tau > 0$. If both are belonging-seeking, as under Assumption 4 in lower-caste rooms, then $\tilde{s}_{ij} = 1$, $B_i = 1$, $R_i = 0$, $S_i = v(1)$ and $N_i = A_i$.

Proof. Desired contact follows from Lemma 1(c). Realized belonging is $B_i = \iota_i + \tau_i\tilde{s}_{ij}$. In (a), $\tilde{s}_{ij} = \min\{1,1\} = 1$ and $B_i = \lambda_M$. In (b) and (c), at least one student wants $\underline{s}$, so $\tilde{s}_{ij} = \underline{s}$ and $B_i = \iota_i + \tau_i\underline{s} = \iota_i(1-\underline{s}) + \lambda_M\underline{s}$. This is strictly below λ_M when $\tau_i > 0$ (because $\underline{s} < 1$) and at least λ_M when $\iota_i \ge \lambda_M$. Same-caste rooms follow from $\lambda_{ij} = 1$, $\iota_i < 1$ (M4) and $\chi_i = 0$ with $c(0;\cdot) = 0$. □

Proof of Proposition 2. Let $C_i = v(1) - S_i^{\text{mixed}}$ denote the cost relative to full belonging, which equals $S_i^{\text{same}} - S_i^{\text{mixed}}$ whenever the student attains full belonging in a same-caste room, as lower-caste students do under Assumption 4. In the mixed room the regime of Lemma 2 holds by the maintained condition of part (ii), so

$$C_i = \big[v(1) - v(B_i)\big] + c(1-B_i;\Delta_i) + \chi_i\tilde{s}_{ij},$$

where the three terms are the belonging loss, the bridging cost (which includes the preparation-gap premium $c(R;\Delta_i) - c(R;0)$) and the friction. All three are nonnegative and the cost is decreasing in B_i at fixed $\tilde{s}_{ij}$ and Δ_i.

Part (i). Let $\iota_i = \iota(c_i, A_i) \mathbb{1}[A_i \geq \underline{A}_{c_i}]$ be effective alignment. Student i is room-oriented if and only if $A_i \in D_c \equiv \{x : \iota(c,x)\mathbb{1}[x \geq \underline{A}_c] < \lambda_M\}$, a down-set since effective alignment is nondecreasing in x. By (M4), $\iota(U,x) \geq \iota(L,x)$ at every advantage level x, so $\iota(L,x) < \lambda_M$ whenever $\iota(U,x) < \lambda_M$, i.e. $D_U \subseteq D_L$. Let F_c be the advantage distribution of caste c. Then

$$F_L(D_L) \geq F_L(D_U) \geq F_U(D_U),$$

the first step because $D_U \subseteq D_L$ and the second by first-order stochastic dominance of $A \mid U$ over $A \mid L$ (Assumption 1), which gives $F_L(D) \geq F_U(D)$ for every down-set D. The share room-oriented is $F_c(D_c)$, so it is weakly larger for lower-caste students.

Part (ii). Fix a caste and a preparation gap Δ; all students below share $\chi = \chi_c(d_M)$. Write C_a, C_b, C_c for the cost in cases (a), (b), (c) of Lemma 2. In (b) and (c), $\tilde{s}_{ij} = \underline{s}$ and C is the same decreasing function of B, with $B_b < \lambda_M \leq B_c$; hence $C_b > C_c$. For (a) against (c),

$$C_a - C_c = \big[v(B_c) - v(\lambda_M)\big] + \big[c(1-\lambda_M;\Delta) - c(1-B_c;\Delta)\big] + \chi(1-\underline{s}) \geq 0,$$

since $B_c \geq \lambda_M$ and $\underline{s} < 1$. For (b) against (a), take a student in case (b), who is belonging-seeking with $\tau_i > 0$, so $G_i'(1) = \tau_i \Phi_i'(\lambda_M) - \chi \geq 0$ and, by concavity of $G_i(s) = \Phi_i(B_i(s)) - \chi s$, G_i is increasing on $[\underline{s}, 1]$. Hence $G_i(1) \geq G_i(\underline{s})$, i.e. $\Phi_i(\lambda_M) - \chi \geq \Phi_i(B_b) - \chi\underline{s}$. Expanding $\Phi_i(B) = v(B) - \pi_{\text{eff},i} B - c(1-B;\Delta_i)$,

$$\big[v(\lambda_M) - v(B_b)\big] + \big[c(1-B_b;\Delta) - c(1-\lambda_M;\Delta)\big] - \chi(1-\underline{s}) \geq \pi_{\text{eff},i}(\lambda_M - B_b).$$

The left side equals $C_b - C_a$. These comparisons hold the student's own type and π_{ij} fixed, varying only the roommate's willingness to match. If $\pi_{\text{eff},i} \geq 0$, that is if $\pi_{c,i}(1-\iota_i) \geq \pi_{ij}(1-\lambda_M)$ (club time carries at least the bridging value of room time), then $C_b - C_a \geq 0$. Together, $C_b \geq C_a \geq C_c$: the cost is weakly larger for room-oriented than for club-oriented students, and largest when the roommate is club-oriented. The inequalities $C_a \geq C_c$ and $C_b > C_c$ do not use $\pi_{\text{eff},i}$. If $\pi_{\text{eff},i} < 0$, room time carries more network value than club time, the student accepts friction in the room partly for network capital, and a roommate's withdrawal saves friction, so $C_b - C_a$, which equals the left side above, is not signed without a bound on the friction $\chi(1-\underline{s})$ relative to the belonging and bridging-cost loss. The same inequality gives the wellbeing statement: since S_i^{same} is unchanged, $S_b \leq S_a - \pi_{\text{eff},i}(\lambda_M - B_b)$, so for $\pi_{\text{eff},i} \geq 0$ a room-oriented student's wellbeing weakly falls, strictly if $\pi_{\text{eff},i} > 0$, once their roommate withdraws. A roommate withdraws if and only if $\tau_j \leq 0$, i.e. $\iota_j \geq \lambda_M$. Network capital changes by $V_b - V_a = (1-\underline{s})\big[\pi_{c,i}(1-\iota_i) - \pi_{ij}(1-\lambda_M)\big] = (1-\underline{s})\tau_i \pi_{\text{eff},i}$, which has the sign of $\pi_{\text{eff},i}$; if $\pi_{c,i} = \pi_{ij}$ it equals $\pi_{ij}(\lambda_M - \iota_i)(1-\underline{s}) \geq 0$.

For a club-oriented student, $1 - B_c = (1-\iota_i) + (\iota_i - \lambda_M)\underline{s} \to \underline{s}(1-\lambda_M)$ as $\iota_i \to 1$, so the belonging loss is bounded by $v'(B_c)(1-B_c)$ by concavity and is first order in $(1-\iota_i)$ and $\underline{s}$, as is the friction $\chi\underline{s}$. By (M2), $c(R;\Delta) \leq \frac{1}{2}\bar{c}\,R^2$ with $\bar{c} = \sup c_{RR}$, so the bridging cost is second order. For high alignment, therefore, the cost is small and arises from the minimum level of forced contact.

The three components cannot be ranked in general for room-oriented students: in case (a) the cost is $[v(1) - v(\lambda_M)] + c(1 - \lambda_M; \Delta) + \chi$, and the model does not order these terms.

Part (iii). Suppose student i's optimum is interior, $\underline{s} < s_i^* < 1$, and realized, $\tilde{s}_{ij} = s_i^*$ (the partner is willing to match). Then $\Phi_i'(B^*) = \chi_i/\tau_i$, i.e. $v'(B^*) + c_R(1 - B^*; \Delta_i) = \chi_i/\tau_i + \pi_{\text{eff},i}$. For any parameter ϑ that shifts only $\pi_{\text{eff},i}$, differentiation gives $\Phi_i'' \, dB^*/d\vartheta = \partial \pi_{\text{eff},i}/\partial \vartheta$, so $dB^*/d\vartheta = (1/\Phi_i'') \, \partial \pi_{\text{eff},i}/\partial \vartheta$ and $dR^*/d\vartheta = -dB^*/d\vartheta$. For wellbeing $S_i = v(B) - c(1 - B; \Delta_i) - \chi_i s$, using $ds^* = dB^*/\tau_i$ and $v' + c_R = \chi_i/\tau_i + \pi_{\text{eff},i}$ at the optimum,

$$dS_i = \big[v' + c_R - \chi_i/\tau_i\big] dB^* = \pi_{\text{eff},i} \, dB^* = -\pi_{\text{eff},i} \, dR_i.$$

At the margin the wellbeing loss therefore equals the effective value of the bridging gained. The sources of variation are $\pi_{c,i}$ and π_{ij}, with $\partial \pi_{\text{eff},i}/\partial \pi_{c,i} = (1 - \iota_i)/\tau_i$ and $\partial \pi_{\text{eff},i}/\partial \pi_{ij} = -(1 - \lambda_M)/\tau_i$. For a room-oriented student ($\tau_i > 0$), bridging rises in the club network value $\pi_{c,i}$ and falls in the roommate's network value π_{ij}, so if $\pi_{\text{eff},i} > 0$ wellbeing falls in $\pi_{c,i}$ and rises in π_{ij}. For a club-oriented student ($\tau_i < 0$) both signs reverse. Network capital is $N_i = A_i + \pi_{c,i}(1 - \iota_i) + \kappa_i s$ with $\kappa_i = -\tau_i \pi_{\text{eff},i}$, so through the reallocation it changes by $\kappa_i \, ds^* = -\pi_{\text{eff},i} \, dB^* = \pi_{\text{eff},i} \, dR^*$, plus the direct effects $(1 - \iota_i)(1 - s^*) \, d\pi_{c,i}$ and $(1 - \lambda_M) s^* \, d\pi_{ij}$. In the club network value this is positive for a room-oriented student with $\pi_{\text{eff},i} \geq 0$; in the roommate's network value the reallocation effect is negative and the sign is ambiguous. The regime requires $\pi_{\text{eff},i} > \underline{\pi}_i$, whose explicit form for each orientation is given in the discussion of regimes.

Part (iv). Under the regime, belonging B_i in Lemma 2 does not depend on θ_j, so

$$\frac{\partial S_i}{\partial \theta_j} = -c_\Delta(1 - B_i; \Delta_i) \mathbb{1}[\theta_j > \theta_i] \leq 0,$$

and the regime persists as θ_j rises, since Φ_i' contains $c_R(1 - \bar{B}_i; \Delta_i)$, which is nondecreasing in Δ_i by $c_{R\Delta} \geq 0$. Wellbeing therefore falls in the roommate's score exactly on pairs in which the roommate outscores the student, which for lower-caste students occurs with probability p and for upper-caste students only on the reversal event, with probability $1 - p$; on all other pairs the slope is zero. The slope is steeper for room-oriented students because $c_\Delta(R; \Delta)$ is nondecreasing in R by $c_{R\Delta} \geq 0$ and $1 - B_i$ is larger in cases (a) and (b) than in case (c): $1 - B_a = 1 - \lambda_M$, $1 - B_b > 1 - \lambda_M$ and $1 - B_c \leq 1 - \lambda_M$. □

Remark (Baseline specification: a single return to bridging)**.** *The main text sets $\pi_{c,i} = \pi_{ij}$, so that $V_i = \pi_{ij} R_i$ and $\pi_{\text{eff},i} = \pi_{ij} \geq 0$. Then the following statements used in the main text hold.*

(a) *Every club-oriented student in a mixed room whose roommate satisfies $\pi_{ij} \leq v'(1)$ is belonging-seeking, so the belonging-seeking condition in Proposition 2(ii) holds automatically for them.*

(b) *The cost ordering in Proposition 2(ii) holds without further condition, including that the cost is largest when the roommate is club-oriented, and a roommate's withdrawal lowers wellbeing and raises network capital by $\pi_{ij}(\lambda_M - \iota_i)(1 - \underline{s}) \geq 0$.*

(c) *For roommates with $\pi_{ij} \leq v'(1)$, the bridging-seeking regime of Proposition 2(iii) is restricted to room-oriented students with $\chi_i/\tau_i > v'(\lambda_M) + c_R(1-\lambda_M;\Delta_i) - \pi_{ij} > 0$. In it, $dB^*/d\pi_{ij} = 1/\Phi_i'' < 0$, $dR^*/d\pi_{ij} > 0$, $dN_i/d\pi_{ij} = R_i - \pi_{ij}/\Phi_i'' > 0$ and $dS_i = -\pi_{ij}\, dR_i < 0$.*

Proof of Remark C.3. (a) For a club-oriented student, $\tau_i < 0$ and $\bar{B}_i = \iota_i(1-\underline{s}) + \lambda_M \underline{s} < 1$, because $\iota_i < 1$ and $\lambda_M < 1$. Hence $\underline{\pi}_i = v'(\bar{B}_i) + c_R(1-\bar{B}_i;\Delta_i) + \chi_i/|\tau_i| \geq v'(\bar{B}_i) > v'(1) \geq \pi_{ij} = \pi_{\text{eff},i}$, using strict concavity of v and $\pi_{ij} \leq v'(1)$. (b) is the proof of Proposition 2(ii) with $\pi_{\text{eff},i} = \pi_{ij} \geq 0$. (c) For a club-oriented student, (a) gives $\pi_{ij} \leq \underline{\pi}_i$. For a room-oriented student, $\bar{B}_i = \lambda_M$, so $\pi_{ij} > \underline{\pi}_i$ requires $\chi_i/\tau_i > v'(\lambda_M) + c_R(1-\lambda_M;\Delta_i) - \pi_{ij} \geq v'(\lambda_M) - v'(1) > 0$, using $c_R \geq 0$ and $\pi_{ij} \leq v'(1)$. The comparative statics are those of the proof of Proposition 2(iii) with $\partial \pi_{\text{eff},i}/\partial \pi_{ij} = 1$. □

Placement.

Lemma 3 (Bridging in same-caste rooms)**.** *Under the single return, in a same-caste room student i desires full contact if and only if $\pi(A_{j'}) \leq v'(1)$. Realized contact is $\tilde{s} = \min\{s_i^*, s_{j'}^*\}$, so the student's bridging $R_i^{\text{same}} = (1-\iota_i)(1-\tilde{s})$ is positive if and only if $\max\{\pi(A_{j'}), \pi(A_i)\} > v'(1)$. Under Assumption 4, $R_i^{\text{same}} = 0$ for lower-caste students, and the probability that $R_i^{\text{same}} > 0$ is weakly higher in upper-caste than in lower-caste rooms.*

Proof. In a same-caste room $\lambda_{ij} = 1$, $\chi_i = 0$, $\tau_i = 1 - \iota_i > 0$ and $\bar{B}_i = 1$, so the threshold of Lemma 1(c) is $\underline{\pi}_i = v'(1) + c_R(0;\Delta_i) = v'(1)$ by (M2), and student i, whose return is $\pi_{ij} = \pi(A_{j'})$, desires full contact if and only if $\pi(A_{j'}) \leq v'(1)$. The roommate's return is $\pi(A_i)$, so the roommate desires full contact if and only if $\pi(A_i) \leq v'(1)$. Realized contact is the minimum of the two, and $B_i = \iota_i + (1-\iota_i)\tilde{s}$ gives $R_i^{\text{same}} = 1 - B_i = (1-\iota_i)(1-\tilde{s})$, which is positive if and only if $\tilde{s} < 1$. Under Assumption 4 both returns are at most $v'(1)$ in a lower-caste room. Let $p_c = \Pr(\pi(A) > v'(1) \mid c)$. Since $A \mid U$ first-order stochastically dominates $A \mid L$ and π is nondecreasing, $p_U \geq p_L$, and with independent draws within a room $\Pr(R^{\text{same}} > 0) = 1 - (1-p_c)^2$ is weakly higher for $c = U$. □

Proof of Proposition 3. Decomposition. Network capital is $N_i = A_i + V_i$, with $V_i = \pi_{ij} R_i$ under the single return. In the mixed room $V_i = \pi(A_j) R_i \geq 0$ whatever the allocation, because $R_i = 1 - B_i \geq 0$ and $\pi \geq 0$. In a same-caste room $V_i^{\text{same}} = \pi(A_{j'}) R_i^{\text{same}} \geq 0$, which is zero for lower-caste students by Lemma 3 and can be positive for upper-caste students. Hence

$$\Delta N_i = \Delta V_i \equiv \pi(A_j) R_i - \pi(A_{j'}) R_i^{\text{same}}.$$

Grades are fixed by the assignment, so $\Delta Y_i = \Delta V_i + \pi_g \Delta g_i$ and $\mathbb{E}_c[\Delta Y_i] = \mathbb{E}_c[\Delta V_i] + \pi_g \bar{g}_c$.

Part (i). For lower-caste students $R_i^{\text{same}} = 0$, so $\Delta V_i = \pi(A_j) R_i \geq 0$ and $\mathbb{E}_L[\Delta Y_i] = \mathbb{E}_L[\pi(A_j) R_i] + \pi_g \bar{g}_L \geq 0$ whenever $\bar{g}_L \geq 0$.

Part (ii). For upper-caste students, $\mathbb{E}_U[\Delta Y_i] = \mathbb{E}_U[\Delta V_i] + \pi_g \bar{g}_U$, which is negative if and only if $\mathbb{E}_U[\Delta V_i] < -\pi_g \bar{g}_U$. The bridging term ΔV_i has no sign: it is positive when mixed-room bridging is valuable, and negative when the same-caste bridging $\pi(A_{j'}) R_i^{\text{same}}$ is large, which requires an advantaged roommate with $\pi(A_{j'}) > v'(1)$, more common in upper-caste rooms (Lemma 3). The

signs of $\bar{g}_L$ and $\bar{g}_U$ are given by Proposition 1 and Corollary 1. If bridging in same-caste rooms is zero for everyone, $\Delta V_i = \pi(A_j)R_i \geq 0$ for all students. □

C.4 Extension: venue-specific returns to bridging

The main text values all bridging at the roommate's network return π_{ij}. This extension lets bridging generated in clubs earn a constant return $\pi_{c,i}$ (written π_c where it does not vary across students) and works out what changes. The notation of Appendix C.1 applies: network capital is $N_i = A_i + V_i$ with $V_i = \pi_{ij}(1-\lambda_{ij})s + \pi_{c,i}(1-\iota_i)(1-s)$, and the return governing time allocation is the effective return $\pi_{\text{eff},i}$, the network value gained per unit of belonging given up. It satisfies

$$\pi_{\text{eff},i} - \pi_{ij} = \frac{(\pi_{c,i} - \pi_{ij})(1-\iota_i)}{\tau_i},$$

so it equals π_{ij} when $\pi_{c,i} = \pi_{ij}$, and equals $\pi_{c,i}$ in a same-caste room.

Condition on club composition. I consider $\bar{\pi}_L \leq \pi_c \leq \bar{\pi}_U$, where $\bar{\pi}_c$ is the mean network value of a caste-c roommate. This is an empirical condition on club composition: club peers are on average a more valuable network than a lower-caste roommate, which is what an upper-caste student has in a mixed room, and a less valuable one than an upper-caste roommate, which is what a lower-caste student has. Each half matters for one caste. The second half can fail if club-access thresholds are high enough that club members are on average more advantaged than upper-caste students, and a direct credential value of club activity for placement would act like a higher π_c and could push it above $\bar{\pi}_U$. For club-excluded students, whose time outside the room does not reach club peers, $\pi_{c,i}$ is lower.

Proposition 6 (Wellbeing under venue-specific returns)**.** *Fix student i with $\tau_i \neq 0$ and suppose $\underline{\pi}_i \geq 0$, so that friction alone would not lead the student to reduce contact below the belonging-maximizing level. Let S_i^* denote wellbeing at the optimal allocation.*

(i) *S_i^* is weakly decreasing in $\pi_{\text{eff},i}$, and constant for $\pi_{\text{eff},i} \leq \underline{\pi}_i$.*

(ii) *$\pi_{\text{eff},i} \leq \pi_{ij}$ if and only if $(\pi_{c,i} - \pi_{ij})\,\tau_i \leq 0$.*

(iii) *Relative to the single-return model ($\pi_{c,i} = \pi_{ij}$), for a given roommate, wellbeing is weakly lower and the mixed-room cost C_i weakly higher if $\pi_{\text{eff},i} \geq \pi_{ij}$, and wellbeing is weakly higher and C_i weakly lower if $\pi_{\text{eff},i} \leq \pi_{ij}$.*

Proof. Write wellbeing as a function of belonging, $\tilde{S}(B) = v(B) - c(1-B;\Delta_i) - \chi_i(B-\iota_i)/\tau_i$, which is concave. The student maximizes $\tilde{S}(B) - \pi_{\text{eff},i}B$ over the feasible interval of belonging, whose upper end is the ceiling $\bar{B}_i$. The derivative of $\tilde{S}$ at the ceiling is $v'(\bar{B}_i) + c_R(1-\bar{B}_i;\Delta_i) - \chi_i/\tau_i = \underline{\pi}_i \geq 0$, so by concavity $\tilde{S}$ is increasing below the ceiling. The objective has decreasing differences in $(B, \pi_{\text{eff},i})$, so the optimal B^* is weakly decreasing in $\pi_{\text{eff},i}$, and by Lemma 1(c) it equals $\bar{B}_i$ for

$\pi_{\text{eff},i} \leq \underline{\pi}_i$. Hence $S_i^* = \tilde{S}(B^*)$ is weakly decreasing in $\pi_{\text{eff},i}$ and constant on $\pi_{\text{eff},i} \leq \underline{\pi}_i$, which is part (i). Part (ii) follows from the identity above, since $1 - \iota_i > 0$. For part (iii), $C_i = v(1) - S_i^*$ moves opposite to S_i^*, and part (i) gives the comparison. □

Effects by type. Combining parts (ii) and (iii) with the condition on club composition gives the following. The signs hold for a roommate on the typical side of π_c: upper-caste students have a lower-caste roommate, with $\bar{\pi}_L \leq \pi_c$, and lower-caste students have an upper-caste roommate, with $\bar{\pi}_U \geq \pi_c$.

Group	$\pi_{\text{eff},i}$ vs π_{ij}	Wellbeing vs single return	Cost C_i
Upper-caste, club-oriented ($\tau_i < 0$)	lower	unchanged (at the corner)	unchanged
Upper-caste, room-oriented ($\tau_i > 0$)	higher	weakly lower	weakly higher
Lower-caste, room-oriented ($\tau_i > 0$)	lower	unchanged or weakly higher	unchanged or weakly lower
Lower-caste, club-oriented ($\tau_i < 0$)	higher	weakly lower	weakly higher

When the difference matters. The venue-specific return changes wellbeing only for students whose allocation changes, that is, whose effective return crosses the threshold $\underline{\pi}_i$. The two types with a lower effective return stay at, or move back toward, the belonging-maximizing corner. The two types with a higher effective return face a stronger conflict between belonging and network value: upper-caste room-oriented students, for whom belonging favors the room but clubs offer a more valuable network than a lower-caste roommate, and lower-caste club-oriented students, for whom belonging favors clubs but an upper-caste roommate offers a more valuable network than club peers. Such students may trade wellbeing for network value, and their total payoff is higher for it. The effects are concentrated among students with τ_i near zero, for whom belonging depends little on the venue and network value decides it. In the main text's single-return model, the effective return equals π_{ij} and the statements of Proposition 2 hold as stated (Remark C.3).

Cost ordering. The claim in Proposition 2(ii) that the cost is largest for a room-oriented student paired with a club-oriented roommate holds under the single return without further condition. Under venue-specific returns, a sufficient condition is $\pi_{\text{eff},i} \geq 0$, which for a room-oriented student is $\pi_{c,i}(1 - \iota_i) \geq \pi_{ij}(1 - \lambda_M)$: club time carries at least the bridging value of room time. It can fail for lower-caste room-oriented students, whose effective return is lower, especially club-excluded

students with $\iota_i = 0$ if $\pi_{c,i} < \pi_{ij}(1 - \lambda_M)$. The condition is sufficient, not necessary: the claim holds whenever the belonging and bridging-cost loss from withdrawal exceeds the friction saved.

Placement. The placement results of Proposition 3 are stated in terms of the net value of bridging ΔV_i and hold for any nonnegative valuation of bridging, provided lower-caste same-caste rooms involve no bridging. Relative to the single return, the value of mixed-room bridging differs from $\pi_{ij}R_i$ by $(\pi_{c,i} - \pi_{ij})(1 - \iota_i)(1 - \tilde{s}_{ij})$, which is zero for students at full room contact or with full club alignment. Under the condition above the single return understates the bridging value of upper-caste students and overstates that of lower-caste students, which does not affect the sign of the lower-caste result.

C.5 Extension: access and competitive placement

Access to internships. Internships differ in how access is allocated. Non-competitive internships are open to all: a student is placed if and only if the index Y_i in equation (7) exceeds a fixed threshold $\bar{y}$. Competitive internships have fixed capacity q: a student is placed if and only if $Y_i \geq \bar{y}^c$, where $\bar{y}^c$ is the $(1 - q)$ quantile of Y in the cohort. Placement is an outcome rather than a choice margin: students do not choose their time allocation in anticipation of winning a rationed, discrete prize. Let Y_i^s denote the index of student i in the same-caste room.

Corollary 2 (Access to internships)**.** *At a fixed threshold $\bar{t} \in \{\bar{y}, \bar{y}^c\}$:*

(i) *mixed assignment creates access if and only if $Y_i^s < \bar{t} \leq Y_i^s + \Delta Y_i$, and removes it if and only if $Y_i^s \geq \bar{t} > Y_i^s + \Delta Y_i$, so students whose same-caste index lies farther than $|\Delta Y_i|$ from the threshold are unaffected;*

(ii) *a positive (negative) expected change in the index in Proposition 3 does not imply that every student in the group gains (loses) access, since grades and roommate draws generate dispersion in ΔY_i;*

(iii) *the competitive threshold is endogenous: absent ties, $\sum_i \mathbf{1}\{Y_i \geq \bar{y}^c\} = qn$ for every room allocation, so changes in competitive placements sum to zero across the cohort.*

Proof of Corollary 2. Student i is placed at fixed threshold $\bar{t}$ if and only if the index is at least $\bar{t}$. The index is Y_i^s in the same-caste room and $Y_i^s + \Delta Y_i$ in the mixed room, so access is created if and only if $Y_i^s < \bar{t} \leq Y_i^s + \Delta Y_i$ and removed if and only if $Y_i^s \geq \bar{t} > Y_i^s + \Delta Y_i$, which proves part (i). Part (ii) follows because the expected change is an average of ΔY_i over students, which varies with grades and roommate draws, and access depends on the realized position of Y_i^s relative to the threshold. For part (iii), $\bar{y}^c$ is the $(1 - q)$ quantile of the cohort distribution of Y, so absent ties a share q of the cohort satisfies $Y_i \geq \bar{y}^c$ under every allocation of rooms. □

Remark (Where the thresholds sit)**.** *Access changes only for students near a threshold, so where the thresholds sit determines who is affected. If the non-competitive threshold sits low in the index distribution,*

where lower-caste students are over-represented, gains in the index matter mainly for them. If the competitive threshold sits in the upper tail, where upper-caste students are over-represented, a fall in the index shows up as lost competitive placements, while a gain for lower-caste students, who sit far below it, is unlikely to move them across it. The model then predicts that lower-caste gains concentrate in non-competitive internships and upper-caste losses in competitive ones. That the thresholds sit where this argument requires is an empirical fact, documented in Section 3 (placement rates by caste and internship type), not an implication of the model. Statements about the probability of placement for the full index require that the joint distribution of roommate advantage and entrance score be ordered by caste in the multivariate stochastic order, which Assumption 1 does not imply, and statements at a fixed competitive threshold are partial-equilibrium.

Organized recruitment. As competition for elite internships increases, recruitment is organized through the placement cell, which changes what bridging is worth. First, information about openings is public, so the roommate's network adds less as a source of information and the value of bridging works instead through the hidden curriculum of recruitment, such as practicing for selection tests, technical interviews and mock interviews, and reviewing CVs. Second, eligibility to apply is restricted by grades and by specialization. I capture the first channel by scaling the information value of bridging by $\rho \in [0, 1]$ and the hidden curriculum by a reduced-form complementarity, and the second by an eligibility rule.

Assumption 9 (Competitive placement)**.** *The competitive index is*

$$Y_i^c = A_i + \rho\, V_i + \pi_g\, g_i + \lambda_{ij}\, A_i\, \pi_{ij}, \qquad \rho \in [0, 1], \tag{15}$$

where V_i is the value of bridging. The last term (a) is proportional to the roommate's network value π_{ij}, so that a more advantaged roommate provides more support; (b) is complementary to the student's own advantage $A_i \geq 0$, which shapes the ability to translate support into competitive outcomes; and (c) is scaled by the alignment of the relationship λ_{ij}, since support from a more aligned relationship is more productive. A student is eligible to apply if and only if the student's specialization is eligible, $E_i = 1$, which is predetermined and does not depend on the room assignment, and grades satisfy $g_i \geq \bar{g}$. An eligible student obtains a competitive internship if and only if $Y_i^c \geq \bar{y}^c$, where $\bar{y}^c$ is the $(1 - q)$ quantile of Y^c among eligible students.

In same-caste rooms $\lambda_{ij} = 1$, whereas in mixed rooms $\lambda_{ij} = \lambda_M < 1$. The terms in Y_i^c and the eligibility rule do not depend on realized contact, and alignment depends on the assignment rather than on contact, so the marginal return to bridging is unchanged and the time-allocation problem is unaffected. The complementarity is a reduced-form supermodularity assumption, not a consequence of the time-allocation model; other mechanisms, such as status competition in recruiting, could generate a similar pattern. Let j and j' denote the roommate in the mixed room and in the counterfactual same-caste room, Δ mixed minus same-caste outcomes, $\bar{\pi}_c \equiv \mathbb{E}[\pi(A) \mid c]$ the mean network value of a caste-c roommate and $\bar{A}_c \equiv \mathbb{E}[A \mid c]$ mean advantage. Because $\bar{\pi}_L \leq \bar{\pi}_U$ ($A \mid U$ first-order stochastically dominates $A \mid L$ and π is nondecreasing), mixed assignment low-

ers the value of the roommate's network for upper-caste students and raises it for lower-caste students. Because the complementarity term does not depend on contact, it is a roommate-advantage exposure effect specific to competitive placement: in same-caste rooms it implies a gradient of the competitive outcome in the roommate's advantage that rises with own advantage, whereas the placement index has none.

Proposition 7 (Competitive internships under organized recruitment)**.** *Under the assumptions of Proposition 3 and Assumption 9:*

(i) *The expected effect of mixed assignment on the competitive index relative to the placement index is*

$$\mathbb{E}_U\big[\Delta Y_i^c - \Delta Y_i\big] = -(1-\rho)\,\mathbb{E}_U[\Delta V_i] - \bar{A}_U\big(\bar{\pi}_U - \lambda_M \bar{\pi}_L\big),$$
$$\mathbb{E}_L\big[\Delta Y_i^c - \Delta Y_i\big] = -(1-\rho)\,\mathbb{E}_L[\Delta V_i] + \bar{A}_L\big(\lambda_M \bar{\pi}_U - \bar{\pi}_L\big).$$

(ii) Lower-caste students. *The information term is nonpositive, since $\Delta V_i \geq 0$, so public information lowers the competitive return to the bridging that mixing induces. The complementarity term is nonnegative if and only if $\lambda_M \bar{\pi}_U \geq \bar{\pi}_L$, that is, if the alignment discount in mixed rooms does not outweigh the greater value of the roommate's network. The sign of the total is therefore not determined by the model.*

(iii) Upper-caste students. *The complementarity term is nonpositive. The information term is nonpositive if $\mathbb{E}_U[\Delta V_i] \geq 0$ and nonnegative if $\mathbb{E}_U[\Delta V_i] < 0$, in which case public information offsets part of the loss from a less valuable network. Hence the expected change in the competitive index is weakly lower than in the placement index whenever $\mathbb{E}_U[\Delta V_i] \geq 0$.*

(iv) Eligibility. *Competitive access can change only for students with $E_i = 1$. Mixed assignment creates eligibility for student i if and only if $g_i^s < \bar{g} \leq g_i^s + \Delta g_i$ and removes it if and only if $g_i^s \geq \bar{g} > g_i^s + \Delta g_i$, where g_i^s is the grade in the same-caste room. Since $\bar{g}_U \leq 0$ and $\bar{g}_L \geq 0$ under the conditions of Corollary 1, upper-caste students near the cutoff lose eligibility and lower-caste students near it gain it.*

Proof. Since $\lambda_{ij} = 1$ in a same-caste room and $\lambda_{ij} = \lambda_M$ in a mixed room, Assumption 9 gives

$$\Delta Y_i^c - \Delta Y_i = -(1-\rho)\,\Delta V_i + A_i\big[\lambda_M \pi(A_j) - \pi(A_{j'})\big],$$

because $Y_i^c - Y_i = -(1-\rho)V_i + \lambda_{ij} A_i \pi_{ij}$ and V_i is the value of bridging in each room type. By random assignment (M5), own advantage is independent of the roommate draw, so $\mathbb{E}[A_i(\lambda_M \pi(A_j) - \pi(A_{j'})) \mid c] = \bar{A}_c(\lambda_M \mathbb{E}[\pi(A_j)] - \mathbb{E}[\pi(A_{j'})])$. For an upper-caste student the mixed-room roommate is lower-caste and the same-caste roommate is upper-caste, so the bracket averages $\lambda_M \bar{\pi}_L - \bar{\pi}_U$; for a lower-caste student it averages $\lambda_M \bar{\pi}_U - \bar{\pi}_L$. This gives part (i). For part (ii), $\Delta V_i \geq 0$ for lower-caste students by Proposition 3, so $-(1-\rho)\mathbb{E}_L[\Delta V_i] \leq 0$ since $\rho \leq 1$. For part (iii), $\bar{\pi}_U - \lambda_M \bar{\pi}_L \geq (1-\lambda_M)\bar{\pi}_U \geq 0$ because $\bar{\pi}_L \leq \bar{\pi}_U$ and $\lambda_M < 1$, and the information term has

the sign of $-\mathbb{E}_U[\Delta V_i]$. Part (iv) follows because the specialization rule is predetermined and the grade rule is a threshold on g_i, which equals g_i^s in the same-caste room and $g_i^s + \Delta g_i$ in the mixed room. □

Implications by group. The three features of organized recruitment affect the two groups differently. For lower-caste students, public information lowers the competitive return to bridging, the hidden curriculum helps only if the alignment discount in mixed rooms is mild, and the eligibility cutoff helps those whose grades rise past it, so competitive gains are smaller than non-competitive gains unless the grade channel dominates. For upper-caste students, the hidden curriculum and the eligibility cutoff both amplify losses, since their grades fall and they lose an advantaged same-caste roommate, while public information dampens the part of the loss that comes from a less valuable network. Competitive losses are therefore larger than non-competitive losses for upper-caste students when the grade and complementarity channels dominate.

C.6 Bonding constraints and welfare

A student is *bonding-constrained* if they would choose more belonging than their ceiling allows, $B_i^\chi > \bar{B}_i$ (Lemma 1). By Lemma 1(c), the belonging-seeking regime of §C.3 is exactly the case $B_i^\chi \geq \bar{B}_i$: bonding-constrained students are those whose effective return to bridging, $\pi_{\text{eff},i}$, is too low to pull them off the ceiling. A student is *floor-bound* if $B_i^\chi < B_i^{lo}$: they would choose less belonging than any feasible contact gives.

Proposition 8 (Who is bonding-constrained)**.** *Fix student i with $\tau_i \neq 0$.*

(i) *The friction-adjusted target B_i^χ satisfies $\partial B_i^\chi/\partial \Delta_i = c_{R\Delta}/|\Phi_i''| \geq 0$, $\partial B_i^\chi/\partial \pi_{\text{eff},i} = 1/\Phi_i'' < 0$ and $\partial B_i^\chi/\partial \chi_i = 1/(\tau_i \Phi_i'')$, negative if $\tau_i > 0$ and positive if $\tau_i < 0$. Moreover $B_i^\chi < 1$ if and only if $\pi_{\text{eff},i} > v'(1) - \chi_i/\tau_i$.*

(ii) *Student i is bonding-constrained if and only if $\pi_{\text{eff},i} < \underline{\pi}_i$, and at their ceiling if and only if $\pi_{\text{eff},i} \leq \underline{\pi}_i$. The threshold $\underline{\pi}_i$ is decreasing in the ceiling $\bar{B}_i$ and increasing in Δ_i, holding the friction term fixed. The friction term $-\chi_i/\tau_i$ lowers the threshold for room-oriented students ($\tau_i > 0$) and raises it for club-oriented students ($\tau_i < 0$).*

(iii) *A room-oriented student is bonding-constrained more often the higher the roommate's network value π_{ij} and the lower the club network value $\pi_{c,i}$; for a club-oriented student the dependence reverses.*

Proof. (i) By Lemma 1, an interior B_i^χ solves $F(B; \Delta, \pi_{\text{eff}}, \chi) \equiv \Phi_i'(B) - \chi/\tau = 0$, with $F_B = \Phi_i'' < 0$, $F_\Delta = c_{R\Delta} \geq 0$, $F_{\pi_{\text{eff}}} = -1$ and $F_\chi = -1/\tau$. The implicit function theorem gives $dB/dx = -F_x/F_B$ for each parameter x, which yields the three derivatives. Next, $B_i^\chi < 1$ if and only if $F(1) < 0$, i.e. $\Phi_i'(1) = v'(1) + c_R(0; \Delta_i) - \pi_{\text{eff},i} < \chi_i/\tau_i$, which by (M2) is $\pi_{\text{eff},i} > v'(1) - \chi_i/\tau_i$.

(ii) The first claim is Lemma 1(c). The threshold is $\underline{\pi}_i = v'(\bar{B}_i) + c_R(1 - \bar{B}_i; \Delta_i) - \chi_i/\tau_i$. The first term is decreasing in $\bar{B}_i$ by concavity of v; the second is decreasing in $\bar{B}_i$ because c_R is increasing

in its first argument (convexity) and $1 - \bar{B}_i$ falls; and it is nondecreasing in Δ_i by $c_{R\Delta} \geq 0$. The friction term is negative if $\tau_i > 0$ and positive if $\tau_i < 0$.

(iii) Student i is constrained if and only if $\pi_{\text{eff},i} < \underline{\pi}_i$, and $\underline{\pi}_i$ does not depend on $\pi_{c,i}$ or π_{ij}. Since $\partial \pi_{\text{eff},i}/\partial \pi_{c,i} = (1 - \iota_i)/\tau_i$ and $\partial \pi_{\text{eff},i}/\partial \pi_{ij} = -(1 - \lambda_{ij})/\tau_i$, a higher π_{ij} lowers $\pi_{\text{eff},i}$ and a higher $\pi_{c,i}$ raises it when $\tau_i > 0$, and the signs reverse when $\tau_i < 0$. □

The cross-caste ordering of thresholds in the earlier version required the two students to face the same return π_{ij}. With a venue-specific return, $\pi_{\text{eff},i}$ differs between a lower-caste room-oriented student and an upper-caste club-oriented student with the same roommate advantage, so the comparison depends on $\pi_{c,i} - \pi_{ij}$ and is not signed by the model.

Corollary 3 (Constraint status). (i) *Bonding-constrained students are at their ceiling: $s_i^* = 1$ if $\tau_i > 0$ and $s_i^* = \underline{s}$ if $\tau_i < 0$. Unconstrained students hold interior contact unless they are floor-bound.*

(ii) *A club-excluded student of either caste has $\iota_i = 0$, so $\tau_i = \lambda_M > 0$ and $\bar{B}_i = \lambda_M$. Such a student has $\pi_{\text{eff},i} = \big[\pi_{c,i} - \pi_{ij}(1 - \lambda_M)\big]/\lambda_M$ and is bonding-constrained if and only if $\pi_{\text{eff},i} < v'(\lambda_M) + c_R(1 - \lambda_M; \Delta_i) - \chi_i/\lambda_M$. Caste enters this condition only through Δ_i, χ_i and the values $\pi_{c,i}$ and π_{ij}.*

(iii) *Holding the other determinants fixed, a student is unconstrained if and only if $\Delta_i \leq \Delta_{\max}$, where $\Delta_{\max}$ solves $\underline{\pi}_i(\Delta_{\max}) = \pi_{\text{eff},i}$ (empty if $\underline{\pi}_i(0) > \pi_{\text{eff},i}$).*

Proof. (i) is Lemma 1(c) together with the definition of the ceiling. (ii) By (M4) a club-excluded student has $\iota_i = 0$, so $\tau_i = \lambda_M > 0$ and $\bar{B}_i = \lambda_M$; substitute into $\underline{\pi}_i$. (iii) By Proposition 8(ii), $\underline{\pi}_i$ is nondecreasing and continuous in Δ_i, so $\{\Delta : \underline{\pi}_i(\Delta) \leq \pi_{\text{eff},i}\}$ is an interval containing 0 if nonempty. □

Moved from the welfare section of the main text. For a student at an interior optimum, the welfare effect of a change in roommate alignment λ is identified by a formula in the contact choice, the friction, the bonding pass-through and the gap between the club and roommate network values, with v and c dropping out through the first-order condition (Proposition 9). When the two network values coincide, or $\iota_i = 1$, the gap term vanishes and the formula depends only on χ_i, s^* and τ_i. For students at a corner, the formula is only a bound: welfare is thus exactly measurable for students whose contact is interior, and only bounded for students at a corner, in particular the bonding-constrained.

Proposition 9 (Sufficient statistic for welfare). *Let $\Omega_i(\lambda) \equiv \max_{s \in [\underline{s},1]} G_i(s; \lambda)$ be student i's welfare, net of terms independent of contact, as a function of roommate alignment λ, where*

$$G_i(s; \lambda) \equiv v(B) - c(1 - B; \Delta_i) - \chi_i s + \pi_{ij}(1 - \lambda)s + \pi_{c,i}(1 - \iota_i)(1 - s), \qquad B = \iota_i + (\lambda - \iota_i)s,$$

holding ι_i, χ_i, π_{ij}, $\pi_{c,i}$ and Δ_i fixed, and let $M_i \equiv \big[\chi_i + (\pi_{c,i} - \pi_{ij})(1 - \iota_i)\big]/\tau_i$.

(a) $\Omega_i'(\lambda) = s^* \big[v'(B^*) + c_R(1 - B^*; \Delta_i) - \pi_{ij}\big]$ *exactly.*

(b) *At an interior optimum,* $\Omega_i'(\lambda) = s^* M_i$.

(c) *At a corner at the ceiling (bonding-constrained,* $\Omega_i'(\lambda) \geq s^* M_i$*; at a corner away from the ceiling (floor-bound,* $\Omega_i'(\lambda) \leq s^* M_i$*. The formula therefore understates the welfare value of alignment, and hence the loss from a less aligned roommate, for bonding-constrained students.*

Proof. G_i is C^1 in (s, λ) and strictly concave in s (for $\tau_i \neq 0$), so $s^*(\lambda)$ is unique, and the feasible set $[\underline{s}, 1]$ does not depend on λ. By the envelope theorem, $\Omega_i'(\lambda) = \partial G_i / \partial \lambda$ at s^*. Since $\partial B / \partial \lambda = s$, this equals $s^* \big[v'(B^*) + c_R(1 - B^*; \Delta_i) - \pi_{ij}\big]$, where the term $-\pi_{ij} s$ comes from the room-sourced bridging $\pi_{ij}(1-\lambda)s$. This holds at corners as well. The slope in s is $\partial G_i / \partial s = \tau_i \big[v' + c_R\big] - \chi_i + \kappa_i$ with $\kappa_i = \pi_{ij}(1-\lambda) - \pi_{c,i}(1-\iota_i)$. At an interior optimum it is zero, so $v' + c_R = (\chi_i - \kappa_i)/\tau_i$. Using $\kappa_i + \tau_i \pi_{ij} = (\pi_{ij} - \pi_{c,i})(1 - \iota_i)$, we get $v' + c_R - \pi_{ij} = \big[\chi_i - \kappa_i - \tau_i \pi_{ij}\big]/\tau_i = M_i$, so $\Omega_i' = s^* M_i$.

At a corner, compare $v' + c_R$ with $(\chi_i - \kappa_i)/\tau_i$ through the sign of $\partial G_i / \partial s$. If $\tau_i > 0$ and $s^* = 1$, then $\partial G_i / \partial s \geq 0$, so $v' + c_R \geq (\chi_i - \kappa_i)/\tau_i$. If $\tau_i < 0$ and $s^* = \underline{s}$, then $\partial G_i / \partial s \leq 0$, and dividing by $\tau_i < 0$ gives the same inequality. These are the ceiling corners, and $\Omega_i' = s^*(v' + c_R - \pi_{ij}) \geq s^* M_i$ because $s^* \geq 0$. For the non-ceiling corners ($\tau_i > 0$ with $s^* = \underline{s}$, and $\tau_i < 0$ with $s^* = 1$) the inequalities reverse and $\Omega_i' \leq s^* M_i$. □

Proposition 10 (The marginal switcher prices wellbeing)**.** *Let a lower-caste student choose ex post between their mixed room and an alternative room, and let* Δ *denote the mixed room minus the alternative. Since* $U_i = S_i + N_i + \pi_g g_i$, $\Delta U_i = \Delta S_i + \Delta N_i + \pi_g \Delta g_i$*. A student who leaves the mixed room reveals* $-\Delta S_i > \Delta N_i + \pi_g \Delta g_i$*, a lower bound on the wellbeing loss in network-capital (placement) units; a stayer reveals the reverse; and a marginal (indifferent) student equates the two, point-identifying the wellbeing loss without a functional form for* c *or* χ_c*, given* π_g *and a measure of* ΔN_i*.*

Proof. U_i is additively separable in S_i, N_i and g_i. A student leaving the mixed room reveals $\Delta U_i < 0$, i.e. $\Delta S_i + \Delta N_i + \pi_g \Delta g_i < 0$, which rearranges to the stated inequality. A stayer reveals $\Delta U_i \geq 0$, the reverse. An indifferent marginal student has $\Delta U_i = 0$, so $-\Delta S_i = \Delta N_i + \pi_g \Delta g_i$, in which only π_g, which is estimable, and the observed or imputed ΔN_i and Δg_i appear. □

Remark (Heterogeneous trade-offs)**.** *The additive form treats consumption and investment as trading off one-for-one for every student. If the true objective is* $\omega_i S_i + (1 - \omega_i)(N_i + \pi_g g_i)$*, the marginal student satisfies* $-\Delta S_i = \frac{1-\omega_i}{\omega_i}(\Delta N_i + \pi_g \Delta g_i)$*, so the wellbeing loss recovered from switchers is contaminated by their own weight* ω_i*, and switchers are selected on proximity to indifference rather than on a representative* ω_i*. This is the logic of heterogeneous returns among the self-selected in Björklund and Moffitt (1987). Baseline mental-health data can speak to whether switchers differ from non-switching constrained students, but cannot by themselves separate a small wellbeing loss from a low weight on wellbeing. This remark is a placeholder for the welfare analysis.*

Theorem 1 (Break-even weight on belief convergence)**.** *Let the planner value* $W = \sum_i U_i + \omega_W \sum_i \alpha_i$*, with* α_i *(Appendix B) entering* W *but not* U_i *(Assumption 7). For a policy with per-student effects* $\Delta U < 0 < \Delta\alpha$ *(*ΔU *in network-capital units, Proposition 10), it is worthwhile if and only if* $\omega_W \geq$

$\bar{\omega}_W \equiv -\Delta U/\Delta\alpha$. Comparing two-person against partially-mixed three-person rooms, with near-identical academic gain and differences mainly in the wellbeing and attitude trade, and with $\Delta\alpha^{(2)} > \Delta\alpha^{(3)}$,

$$\bar{\omega}_W^{2v3} = \frac{\Delta U^{(3)} - \Delta U^{(2)}}{\Delta\alpha^{(2)} - \Delta\alpha^{(3)}}.$$

The threshold is informative without committing to a value of ω_W: if implausibly large, two-person rooms are not worth their wellbeing cost regardless of one's view on the value of the attitude shift; if small, the reverse.

Proof. Per student, $\Delta W = \Delta U + \omega_W \Delta\alpha$. Setting $\Delta W \geq 0$ and dividing by $\Delta\alpha > 0$ gives $\omega_W \geq -\Delta U/\Delta\alpha$. For the two-versus-three-person comparison, preferring configuration (2) requires $\Delta U^{(2)} + \omega_W \Delta\alpha^{(2)} \geq \Delta U^{(3)} + \omega_W \Delta\alpha^{(3)}$, which rearranges to the stated threshold after dividing by the positive quantity $\Delta\alpha^{(2)} - \Delta\alpha^{(3)}$. □